Supplementary information for

Deep time diversity and the early radiations of birds

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A. Supplemental Text A. Phylogenetic definitions used in this study.

Theropoda Marsh, 1881: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Saltasaurus loricatus* Bonaparte and Powell, 1980 and *Triceratops horridus* Marsh, 1889; Tetanurae: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Ceratosaurus nasicornis* Marsh 1884*;* Avetheropoda: the least inclusive clade containing *Passer domesticus* Linnaeus, 1758 and *Allosaurus fragilis* Marsh, 1877; Coelurosauria: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Allosaurus fragilis* Marsh 1877; Tyrannoraptora: the least inclusive clade containing *Tyrannosaurus rex* Osborn, 1905 and *Passer domesticus* Linnaeus, 1758; Maniraptoriformes: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Tyrannosaurus rex* Osborn 1905; Maniraptora: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Ornithomimus velox* Marsh 1890; Pennaraptora: the least inclusive clade containing *Oviraptor philoceratops* Osborn 1924, *Deinonychus antirrhopus* Ostrom 1969, and *Passer domesticus* Linnaeus, 1758; Paraves: the most inclusive clade containing *Deinonychus antirrhopus* Ostrom 1969*, Passer domesticus* Linnaeus, 1758, but not *Oviraptor philoceratops* Osborn 1924; Eumaniraptora: the least inclusive clade containing *Deinonychus antirrhopus* Ostrom 1969 and *Passer domesticus* Linnaeus, 1758; Avialae: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Deinonychus antirrhopus* Ostrom 1969; Pygostylia: the least inclusive clade containing *Passer domesticus* Linnaeus, 1758 and *Confuciusornis sanctus* Hou *et al.*, 1995; Ornithothoraces: the least inclusive clade containing *Passer domesticus* Linnaeus, 1758 and *Gobipteryx minuta* Elżanowski, 1974; Enantiornithes: the most inclusive clade containing *Gobipteryx minuta* Elżanowski, 1974, but not *Passer domesticus* Linnaeus, 1758; Euornithes: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Gobipteryx minuta* Elżanowski, 1974; Ornithurae: the least inclusive clade containing *Passer domesticus* Linnaeus, 1758, *Ichthyornis dispar* (Marsh 1872), and *Hesperornis regalis* Marsh 1872; Aves: the least inclusive clade containing *Passer domesticus* Linnaeus, 1758 and *Struthio camelus* Linnaeus, 1758; Palaeognathae: the most inclusive clade containing *Struthio camelus* Linnaeus, 1758, but not *Passer domesticus* Linnaeus, 1758; Neognathae: the most inclusive clade containing *Passer domesticus* Linnaeus, 1758, but not *Struthio camelus* Linnaeus, 1758

B. Supplemental Text B. Previous quantitative studies of evolutionary transition to birds.

The past decades have witnessed significant advances in our understanding of bird evolution owing a great deal to recent spectacular fossil discoveries, newly collected genomic data as well as new methods and analytical tools for large scale macroevolutionary analyses (1-6). In particular, several quantitative morphological evolution rate analyses have revealed sustained miniaturization (7, 8) and rapid morphological evolution along bird stem lineage (2, 9-11), and heterogenous and high morphological evolutionary rates among stem birds (12-14); one of the study has also revealed a higher lineage diversification rate in stem birds than in other theropods using a topological method (12, 15, 16). These studies thus provide support for the presence of a radiation among stem birds as suggested by a number of studies based on qualitative analyses (17-23). However, one study suggested that there was no statistically solid evidence for a significant difference between stem birds and other coelurosaurian theropods (2); another study using generalized linear mixed model also was unable to reveal a higher speciation rate in Mesozoic birds than in other theropods (24). Thus, the published quantitative analyses are in debate on the macroevolutionary process along the line to birds.

C. Supplemental Text C. Details of time-scaled supertree assembly.

Informal supertrees, which are assembled by manually joining different the source trees, have been widely used for various analyses including the lineage diversification rate through time (DRTT) analyses in paleontological studies (11, 12, 25). The alternative supermatrix approach for DRTT analysis performs a total-evidence analysis to infer the timetree and diversification rates simultaneously (6, 26, 27). It requires extensive overlapping data for the sampled taxa, but this is often practically difficult for the clade including both stem and crown taxa, because the available data for stem taxa is normally derived from morphology and that for crown taxa is mostly derived from DNA sequences. This is also true for bird diversification analyses, and we thus choose supertree approach in the present paper.

We assembled three time-scaled supertrees of birds combing published phylogenies of both stem and crown birds from six studies (3, 4, 13, 28-31) (Supertrees 1-3; **Figs. S1-3**). First, we assembled three backbone trees by combining one stem bird phylogeny to three crown bird phylogenetic trees, respectively (**Fig. S4C and D**; also see Methods). The three crown bird trees are all Bayesian molecular-clock calibrated trees and their median ages of divergence times were used.

The main stem bird tree is derived from ref.(13). Because Mrbayes requires bifurcating trees as input, one polytomies that includes *Vescornis*, *Neuquenornis*, *Gobipteryx* and *Eoalulavis* was resolved randomly (see https://figshare.com/s/53314d5f11da4106d173 for the original topology and **Figs. S1–3** for the result). The source tree has relatively good resolution and this operation should have neglectable impact. To date the divergence times under Bayesian tip dating, we used the dataset that contains 262 unpartitioned characters and 58 taxa (55 fossils and 2 extant birds). The Mkv model (32) with gamma rate variation across characters (33) was used in the inference. The divergence times were assigned a uniform tree prior (26) which is diffuse and relatively uninformative. We did not use the FBD prior (6, 34) at this stage to avoid introducing conflicting signal about the diversification and sampling processes and could potentially mislead the subsequent diversification rate analyses. The fossil ages were assigned uniform priors with lower and upper bounds from the corresponding stratigraphic ranges, which were revised from the data in the Paleobiology Database (PBDB) with updated isotopic dating data (13). The root age was assigned an offset exponential prior with a mean of 171 Ma and a minimum of 168 Ma. Two relaxed clock models, the white noise (WN) (35) and independent lognormal (ILN) (36) were used, and the one with a higher marginal likelihood (approximated by the harmonic mean estimator) was chosen for the supertrees. We executed two independent MCMC runs with four chains (one cold and three hot) per run for 20 million iterations and sampled every 1000 iterations. The first 25% samples were discarded as burn-in for each run, and the remaining samples from the two runs were combined after checking convergence between runs using Tracer (37). The posterior

median estimates of the node ages were used in this stem tree to combine with each of the three crown trees to form the backbone supertrees 1–3. The two extant birds (*Anas* and *Gallus*, which represent Neornithes) and *Vegavis* were pruned as their divergence estimates were unreliable due to lack of extant and fossil crown bird samples. The molecular timetree was then grafted onto the remainder branch while maintaining all branch lengths and node ages (**Fig. S4C and D**).

Then we added 18 additional Late Cretaceous birds from the Adams consensus tree in ref.(28) to the backbone trees (**Fig. S4E**). The data matrix of ref.(28) has 227 unpartitioned characters and 45 species. The root and tips that are identical with those in the tree of ref.(13) were given fixed ages as the posterior medians were estimated already (median root age is 168.47 Ma). The other fossil ages were assigned uniform distributions with lower and upper bounds collected from PBDB. The polytomies in the tree were also randomly resolved. We used the same MCMC settings to estimate the divergence times (same for below). Then, we manually inserted the 18 Late Cretaceous birds to each backbone supertree.

To add more fossils to the crown bird trees, we then dated the tree in ref.(29) and sampled 12 fossil species from the tree (**Fig. S4F**). The data matrix used contains 297 unpartitioned characters of 39 species. The root was given fixed ages as the posterior medians of *Ichthyornis*+Neornithes (i.e., Aves in present paper) node in the dated tree in ref.(13). The tips that are identical with those in the tree of ref.(13) were given fixed ages as the posterior medians were estimated already. The other fossil ages were assigned uniform distributions with lower and upper bounds collected from PBDB. We treated *Vegavis* as an outgroup of Neornithes, and the node age of *Vegavis*+Neornithes was limited between the root age of molecular time tree and the age of *Cimolopteryx petra*+Neornithes node. The node age of *Vegavis* + Neornithes estimated here was used in the supertrees rather than the result from ref.(13), because there are more fossils and extant Neornithes in ref.(29) than in ref.(13) and the node age of *Vegavis* + Neornithes should be inferred more accurately. In order to avoid contradictions with the molecular timetrees, we enforced several additional node constraints referring to the molecular timetrees. For supertrees 1 and 3, we constrained the ages of Neornithes, crown Anseriformes, crown Galliformes and Galloanserae to those in molecular timetrees (**Table S5**). For supertree 2, the node ages of Galloanserae and crown Anseriformes and *Anseranas semipalmata*+*Anas superciliosa* node in the molecular timetree are 54.61 Ma, 46.85 Ma and 39.99 Ma, which are younger than several fossil Galloanseraes in the tree of ref.(29), thus we manually adjusted the ages of these nodes in the molecular timetree to match the tip-dating estimates.

Additionally, we sampled 11 fossil taxa from the tree in ref.(31) (**Fig. S4A**). That data matrix has 290 unpartitioned characters of 48 taxa. We grafted *Vegavis* to the outgroup of Neornithes, and the tips that are identical with those in the tree of ref.(29) were given fixed ages as the posterior medians were estimated already. The other fossil ages were assigned uniform

distributions with lower and upper bounds collected from PBDB. In order to avoid the contradictions with previous dating results (molecular timetrees and time calibrated tree in ref.(29)), we enforced several node constraints (Neornithes, Anseriformes, Galliformes and Galloanserae) referring to the molecular timetree and dated phylogeny in ref.(29) (**Table S5**).

Finally, we sampled 7 fossil taxa from the tree in ref.(30) (**Fig. S4B**). The data matrix used in the dating analysis has 111 unpartitioned characters and 43 taxa. The fossil ages were assigned uniform distributions with lower and upper bounds collected from PBDB. We also enforced several additional node constraints (Passeriformes, Passeriformes+Psittaciformes, Coraciiformes, Neoaves and Neornithes) referring to the molecular timetrees to avoid contradictions (**Table S5**).

All these fossil lineages were grafted onto the backbone timetrees manually, with the phylogenetic positions, node ages and branch lengths referring to the time calibrated source trees. The medians of the posterior age estimates were used in the supertrees. The sampled fossil lineages are shown in **Fig. S5**. Other information of supertree construction is given in **Table S5**. There are 331 species in supertree 1 (102 fossil birds and 229 extant birds), 300 for supertree2 and supertree3 (102 fossil birds and 198 extant birds) (**Figs. S1-3**).

In our supertrees, *Vegavis* is treated as the outgroup of all Neornithes following ref.(29). *Cimolopteryx rara*, *Cimolopteryx minima*, *Cimolopteryx maxima*, *Cimolopteryx petra*, *Ceramornis major*, *Iaceornis marshii*, Ornithurine A, Ornithurine B, Ornithurine C, Ornithurine E, and Ornithurine F form a monophyletic group, which is closer to the crown birds than *Ichthyornis* and forms an outgroup of *Vegavis*+Neornithes. Ornithurine D is recovered as a sister taxon of *Ichthyornis*. *Palintropus retusus* is recovered as a sister taxon of *Apsaravis*. Hesperornithiform A and Hesperornithiform B form a clade, which is closer to *Baptornis advenus* than *Enaliornis*. Enantiornithine A and Enantiornithine B are closer to *Protopteryx* and *Avisaurus archibaldi* than *Pengornis* and *Eopengornis*. *Lithornis promiscuus*, *Lithornis plebius* and *Paracathartes howardae* form a monophyletic group, which is the outgroup of other crown Palaeognathae. *Asteriornis maastrichtensis* is an outgroup of Galloanserae. *Gallinuloides wyomingensis*, *Megavitiornis altirostris* and *Sylviornis neocaledoniae* form a paraphyletic group and placed outside of crown Galliformes. *Wilaru tedfordi* and *Presbyornis pervetus* form a clade, which is the sister group of *Anseranas semipalmata*. *Pelagornis chilensis* and *Protodontopteryx ruthae* form a clade, and *Conflicto antarcticus* and *Anatalavis oxford* form another clade. Both clades are placed outside of crown Anseriformes. *Gastornis giganteus*, *Gastornis parisensis*, *Barawertornis tedfordi*, *Ilbandornis lawsoni*, *Ilbandornis woodburnei*, *Dromornis murrayi*, *Dromornis planei*, *Dromornis stirtoni* and *Genyornis newtoni* form a monophyletic group, which is the sister group of Galliformes. *Zygodactylus luberonensis* and *Eozygodactylus americanus* form a clade and is the outgroup of Passeriformes. *Pseudasturides macrocephalus* and *Cyrilavis colburnorum* form a clade and is the outgroup of crown Psittaciformes. *Plesiocathartes kelleri*

is the sister taxa of *Leptosomus discolor*. *Palaeotodus itardiensis* is recovered as a sister taxon of *Todus angustirostris*. *Eocoracias brachyptera* is the sister taxa of the clade *Brachypteracias leptosomus*+ *Coracias caudata*. All supertrees were assembled by manually joining the stem and crown trees in Mesquite 3.6 (38). The supertrees are shown in **Figs. S1–3** and also provided as NEXUS tree files in Supplementary material 2.

D. Supplemental Text D. Details of the methods and results of diversification rate analyses.

We applied the skyline fossilized birth-death (SFBD) model (6, 34) in Mrbayes 3.2.8(39) to infer the diversification and fossil sampling rates changing through time, taking each of the three supertrees as input. We modified the source code of mrbayes to have more tolerance of decimal imprecisions (up to 6 digits) so that the supertrees can be read in, and the code is available from authors. Both the topology and times of each supertree were fixed in each analysis. The geological time was binned into 33 intervals of 5 Myr each.

To examine the robustness of rate estimates to the prior changes, we used exponential (50), exponential (20) (default, referring to the net speciation rates in ref.(4) which fluctuate around 0.05), and exponential (10) for the net speciation rate (speciation rate minus extinction rate). The range of net speciation rate is from zero to positive infinity, and the exponential prior puts less weight on higher net speciation rate. We used beta (1,1) (default, uniform prior), beta (2,2) (favors mid values with mean 0.5), beta (1,4) (favors small values with mean 0.2) and beta (4,1) (favors large values with mean 0.8) for the relative extinction rate (extinction rate over speciation rate, which is between 0 and 1), and also used beta (1,1) (default), beta(2,2), beta (1,4) and beta (4,1) for the relative fossil-sampling rate (fossil-sampling rate over the sum of extinction rate and fossil-sampling rate, which is also between 0 and 1). To reduce the number of parameter combinations, we varied one prior parameter at a time and kept the others to the default.

In all analyses, extant taxa were assumed sampling uniformly at random and the sampling proportion was set to 0.023 for Supertree 1 and 0.02 for Supertrees 2 and 3, based on the number of described extant species of birds at around 10000. The random sampling assumption is more suitable for supertrees 2 and 3 than for supertree 1. The extant sampling for supertree1 appears closer to diversified sampling. However, the diversified sampling implemented in MrBayes assumes that there is a cut-off time after which only one representative extant taxa per clade is sampled and no fossil sampling between the cut-off time and the present (6). It simply means that the unsampled extant taxa are all diverged after the cut-off time, leading to an artifact of decreasing the net diversification rate before the cut-off time and increasing it afterwards. The diversified sampling assumption appears too extreme and thus we did not include this result in this study. Although the random sampling assumption is the other end of the extreme, and the actual sampling lines in between these two assumptions, we argue, however, it should not affect divergences deep in the tree, as unsampled extant taxa only contribute to recent divergences (thus the diversification rates in the last few intervals).

Each analysis was run for 40 million iterations and sampled every 1000 iterations (see details in Mrbayes command files). The first 10% samples were discarded as burn-in. Convergence was examined using Tracer (37). The skyline plots were drawn using "plotSkylinePretty" in package "bdskytools" for R3.6.1 (40). This package was still in development and had to be installed over Github (41) (https://taming-the-beast.org/tutorials/Skyline-plots). All results of net diversification rate are given in **Figs. S6-8**, and results of relative extinction rate and relative fossil-sampling rate are given in **Tables S6-7**. We also estimate the value of speciation, extinction and fossilization rates under the default prior assumptions, and results are given in **Figs. S9-11**.

We also applied the lineage-specific multitype birth-death (MTBD) model (42) to estimate the diversification rate variation based on Supertrees 1-3, using the MSBD package in Beast v2.6 (43), which claimed to overcome a few shortcomings in the BAMM implementation (44), with the extant species sampling proportion set as 0.023 for supertree 1 and 0.02 for supertree 2 and 3 (same as in the SFBD analyses before). In consideration that many extinct species are not represented in the fossil record and are not possible to be included in our study, we used 0.05 as the sampling rate of extinct species for all analyses. For speciation and extinction rates, we used the same lognormal(–3, 2.0) prior distribution (mean = 0.37 and median = 0.05) which fluctuates around 0.1 (referring to previous SFBD analyses and in ref.(45)). The method was reported sensitive to the prior for the overall rate of change (γ) (42), thus, we repeated our analyses for each supertree with three sets of parameters in this prior, which include lognormal(4,1) (favoring large rate), lognormal(–4,1) (favoring medium rate) and lognormal(–8,1) (favoring lower rate). Results are given in **Figs. S12-20**

The analyses detected three diversification rates in the tree, but failed to detect any rate shift in stem birds. Similar behavior has also been reported in other studies (44, 46), and whether the lineage-specific model is applicable to extinct clades has been questioned by some studies (46). The lineage-specific model has been observed sensitive to the prior and tends to detect few large rate shifts in the tree (47, 48). However, unlike the empirical datasets which contain only extant taxa in the trees where the number of rate shifts is quite sensitive to the prior (especially for γ, the overall rate of change) (42), our analyses revealed the same pattern of major diversification rate changes in the tree, with minor differences among different priors. This is probably due to the abundant fossils included in our supertrees which contain much more information to inform the diversification rates.

Another reason for the failure of detecting rate shifts among stem birds in our analyses might be that the extinct species sampling rate is assumed constant over time in the model and the software implementation, which is against the fact that the bird fossil record is clearly not uniformly sampled over time, and thus the change in diversity may be inaccurately inferred (44).

As to the different results from our study and some previous studies of the lineage diversification rate for stem and extant birds, we provide the following interpretations. The topological method used in ref.(12) to detect shifts of lineage diversification rates in stem birds is not suitable for extinct taxa, because it compares the species diversity to expectations under the equal-rate Markov model, which is a pure-birth Markov process (16). Thus, the terminal taxa branching from the ancestral node should at the same age (ultrametric tree). If extinct (fossil) taxa are used, all those taxa should be pruned to the same age, which might lead to loss of information. Topological method also cannot estimate the values of net diversification rate in different time intervals. Thus, it is unlikely to reveal the down turn period of net diversification rate during 125-95 Ma.

Ref. (24) discussed the changing of diversification rate of dinosaurs by using generalized linear mixed model and concluded that the speciation dynamics in Mesozoic birds are not distinct from those of non-avian theropods. They used a Bayesian approach to estimate the regression parameters (intercept and slope) in the regression equation of node count against time, and predicted the node count using the parameters of the regression model and times taken at 1Myr interval and took the difference between adjacent predictions. This gives the change of node count per Myr as a reference to the net speciation rate in ref.(24). In other words, the net speciation rate is not a direct parameter in the model. In our approach instead, the net speciation rate is a parameter in the SFBD model and can be estimated directly under the Bayesian approach.

E. Supplemental Text E. Additional discussions on methological validity of supertree-based diversification rate analyses.

A number of studies, mostly using total-evidence approach and testing the correlations between morphological evolution, molecular evolution, and lineage diversification, have discovered variable relationships between the estimated evolutionary rates from different types of data (molecular vs. morphology) and their relationships to lineage diversification (49-55). In general, it seems that there are more cases in plants (51, 53, 56) than in animals (50, 51, 57) showing a correlation between morphological and molecular evolutionary rates. A number of studies find correlations between molecular evolutionary rates and diversification rates or species richness (53, 58-60) or between morphological evolutionary rates and diversification rate (61) in plants. Among animals, a correlation has been found between morphological evolution and speciation rate or species richness in ray-finned fishes (62) and birds (63, 64), between morphological evolution and species richness in salamanders (65), and between molecular evolution and species richness and diversification rate in birds and reptiles (66, 67), but a correlation has not be detected between morphological and molecular evolution in flatworms (57) and mammals (68), between morphological evolution and diversification rate in salamanders (55), squamates (54) and mammals (69) and between molecular evolution and diversification rate in mammals (70). For the morphological evolution, one study also noticed different evolutionary rates for different morphological features (69).

Although these studies demonstrate the complexity in the relationships of morphological evolution, molecular evolution, and lineage diversification, there seems to be empirical evidence supporting the correlations between morphological and molecular evolution and lineage diversification in some reptiles and birds (63, 64, 66, 67), which provide some justifications for our estimating diversification rate based on information derived from different data types.

Methologically, we applied different (and independent) evolutionary models for morphological and molecular evolution when building Supertrees 1-3. For the stem group which contains only fossils and morphological characters, we used the Mkv model (32), the WN and ILN clocks (35, 36), and the Bayesian tip-dating approach (using fossil ages as tip calibrations) to infer the branch lengths (measured in myr). For the crown group which contains only molecular data (fossil data were used as node calibrations), the trees from previous studies (3, 4) were inferred with Bayesian node-dating approach (calibrating internal nodes using fossil information), ILN relaxed clock, and GTR+Γ substitution models (3, 4, 33, 71, 72). Thus, each model captures the nature of different data types (morphological vs molecular, stem vs crown/extant) and the models have been widely adopted to suite these types of analyses (73-75).

The diversification rates were inferred from the supertrees and corresponding node and tip ages in the trees, independent of the morphological and molecular data. As long as the supertrees and ages in the trees are inferred with well justified methods, the diversification rate estimates, which reflect the patterns of speciation and extinction over time, should not be misled by the difference or the transition of data types.

The supertree can be viewed as a reconstructed history of the included taxa which have undergone the process of speciation, extinction and subsequent sampling of extinct and extant taxa. This is a unified process started from the most recent common ancestor of these taxa and continued until the present, and is modeled explicitly in the SFBD model. Because the unit of the diversification rate is the same over time (as a unified process), direct comparisons of the diversification rates through time and between the stem and crown groups can be made.

F. Supplemental Text F. Details of the methods and results of morphological evolution rate analyses.

For the morphological evolution analyses, we used a modified TWIG (Theropod working group) discrete character matrix (2), with five wildcard taxa (*Kinnareemimus*, *Epidendrosaurus*, *Pyroraptor*, *Hesperonychus*, and *Limenavis*) deleted. Following the note of ref.(2), the character coding of these five taxa is highly incomplete and might induce large uncertainties in the Bayesian phylogenetic analyses. We also added *Caihong Juji* (76) into our data matrix. This taxon is a newly discovered Yanliao theropod with a nearly complete skeleton and bears a combination of features consistent with proposed rapid character evolution and significant diversity in locomotor strategies near bird origins (76), which is important for our understanding of morphological evolution of the bird stem lineages. The dataset was partitioned into six anatomical regions including skull, axial (including vertebrae, ribs and gastralia), pectoral girdle (including pectoral girdle, sternum and furcula), forelimb, pelvis, and hindlimb. The characters of each region are given in the MrBayes command files. Since Mrbayes only supports ordered characters up to six states, character 108 with 8 states had to be defined as unordered. The missing and inapplicable characters are averaged over in the likelihood calculation.

The tree topology and divergence times were co-estimated from the data. A uniform prior was used for the time tree. We also performed analyses on FBD prior, however, the MCMC was not convergence. The root age was assigned an offset-exponential prior with mean 181 Mya and minimum 177 Mya, which is slightly older than the unequivocal Avetheropoda (=Neotetanurae) (77). Each fossil age was assigned a uniform distribution with the data collected from the PBDB. Both the WN and ILN relaxed clock models were used in our analyses. The base rate of the clock was assigned a diffuse normal distribution prior with mean 0.01 (expecting one change per 100 characters per Myr) and very large standard deviation 0.05. The variance parameter of the clock was given default exponential priors (see MrBayes commands for details). The variance parameter under the WN or ILN model was unlinked to allow evolutionary rate heterogeneity across partitions. Each analysis was run 300 million iterations and sampled every 2000 iterations. The first 25% samples were discarded as burn-in. We also conducted analyses based on the unpartitioned matrix with 60 million iterations (25% burn-in). Convergence of parameters was identified using Tracer (37). All original tree figures were drawn in Figtree (http://tree.bio.ed.ac.uk/software/figtree), and the tree topology and results of parameter estimation are given in supplementary tree files.

In consideration of the phylogenetic uncertainty in Paraves, we repeated our analyses under both the monophyletic Avialae + Dromaeosauridae and Avialae + Deinonychosauria phylogenetic constraints. For the partitioned models, our analyses show that birds and

troodontids form a monophyletic group when the search is performed without any constraint within Paraves. The results of our analyses are briefly described below. For the convenience of the description, we use stem lineage here referring to the first branch of stem-based clades or the immediate stemward branch of the node-based clades described.

In general, rates of morphological evolution in different anatomical regions are extremely heterogenous. The highest skull rates are seen along the branches between Tyrannoraptora and Ornithomimosauria + Pennaraptora nodes and between Eumaniraptora and Troodontidae + Avialae nodes, slightly lower ones seen along the oviraptorosaurian, troodontid, and neognath stem lineages and also several internal branches in late-diverging tyrannosauroids, and relatively high ones along the Pygostylia stem lineage. However, the skull rates are low along the branches between the Eumaniraptora and *Archaeopteryx* + Pygostylia nodes when the search is implemented with a monophyletic constraint of Avialae + Dromaeosauridae or Avialae + Deinonychosauria. In these cases, high rates can be detected along troodontid stem lineage or the deinonychosaurian stem lineage. This suggests that a rapid skull evolution along the Avialae + Troodontidae stem lineage detected by our earlier analysis without phylogenetic constraints is not induced by cranial modifications in birds, but by those in troodontids.

Similar to the skull evolution, the highest axial rates are seen along the maniraptoriform stem lineage. The axial evolutionary rates are still high along the Maniraptora and the *Sapeornis* + Ornithothoraces clade stem lineages and relatively high along the branches between the Alvarezsauria + Pennaraptora and Eumaniraptora nodes.

The highest pectoral rates are present along the *Jeholornis* + Pygostylia stem lineage, and slightly lower rates are seen along the branches between the Alvarezsauria + Pennaraptora and Eumaniraptora nodes. Relatively high rates are also seen along the phylogenetically earliest internal branches of Ornithothoraces.

The highest forelimb evolutionary rates are seen within early stem birds, i.e., along the branches between the *Jeholornis* + Pygostylia and Ornithothoraces nodes. The forelimb also evolved fast along the Manirapora and Pennaraptora stem lineages.

The highest pelvic rates are seen along the Maniraptora and Therizinosauria + Pennaraptora stem lineages, and are lower but still high along the Pennarapora and Paraves stem lineages. High evolutionary rates in pelvis also are seen within the Therizinosauria, and a slightly higher one along the *Sapeornis* + Pygostylia stem lineage.

The hindlimb evolved extremely fast within stem birds, i.e., along the *Sapeornis* + Pygostylia, Pygostylia, and Euornithes stem lineages, and along an early therizinosaurian branch. High

hindlimb evolutionary rates are also present along the maniraptoran, pennaraptoran, and paravian stem lineages and an alvarezsaurian internal branch than along most other lineages.

In general, there are two approaches to estimating morphological evolutionary rates in terms of data types used in the analyses, i.e., using data from a morphological feature or using a phylogenetic data matrix, and both approaches have been used in understanding the evolution of stem birds and their dinosaurian relatives (2, 8-14). For example, several studies estimated the morphological evolutionary rates of theropods including birds based on limb or body size measurements (8, 10-12) and this approach has the advantage of using the same morphological character to quantitatively compare the morphological evolution of different lineages.

The data-matrix-based approach has also been used in morphological evolutionary rate studies pertaining to the transition to birds (2, 8, 9, 13, 14, 78), and it has the advantage of holistically examining the morphological evolution, and is able to give an unbiased result given the prevalence of mosaic evolution. The partitioned model of data-matrix-based approach has the advantage of detecting more specifically which lineage evolves faster in a particular body region. However, if the comparison is made between the evolutionary rates of one body region that contains more characters (with more signals) and another body region with fewer characters, the result is likely to be biased toward the body region with more characters. However, in our case, the comparison of the evolutionary rates is made among different lineages within the same body region and thus the bias can be avoided.

G. Supplemental Text G. Additional discussion on coelurosaurian morphological evolution.

The morphological evolution in general is rapid along the branches between the Tyrannoraptora and Ornithothoraces nodes, and the highest rates are seen along the branches between the Tyrannoraptora and Ornithomimosauria + Pennaraptora nodes and between the Therizinosauria + Pennaraptora and Oviraptorosauria + Pennaraptora nodes. Relatively rapid morphological evolution also is seen along the internal branches within the Tyrannosauroidea, Therizinosauria, and Avialae.

Based on the maximum evolutionary rates, the fastest evolving anatomical regions are the forelimb (along the branch between the Pygostylia and Ornithothoraces nodes) and the shoulder girdle (along the branch between the *Sapeornis* + Pygostylia and *Jeholornis* + Pygostylia nodes), and then followed by the cranial skeleton (along the branches between the Tyrannoraptora and Ornithomimosauria + Pennaraptora nodes and between the Eumaniraptora and Troodontidae + Avialae nodes) and the axial skeleton (along the branches between the Tyrannoraptora and Ornithomimosauria + Pennaraptora nodes). The hindlimb evolved slower than other body regions among coelurosaurians, but it evolved considerably faster independently along several branches among early stem birds than other coelurosaurian lineages.

These results suggest that bird-line coelurosaurians had a relatively slow beginning of morphological evolution but evolved much faster subsequently, continuing into the early part of bird phylogeny. This fast-evolving part of coelurosaurian phylogeny is linked with a more diverse body-size, particularly toward the small-size end (79), and a more diverse diet from the strictly carnivorous one (80, 81), suggesting that maniraptoriforms invaded new niches compared with non-maniraptoriform theropods (81).

These results also reveal different evolutionary behaviors of different anatomical regions across the coelurosaurian phylogeny. Comparatively, the quick cranial evolution occurred more independently in different coelurosaurian lineages than the shoulder girdle and pelvis, which had a more continuous rapid evolution along the line to Paraves; and the other regions display an intermediate condition.

Moreover, the results highlight two important evolutionary events: the origins of Maniraptoriformes and Pennaraptora. The origin of Pennaraptora is considerably different from that of the Maniraptoriformes in which skeletal regions were involved and how fast the skeletal regions evolved. The origin of Maniraptoriformes corresponds to accelerated evolution in cranial and axial skeletons. Previous studies demonstrated significant

modifications of cranial and axial skeletons and the appearance of omnivory at the base of Maniraptoriformes (80-82), and suggested that the relaxion of body size and pneumatization and the evolution of higher metabolic rates might be related to the axial skeleton evolution along the branches around the Maniraptorformes (83-85).

The origin of Pennaraptora corresponds to quick evolution in axial, shoulder girdle, forelimb, pelvis, and hindlimb skeletons. None of them evolved particularly fast near the origin of Pennaraptora, but nearly the whole skeleton except the cranial part evolved at a relatively high rate compared to the average morphological evolutionary rate seen in coelurosaurians. The origin of Pennaraptora is associated with a number of important modifications pertaining to flight (1) including a significant physiological change (86).

The morphological modifications among early stem birds are featured by rapid evolution related to locomotory systems such as shoulder girdle, forelimb, and hindlimb, but by slow evolution in the cranial and pelvic skeletons, and by an intermediate one related to axial skeleton; early neognath birds display a rapid evolution in cranial skeleton. Previous studies highlighted the increase of cervical numbers by homeosis and the fusion and modification of posterior caudals into a pygostyle (87, 88), several modifications related to the anterior shift of gravitational center and the unique crouched legs (89-91), and the evolution of wings (92-94) among early stem birds.

H. Supplemental Text H. Details of the methods and results of morphospace analyses.

We performed distinct morphospace comparisons apart from the evolutionary rate analyses using the same modified character matrix (2). We considered the following phylogenetic uncertainty. *Coelurus* and *Tanycolagreus* formed a sister group at the base of Tyrannosauroidea in several previous studies (2, 95, 96). This result is also supported in almost all of our analyses except under the partitioned-matrix WN model. *Epidexipteryx* and *Pedopenna* were at the base of Oviraptorsauria, similar to ref.(2) but different from other studies (97, 98). *Ornitholestes hermanni* was placed at the base of Compsognathidae lineage in all WN clock and unpartitioned-matrix ILN clock analyses, but was placed elsewhere in other studies (95, 96). Finally, *Anchiornis*, *Aurornis*, *Eosinopteryx*, *Caihong juji*, and *Xiaotingia* form a monophyletic group at the base of Troodontidae, except for the analyses performed under the Avialae + Deinonychosauria constraint, where they were placed as the outgroup as other Deinonychosauria. These phylogenetic uncertainties were taken into account as sensitivity tests when comparing birds with other coelurosaurian clades (**Tables S1-3**).

The analyses were performed in R 3.6.1 (40). The dataset was divided into six cladistic matrixes and each cladistic matrix (including the cladistic matrix of all characters) was transformed into a distance matrix for subsequent principle coordinate analyses. Several previous studies incorporated reconstructed ancestral states for internal nodes or recovered missing data for tips into the subsequent disparity analyses (99, 100). However, these approaches might have introduced erroneous phylogenetic signals and led to a compact morphospace (101). Thus, we did not reconstruct these unknown data in our analyses. Instead, we used a better solution which can provide a good fidelity with high percentage of missing data.

We ran our analyses on the Maximum Observable Rescaled Distance (MORD) matrix (102) and repeated all analyses on the Generalized Euclidean Distance (GED) method (103) as a robustness test. Recent research has revealed that GED matrix creates a systematic bias in cases of high percentage of missing data and MORD matrix can provide a better fidelity under these circumstances (102, 104). The principle coordinate analyses were performed using the function “MorphMatrix2PCoA” in package “Claddis” (102), which can trim a distance matrix beforehand by removing the taxa that cause empty cells. Two disparity metrics were used to evaluate the volume of morphospace, including the sum and product of the variances (105). We did not use those range metrics (sum of ranges and product of ranges) in these analyses because these metrics are sensitive to uneven sample sizes between the two test groups. The product metrics were normalized by taking the nth root (n equals to the number of axes used for calculating disparity metrics). We used the scores on all axes that together comprise 90% of total variance to calculate those disparity metrics (whole skeleton

76 axes; skull 62 axes; axial 63 axes; pectoral girdle 58 axes; forelimb 60 axes; pelvis 64 axes; hindlimb 71 axes).

To test the hypothesis that the disparity of birds is different from that of other coelurosaurian clades against the null hypothesis of no difference, each permutation test (two-tailed) was run 2000 replications (by randomly permuting taxa into different groups). The test statistic was obtained using the disparity metric of birds minus that of other coelurosaurian clades. The value in **Table S1** were the proportion greater than the observed value of the test statistic in the null distribution. If the value is smaller than 0.025, it means the disparity of birds is significantly larger than other coelurosaurian clades. On the other hand, if the value is larger than 0.975, it means the disparity of coelurosaurian clades is significantly larger than that of birds. All observed values of test statistics are given in **Table S2**.

Most tests based on unpartitioned data matrix are significant regardless of whether or not including the six Cenozoic species and the selection of distance matrix. This provides strong evidence for the hypothesis that Mesozoic birds occupied a larger morphospace than other coelurosaurian clades.

The tests based on partitioned data matrix produced different results for different anatomical regions. The difference in skull disparity of birds and other coelurosaurian clades is generally insignificant statistically. This pattern is robust in tests based on both MORD and GED matrix, leading to a conclusion that no evidence can confirm that birds have a larger skull disparity compared with other coelurosaurian clades. For the axial region, deletion of Cenozoic species leads to more insignificant results, suggesting that Mesozoic birds do not have greater axial disparity than other coelurosaurian clades. The difference in pectoral girdle disparity of birds and other coelurosaurian clades is generally significant statistically. The results remain robust under different sampling strategies (whether or not including six Cenozoic birds) or using different distance matrices, supporting that the hypothesis that Mesozoic birds occupied a larger pectoral girdle morphospace than non-avialan coelurosaurian clades. The difference in forelimb disparity of birds and other coelurosaurian clades is also significant statistically. For the pelvis disparity, the difference between birds and non-maniraptoran coelurosaurian clades is significant, but that between birds and other maniraptoran clades is insignificant when the analyses are conducted on a MORD matrix. However, our analyses based on a GED matrix generated more insignificant results. This suggests that no evidence can confirm that Mesozoic birds have a larger pelvis disparity than other coelurosaurian clades. When the hindlimb disparity is tested based on a MORD matrix, most tests are significant and a few insignificant results are derived from comparisons between birds and Oviraptorsauria. However, this result is not robust under the deletion of six Cenozoic birds and the usage of GED matrix. GED matrix creates a systematic bias in cases with high percentage of missing data, and MORD matrix can provides a better fidelity under these circumstances (102, 104). Given that our matrix has a relatively high percentage of

missing data, a feature present in most paleontological dataset, the MORD results are preferred here, i.e., that Mesozoic birds have a larger hindlimb disparity compared with non-avialan coelurosaurian clades.

We also performed permutation tests with sample size corrected and added those range metrics. For two designated groups with different sample sizes, we first performed random subsampling of birds to obtain equal size of samples as in the dinosaur group before each permutation to take the unequal into account (similarly, if the sample size of dinosaur group is larger than birds, we performed subsampling of the dinosaur group). Based on the newly obtained two groups with equal sample sizes, we calculated the observed value of the test statistic. Then we randomly permuted those species into different groups one time and calculated a test statistic. We repeated this procedure (subsampling and permutation) 10000 times and obtained a null distribution and a group of observed values. We then calculated a set of proportions greater than the observed values in the null distribution. Analogically, if the median of the proportions is equal or smaller than 0.025, it means the disparity of birds is significantly larger than other coelurosaurian clades; if the median value is larger than 0.975, it means the disparity of coelurosaurian clades is significantly larger than that of birds. All results are given in **Figs. S26-39**. The conclusions of these new tests were generally the same as our previous tests.

To test the hypothesis that birds have a morphospace different from other coelurosarian clades, we ran the permutational multivariate analysis of variance (PERMANOVA) (106) test using the original distance matrix (MORD and GED matrix) as input and ran 3000 permutations to calculate the null distribution of pseudo F ratio. These tests were also performed in R 3.6.1, using the "adonis2" function in package "vegan"(107). All results are given in **Table S3**. All plots of 3D morphospaces were drawn using package "scatterplot3d" (108) in R 3.6.1(40).

Most results of PERMANOVA are significant, indicating that birds occupy a different area in the multivariate morphospace compared with other coelurosaurian clades in all anatomical regions. This conclusion remains robust when the analyses are conducted with Cenozoic birds deleted, several taxa being placed in different systematic positions, and different distance matrices being used.

To test whether Mesozoic birds differ from their closely-related clades significantly more than other coelurosarian groups differ from their closely-related clades, we performed comparative distance test with the distance matrix (the MORD and GED matrix for whole skeleton and all anatomical regions) as the original input in R 3.6.1 (compare the distances between each Mesozoic bird and each of the species in their close relatives to the distances between each species in non-avian coelurosaurian clades and each one in their closely related clades). The Mann-Whitney-Wilcoxon test was used because some data doesn't obey normal distribution.

We carried out three analyses as in ref.(2) for all anatomical subregions and the whole skeleton. Nearly all results were significant according to the Mann-Whitney-Wilcoxon tests, and Mesozoic birds differ from their closely-related clades significantly more than other coelurosarian groups differ from their closely-related clades in all postcranial subregions and the whole skeleton, but not in cranial subregion. In conclusion, we have found that the differences between Mesozoic birds and their close relatives were significantly different from the differences between other coelurosarian groups and their close relatives in all anatomical subregions (**Table S4**), which implies that birds are an outlier compared to other coelurosarian groups in morphospace.

We note two differences of the morphospace analyses between ref.(2) and our study. One is that they did not perform analyses on different anatomical subregions, and the other is that ref.(2) used a method different from ours. Specifically, ref.(2) calculated the Mahalanobis distance between the two group means and compared with a null distribution of between-group distances obtained by randomly permuting taxa into different groups. The variance-covariance matrix in the Mahalanobis distance calculation is pooled from the variance-covariance matrices of the two groups (109, 110). This metric is only valid when the two groups have similar variance–covariance matrices ((110) and (109) part 3.2). In another word, testing the similarity of variance-covariance matrices of the two groups is required (which can be done, e.g., in the Box M's Test (111-113), otherwise, the Mahalanobis distance calculated this way may be misleading. However, they did not perform such test and their permutation test is likely biased by the dissimilarity of variance–covariance matrices between different coelurosaurian clades. In our study, PERMANOVA was used to test the overlap in morphospace occupation between birds and other coelurosaurian clades. This method is widely used in testing morphospace occupation (114, 115), and does not require assessing the similarity of variance–covariance matrix between groups.

I. Other material on Figshare.

Supplementary material 1 MrBayes tip dating

Supplementary material 2 Supertrees and source tree topology

Supplementary material 3 MrBayes diversification rates

Supplementary material 4 Code and result of lineage specific model (MTBD implement in Beast 2.6)

Supplementary material 5 MrBayes morphological rates

Supplementary material 6 MORD and GED matrices for principle coordinate analysis, PERMANOVA and comparative distance test

Supplementary material 7 R scripts

J. Supplemental Figures.

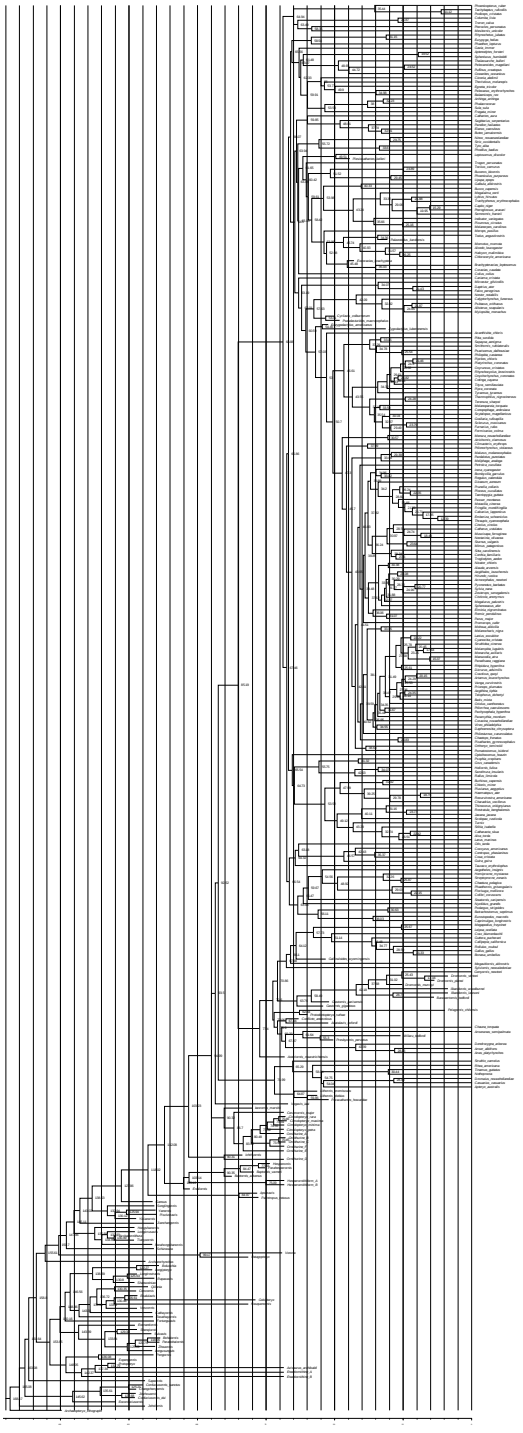


Figure S1. Time-scaled informal supertree 1. Numbers at each node are ages (posterior medians).

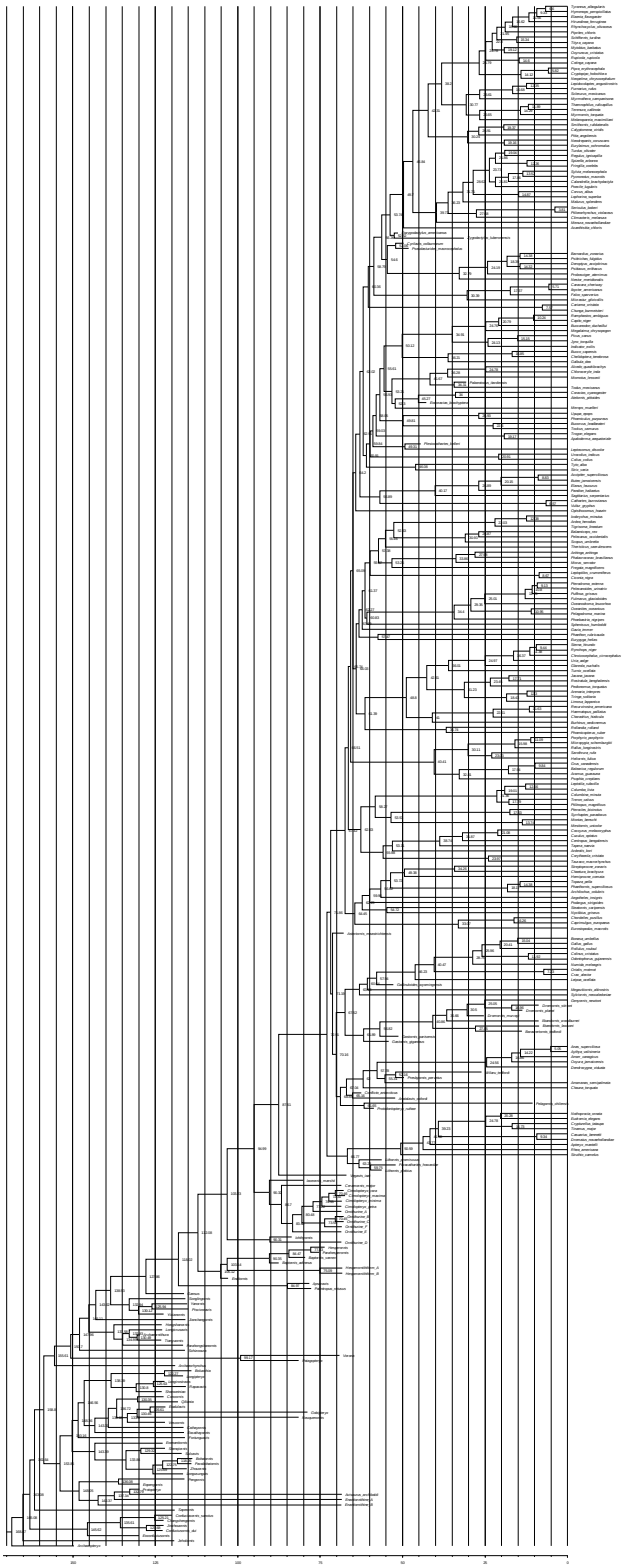


Figure S2. Time-scaled informal supertree 2. Numbers at each node are ages (posterior medians).

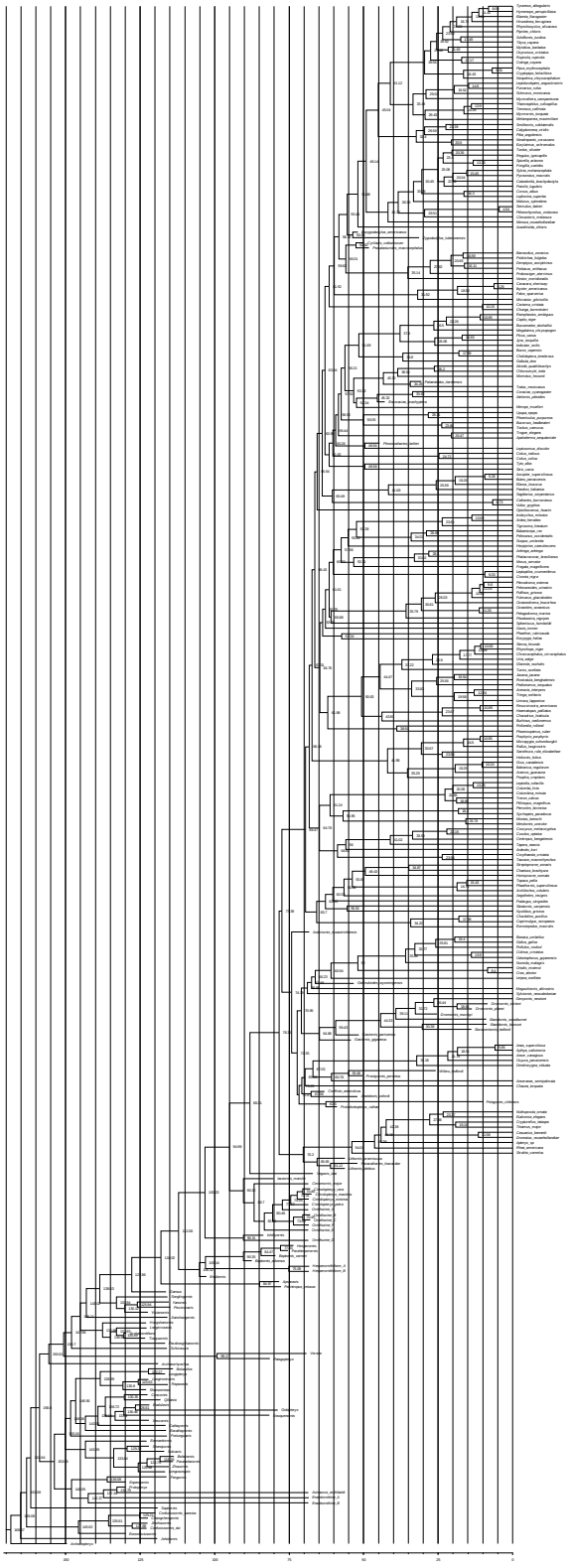


Figure S3. Time-scaled informal supertree 3. Numbers at each node are ages (posterior medians).

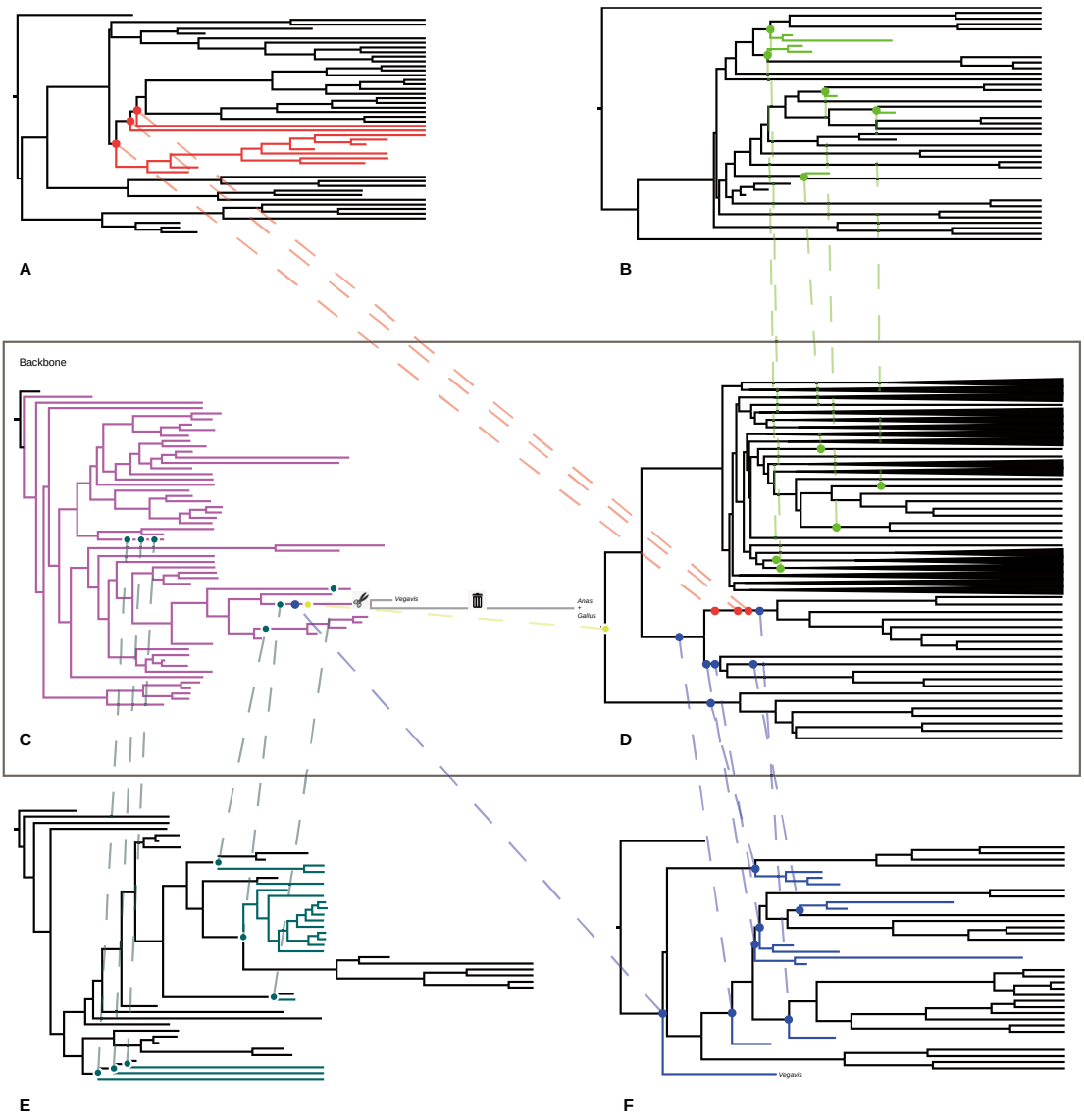


Figure S4. Sketch map of joining a supertree. (A) Dated tree in ref.(31). (B) Dated tree in ref.(30). (C) Dated tree in ref.(13). (D) A molecular timetree in ref.(4) or ref.(3). (E) Dated tree in ref.(28). (F) Dated tree in ref.(29). The scissors mean pruning branches, and the trash can means discarding branches. Fossil lineages sampled for supertree construction are painted into different colors.

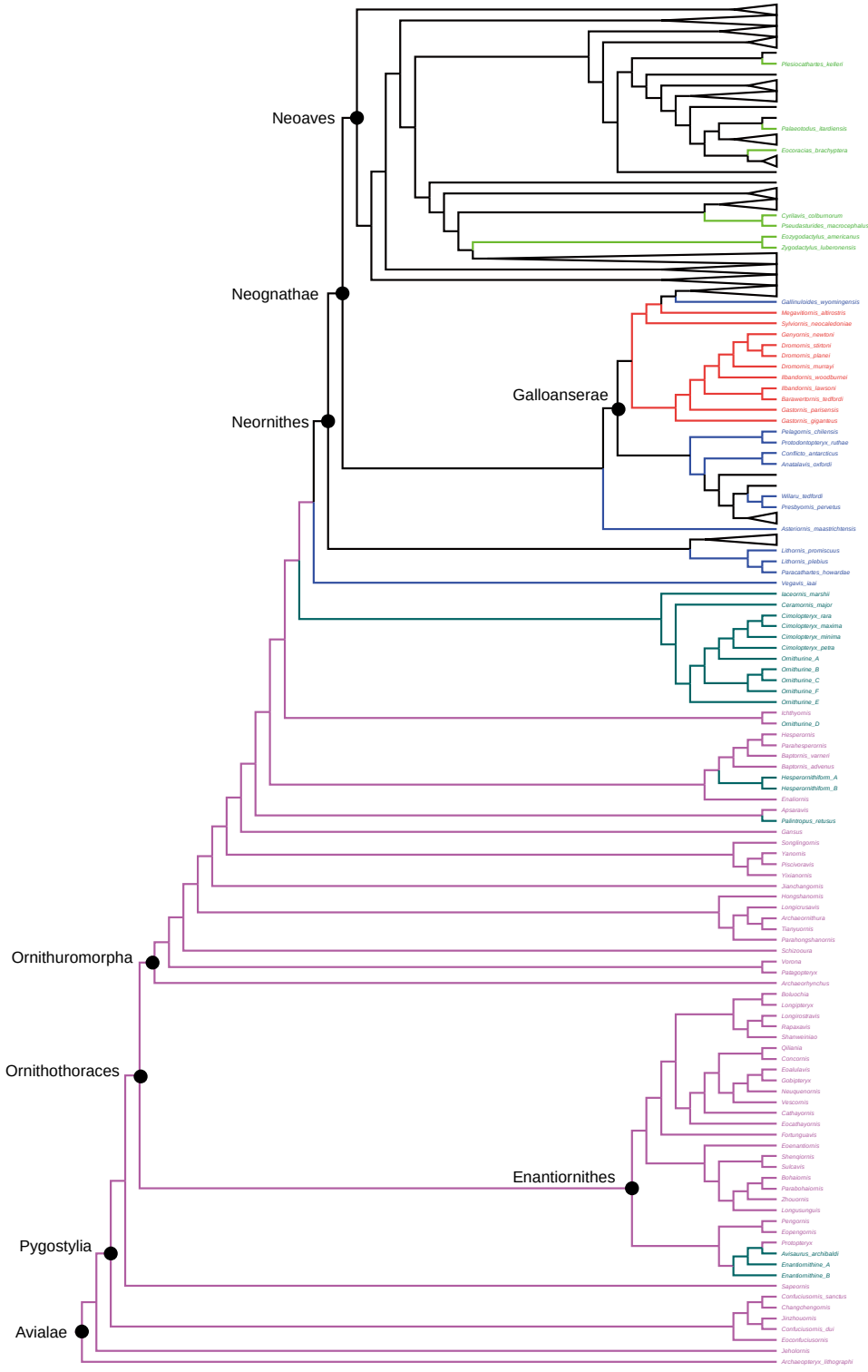


Figure S5. Cladogram of a supertree. Fossil lineages colored purple are sampled from the tree in ref.(13), deep green from that in ref.(28), blue from that in ref.(29), red from that in ref.(31), and light green from that in ref.(30). Also see **Figure S4**.

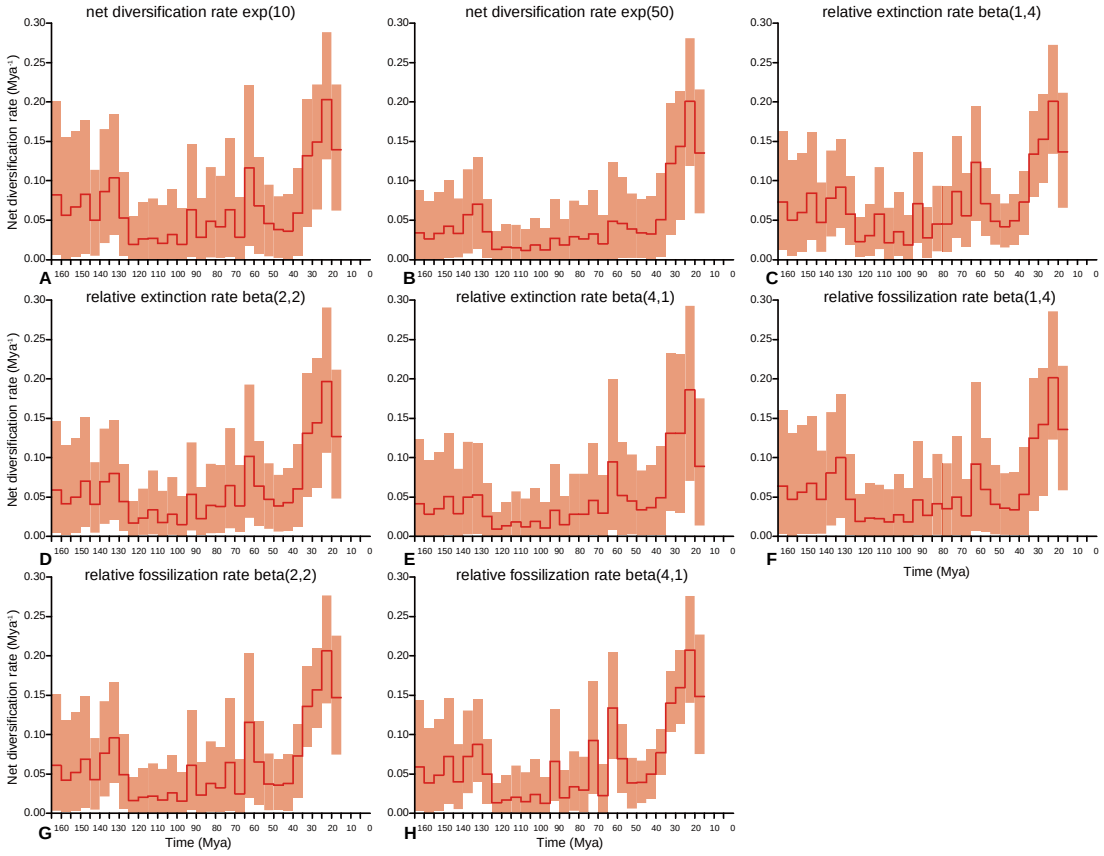


Figure S6. Diversification of birds through time estimated by using Supertree 1 based on different prior assumptions. Net diversification rate with exponential (10) (A) and exponential (50) (B); Relative extinction rate with beta (1,4) (C), beta (2,2) (D) and beta (4,1) (E); Relative fossilization rate with beta (1,4) (F), beta (2,2) (G) and beta (4,1) (H). Red boxes around medians are 95% HPD intervals.

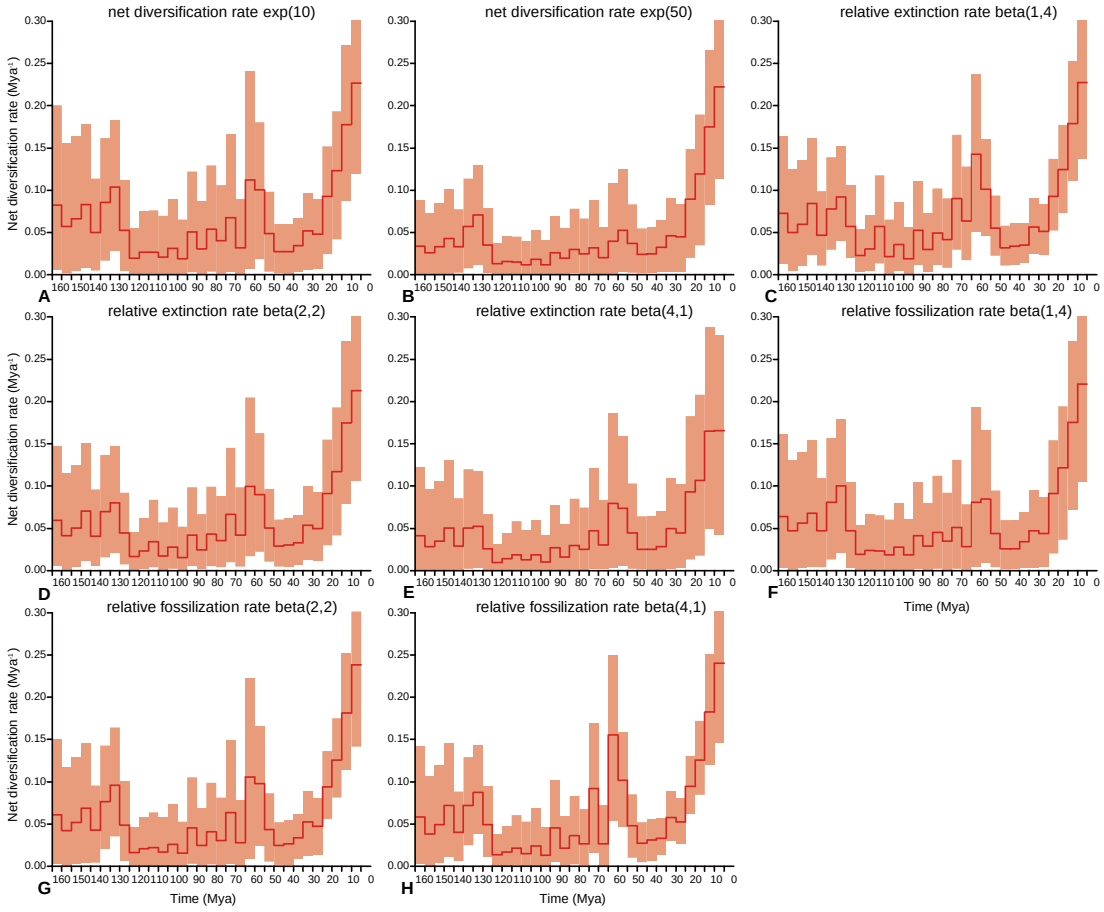


Figure S7. Diversification of birds through time estimated by using Supertree 2 based on different prior assumptions. See legend of **Figure S6**.

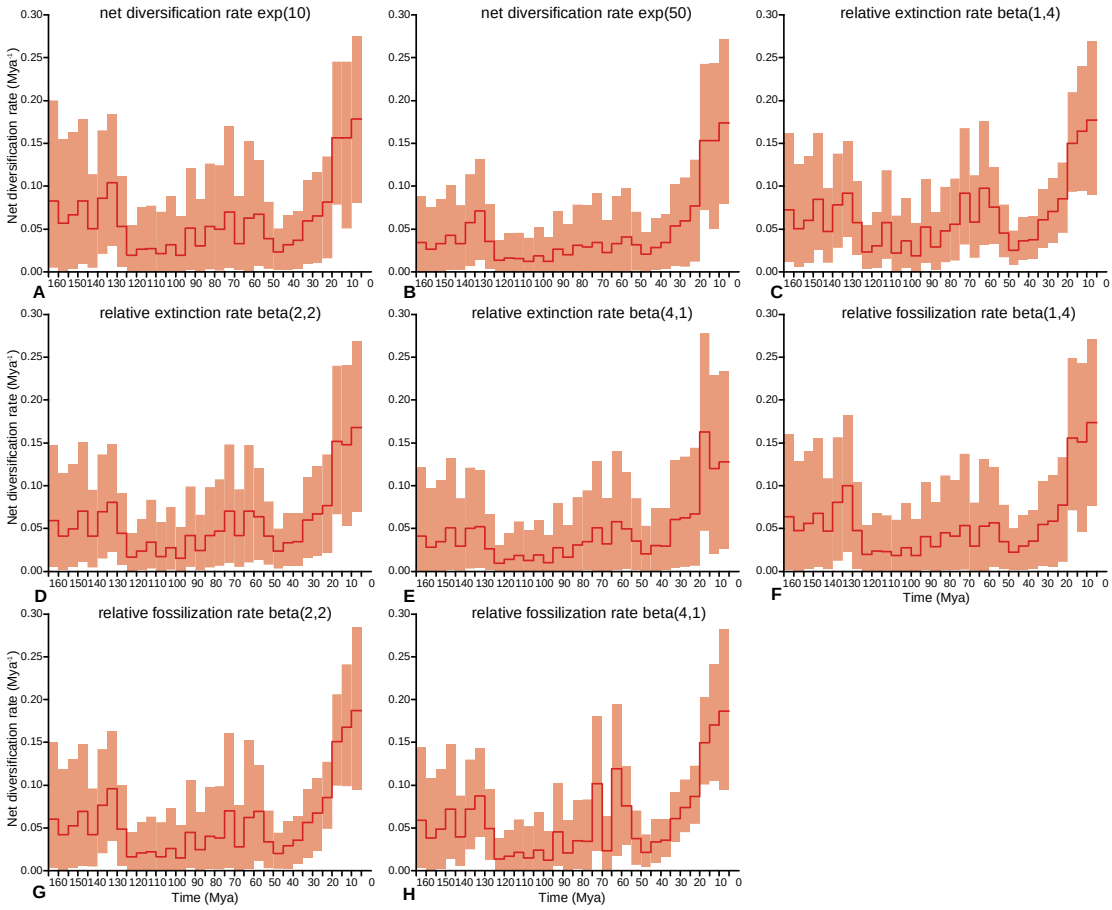


Figure S8. Diversification of birds through time estimated by using Supertree 3 based on different prior assumptions. See legend of **Figure S6**.

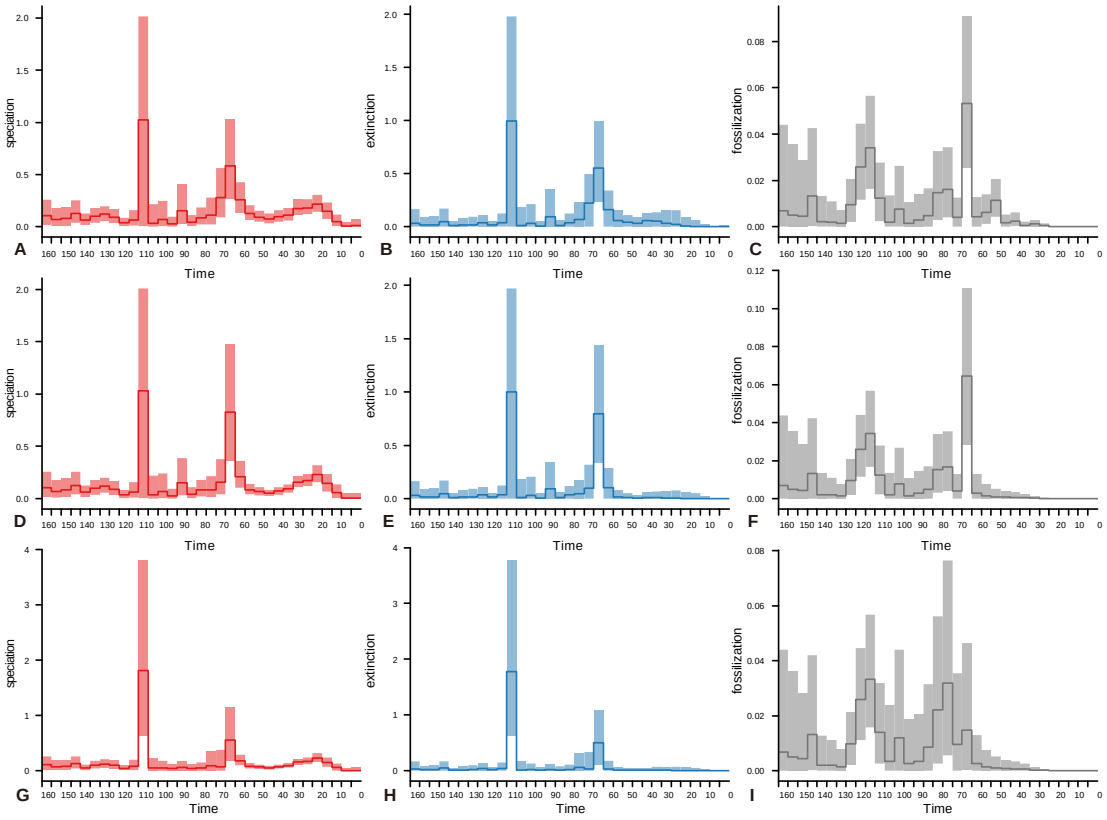


Figure S9. Speciation, extinction and fossilization rates of supertree1 estimated based on default prior assumptions. (A)-(C) are based on supertree with all fossil species included. (D)-(F) are based on supertree with fossil crown birds deleted. (G)-(I) are based on the backbone supertree. Boxes around medians are 95% HPD intervals.

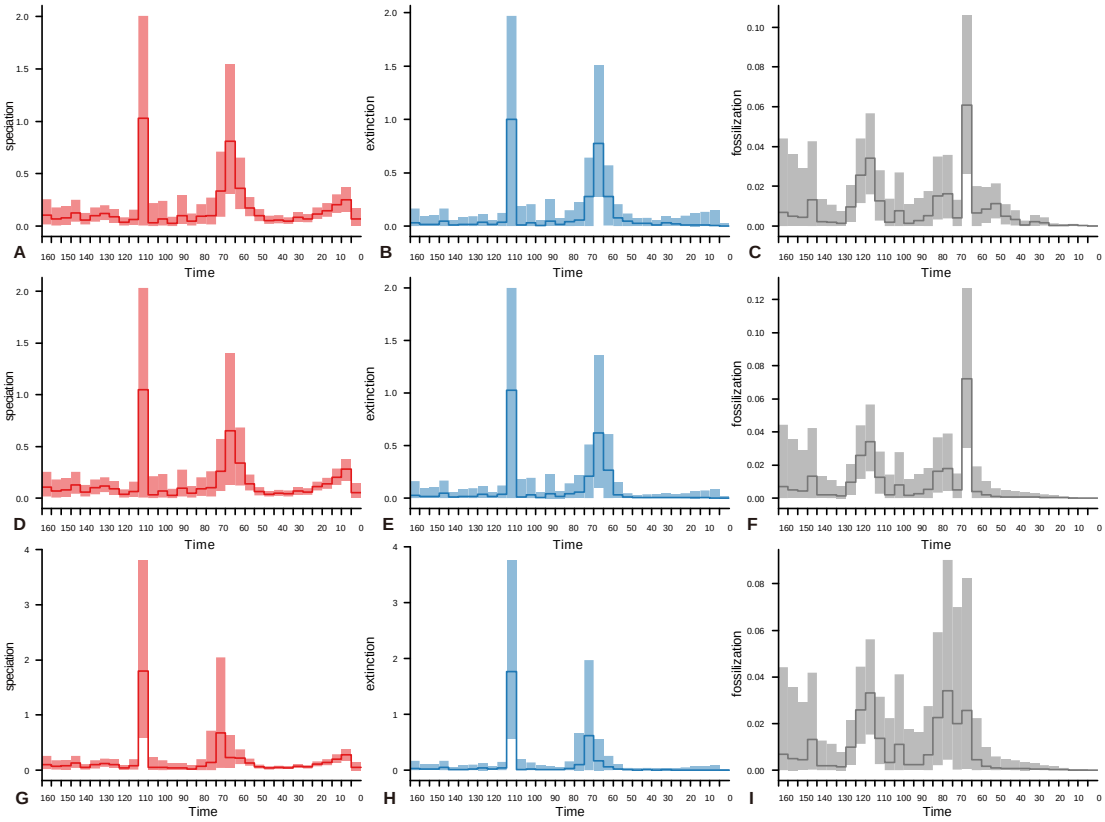


Figure S10. Speciation, extinction and fossilization rates of supertree2 estimated based on default prior assumptions. See legend of **Figure S9**.

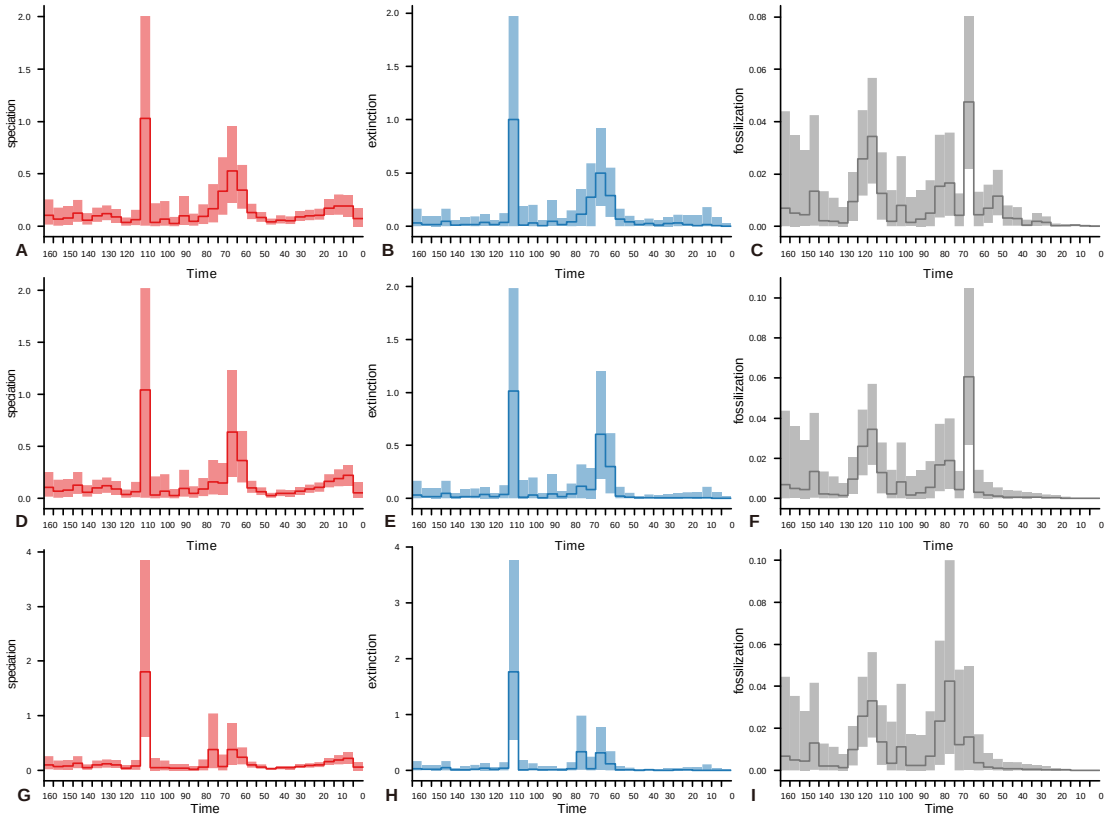


Figure S11. Speciation, extinction and fossilization rates of supertree3 estimated based on default prior assumptions. See legend of **Figure S9**.

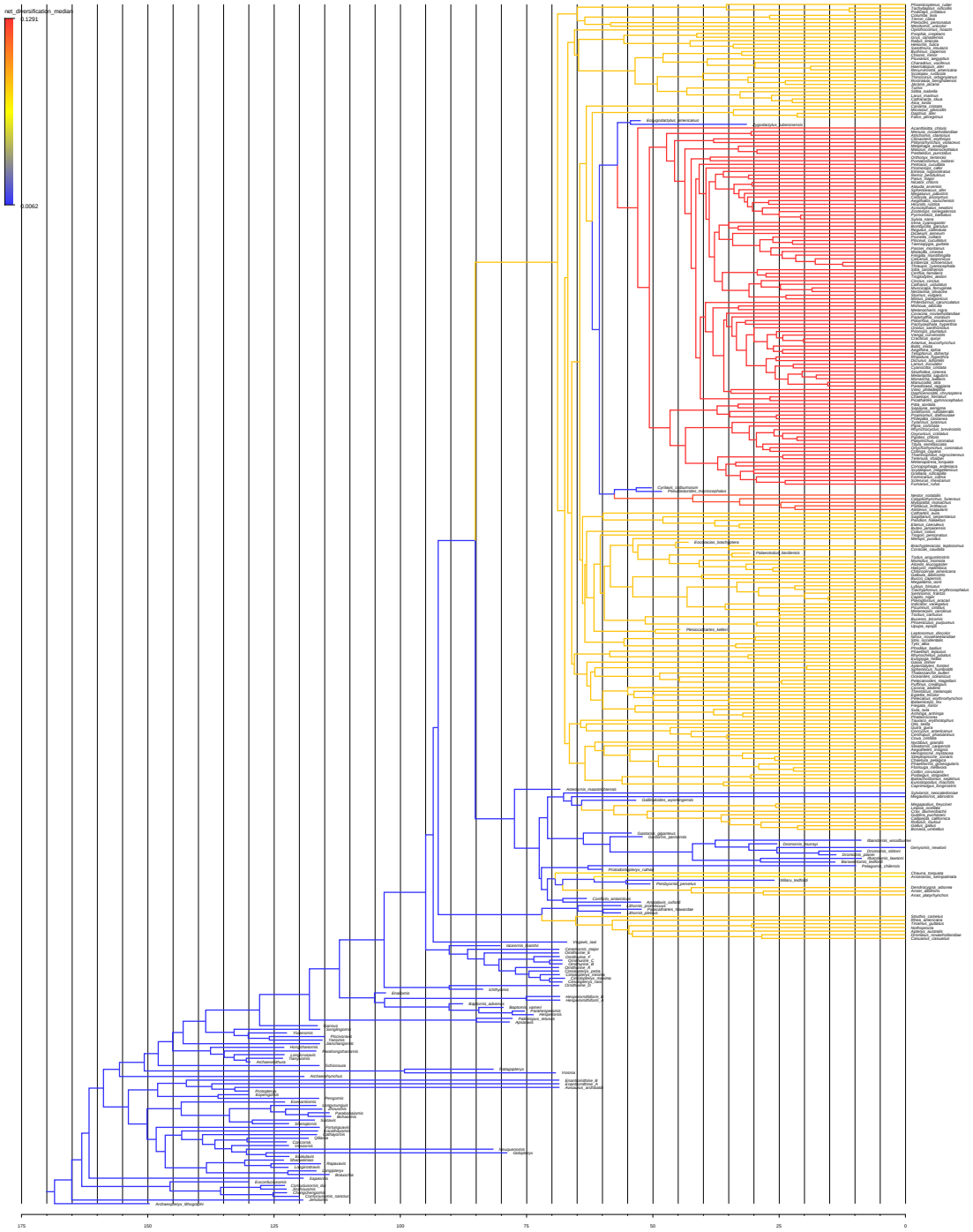


Figure S12. Net diversification rate variation in supertree 1 from the MTBD analyses with a large rate of change (lognormal(4,1) prior for γ**).**

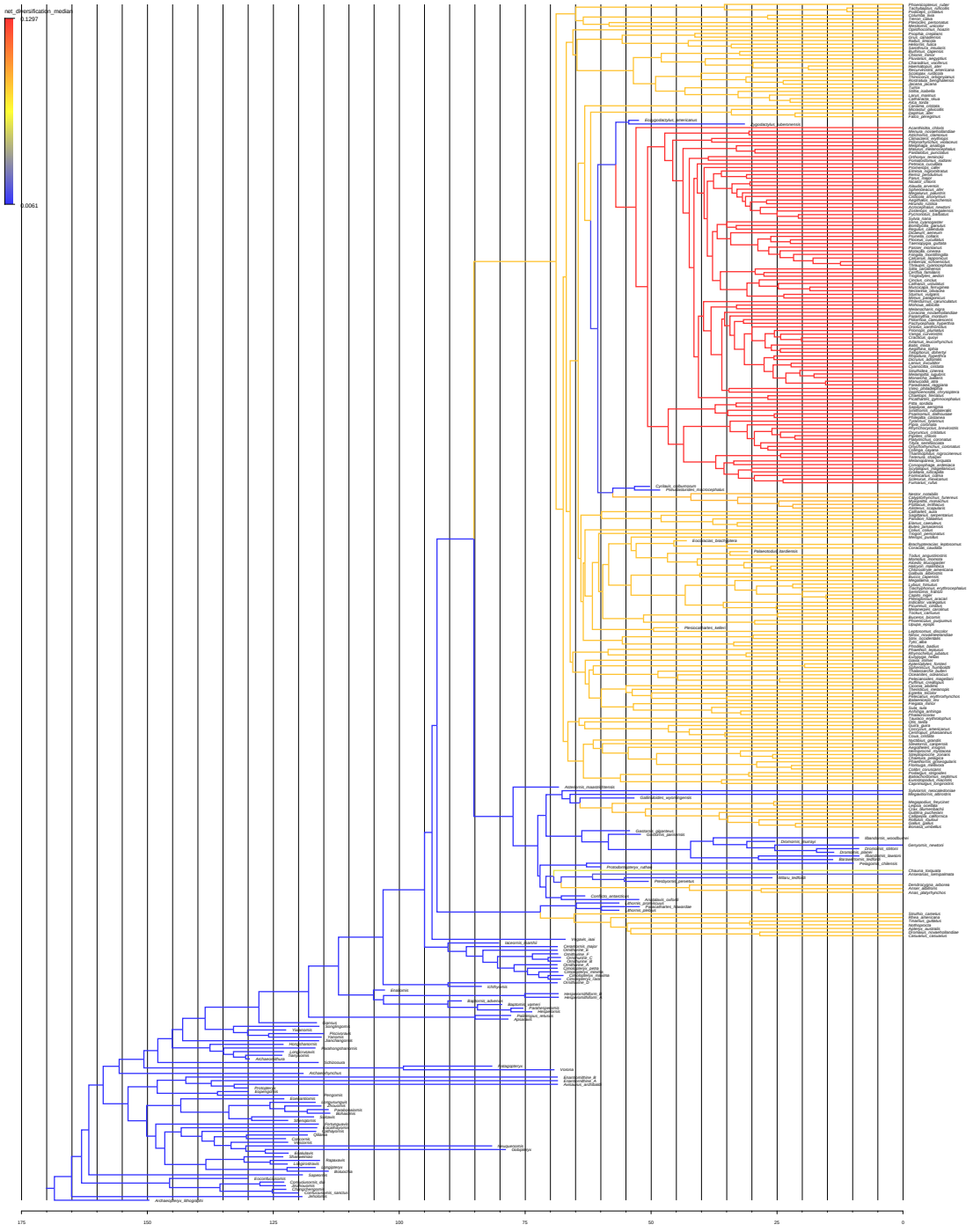


Figure S13. Net diversification rate variation in supertree 1 from the MTBD analyses with a medium rate of change (lognormal(–4,1) prior for γ**).**

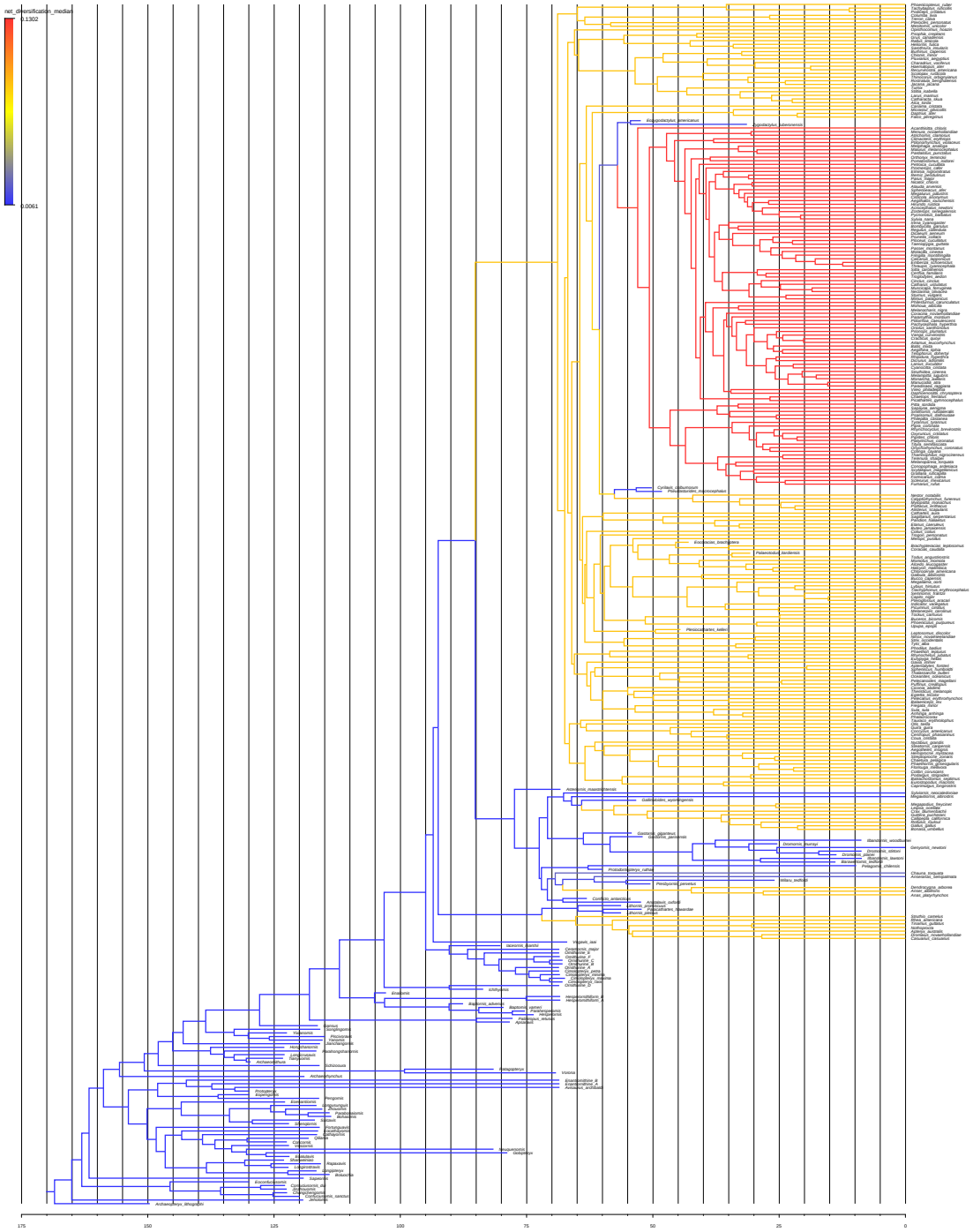


Figure S14. Net diversification rate variation in supertree 1 from the MTBD analyses with a low rate of change (lognormal(–8,1) prior for γ**).**

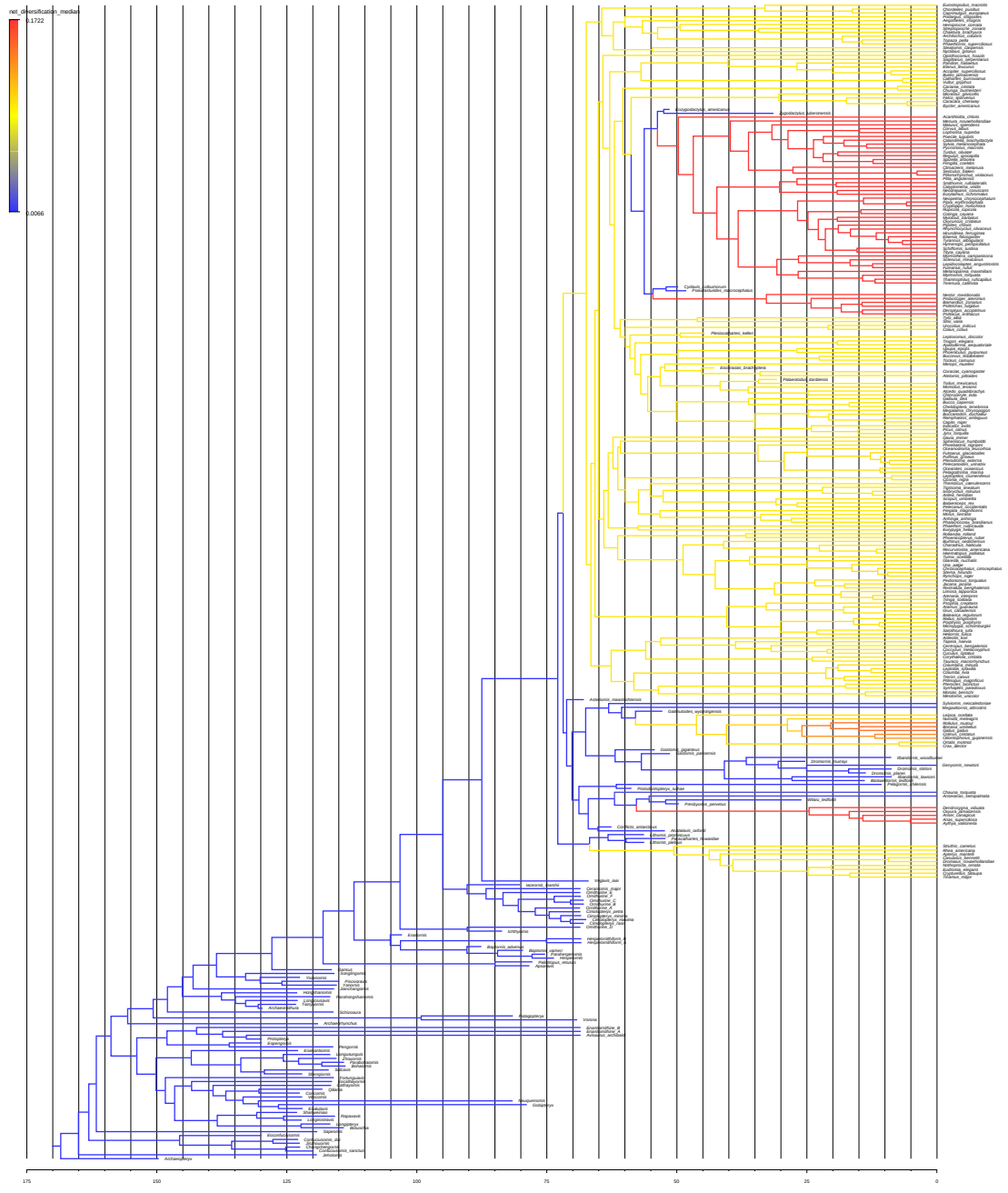


Figure S15. Net diversification rate variation in supertree 2 from the MTBD analyses with a large rate of change (lognormal(4,1) prior for γ**).**

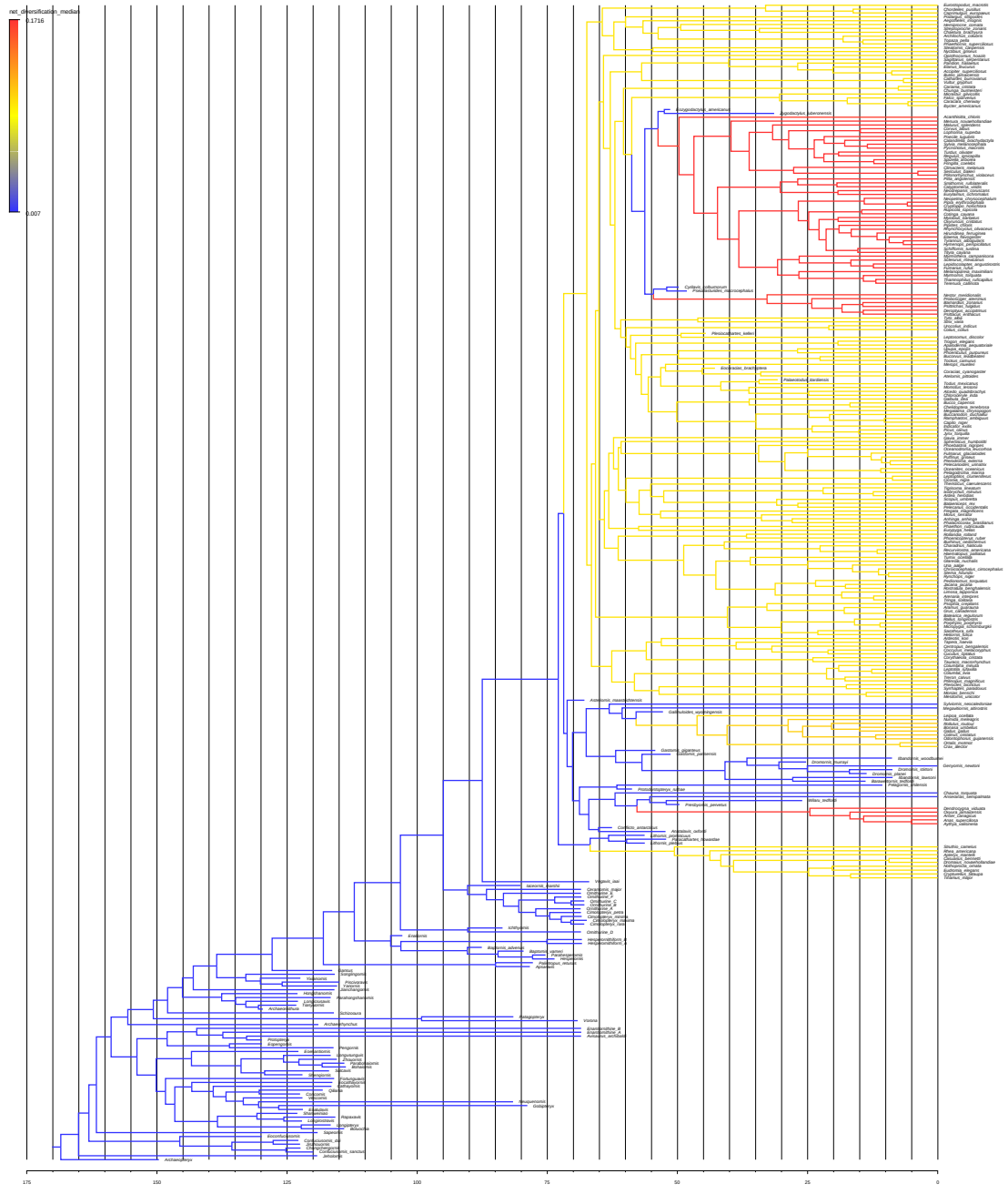


Figure S16. Net diversification rate variation in supertree 2 from the MTBD analyses with a medium rate of change (lognormal(–4,1) prior for γ**).**

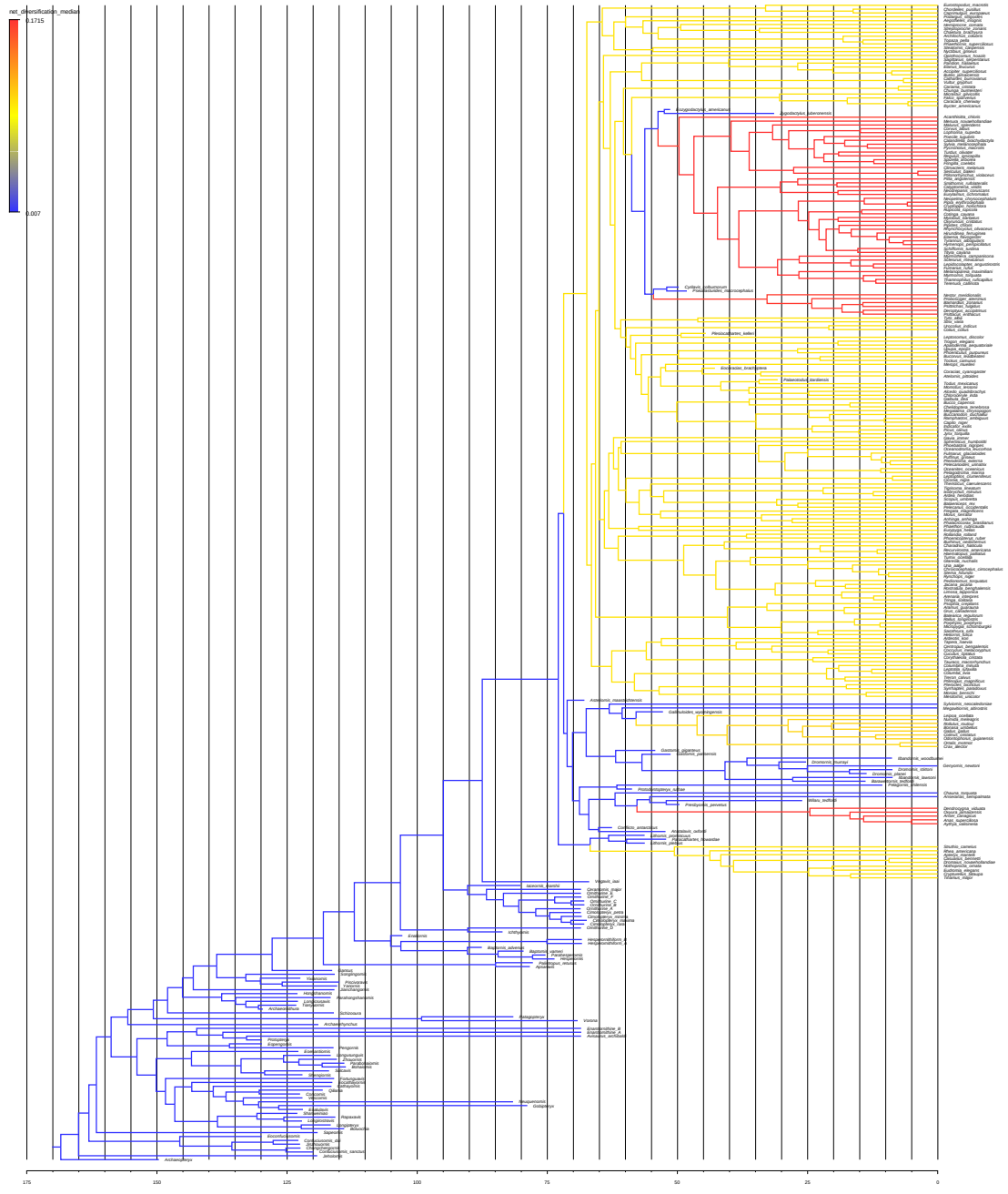


Figure S17. Net diversification rate variation in supertree 2 from the MTBD analyses with a low rate of change (lognormal(–8,1) prior for γ**).**

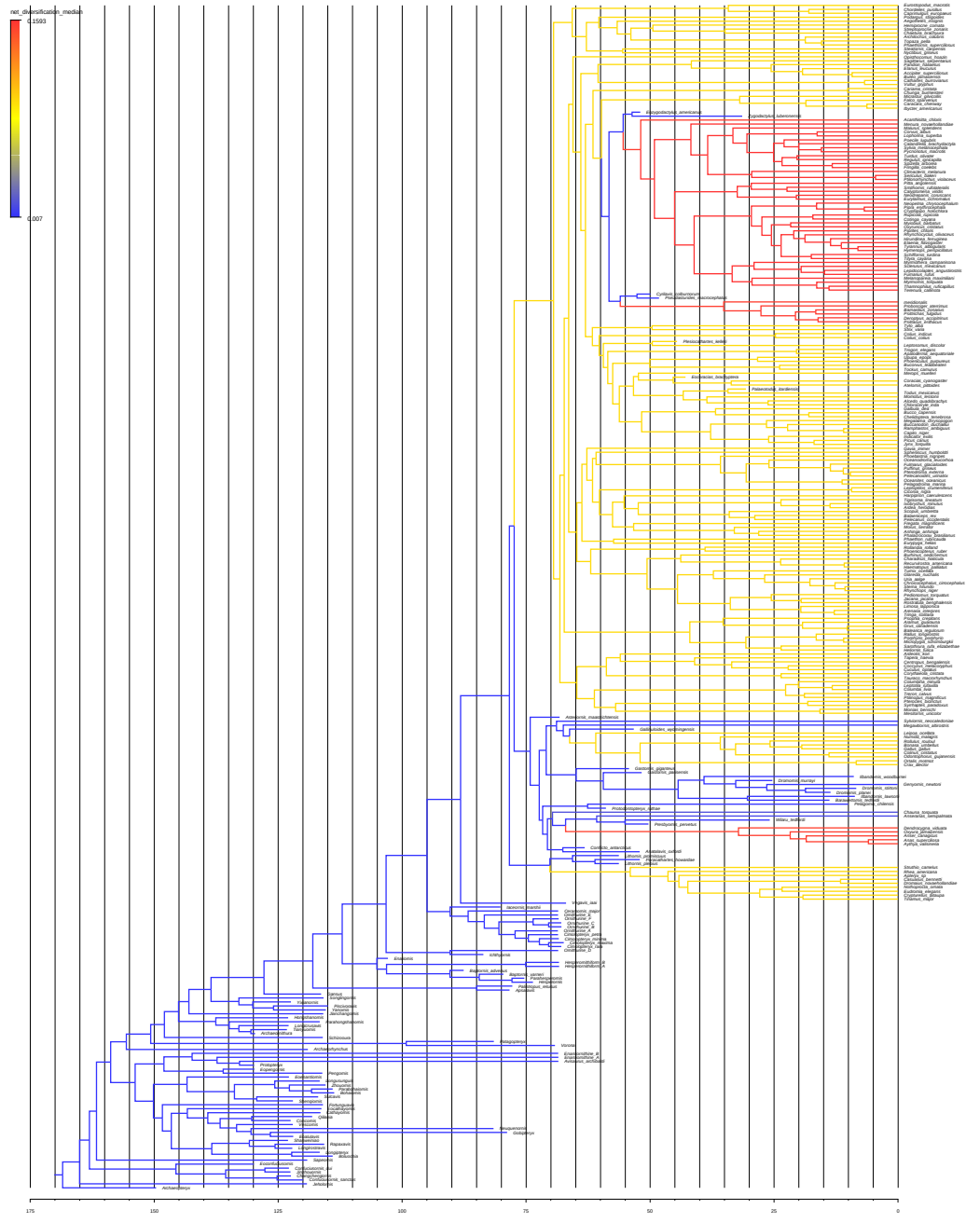


Figure S18. Net diversification rate variation in supertree 3 from the MTBD analyses with a large rate of change (lognormal(4,1) prior for γ**).**

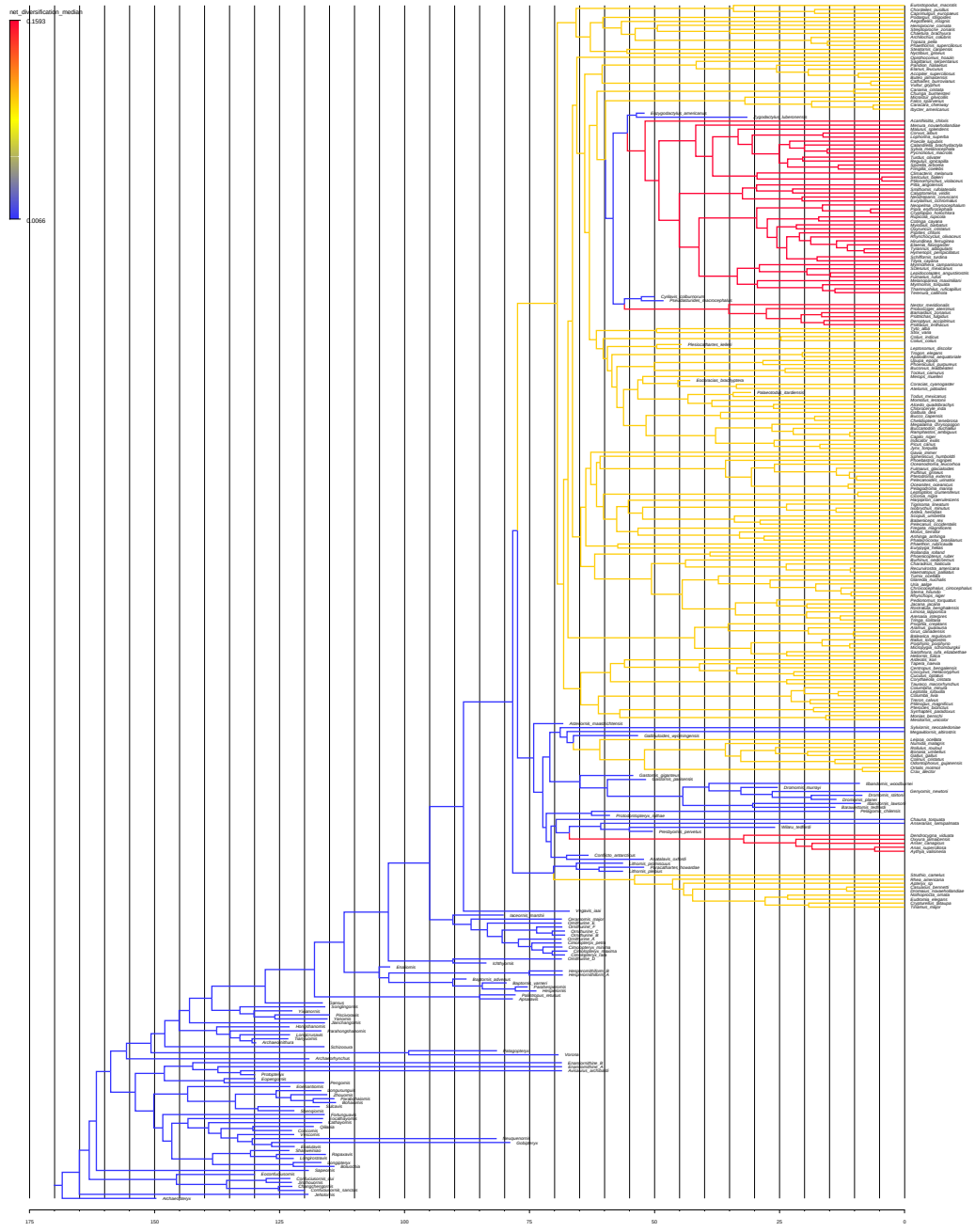


Figure S19. Net diversification rate variation in supertree 3 from the MTBD analyses with a medium rate of change (lognormal(–4,1) prior for γ**).**

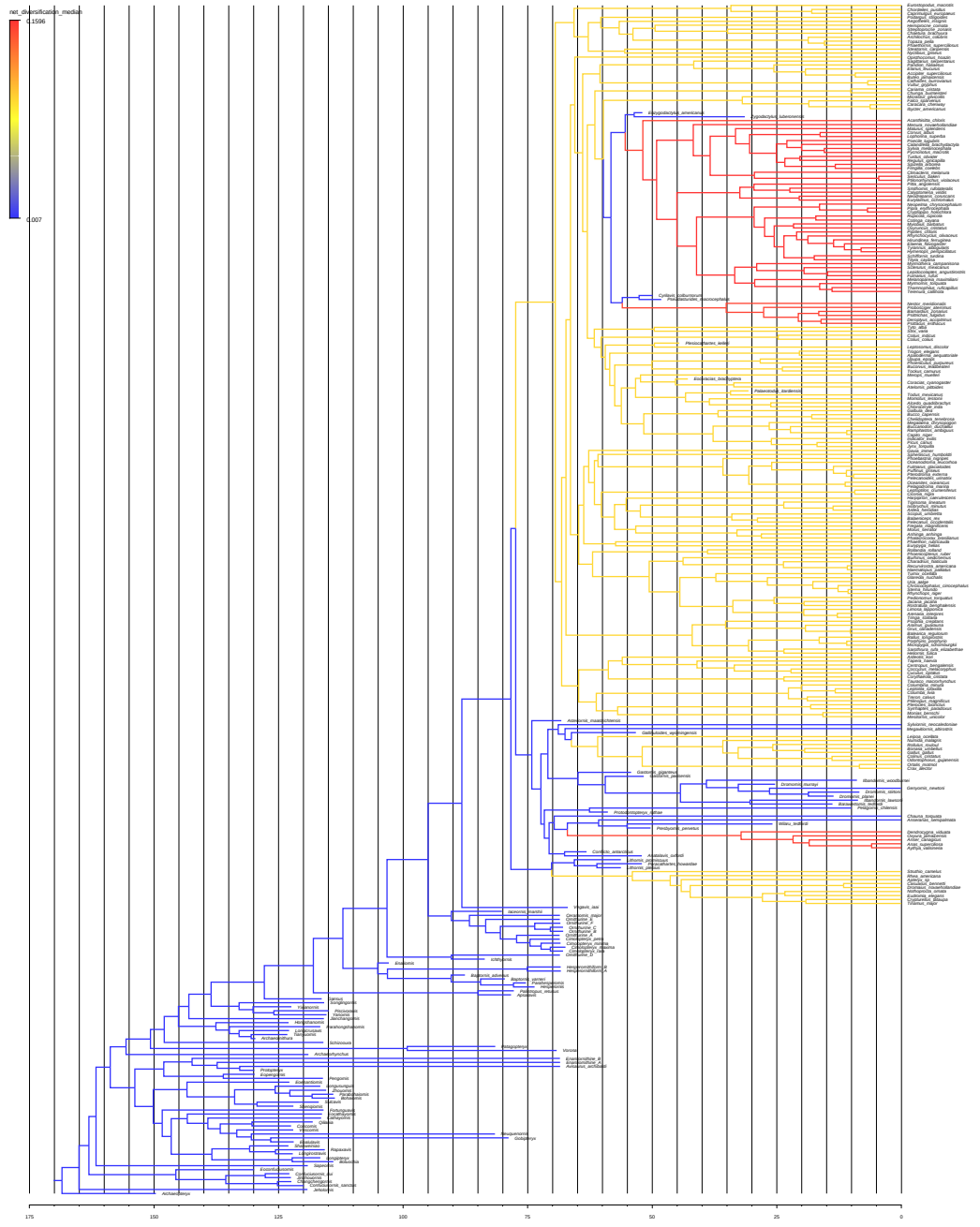


Figure S20. Net diversification rate variation in supertree 3 from the MTBD analyses with a low rate of change (lognormal(–8,1) prior for γ**).**

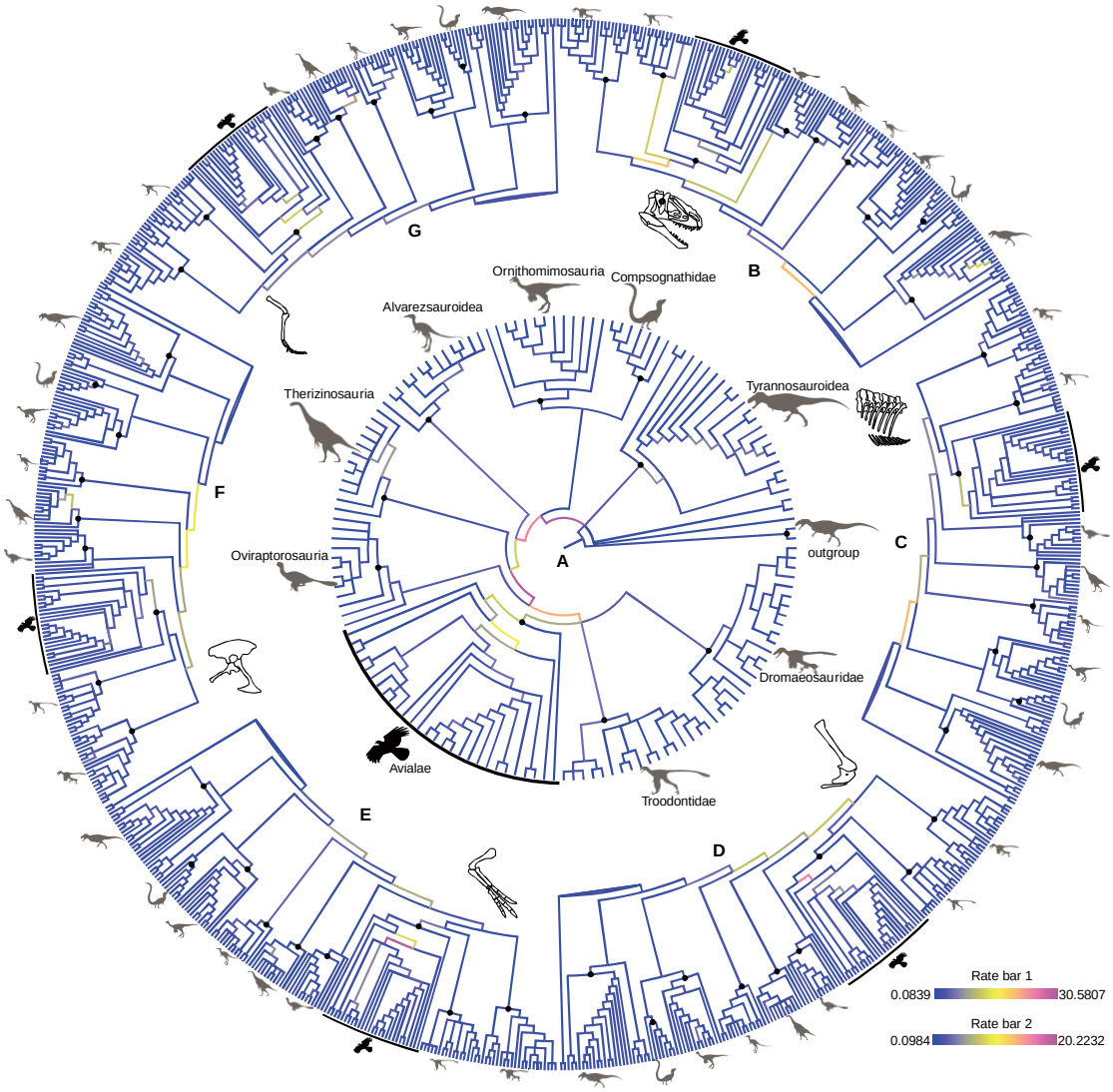


Figure S21. Morphological evolutionary rates of the coelurosaurian dinosaurs including birds based on WN relaxed clock model without any extra phylogenetic constraint. Black circles denote coelurosaurian clades. The rates presented in the cladograms were estimated from an unpartitioned character-matrix (A) and a partitioned one of skull (B), axial skeleton (C), pectoral girdle (D), forelimb (E), pelvis (F), and hindlimb (F) (see Supplementary information for details). Rate bar 1 for (A) and Rate bar 2 for (B-G). Silhouettes are from PhyloPic.org.

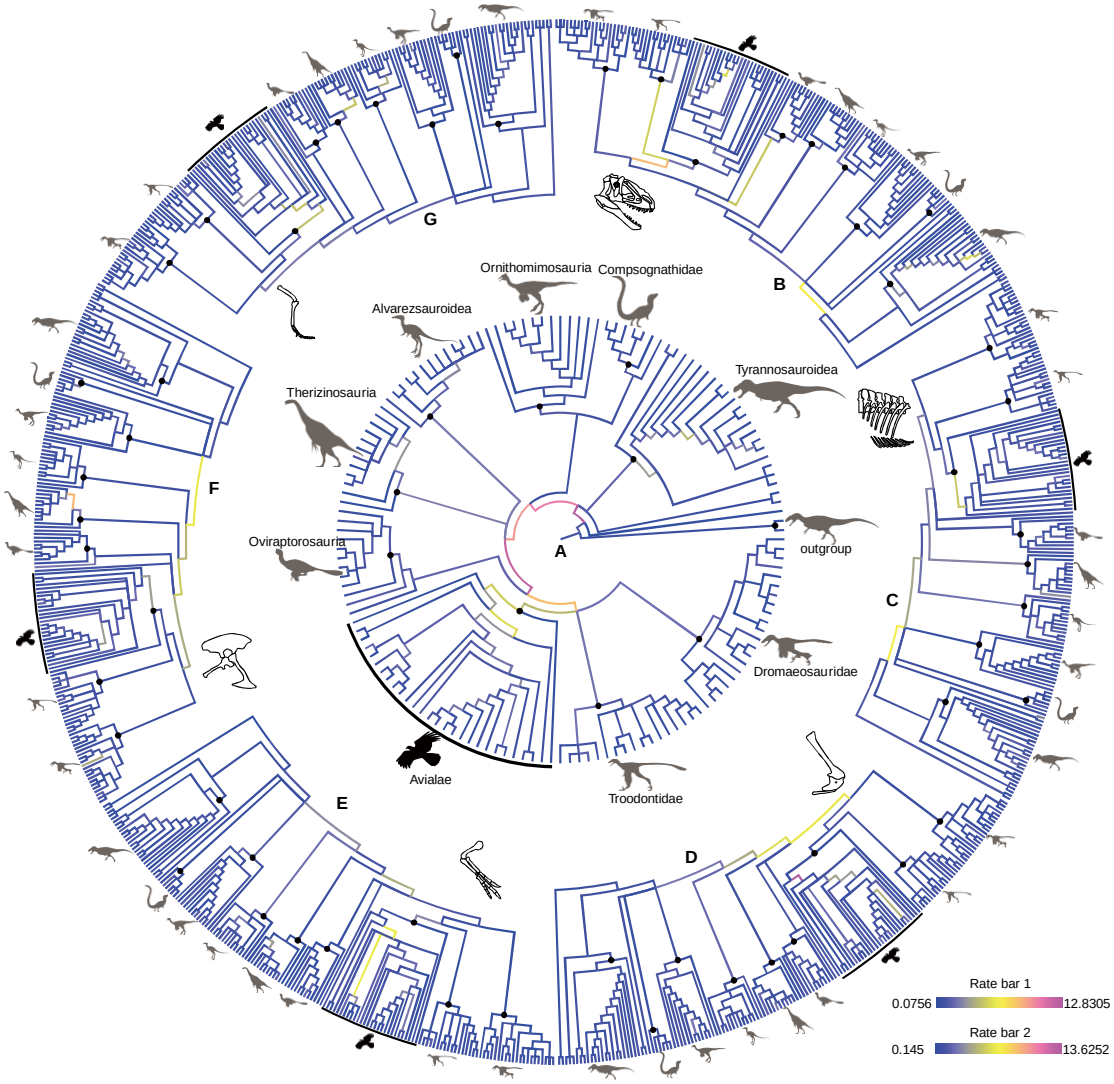


Figure S22. Morphological evolutionary rates of the coelurosaurian dinosaurs including birds based on ILN relaxed clock model without any extra phylogenetic constraint. Black circles denote coelurosaurian clades. The rates presented in the cladograms were estimated from an unpartitioned character-matrix (A) and a partitioned one of skull (B), axial skeleton (C), pectoral girdle (D), forelimb (E), pelvis (F), and hindlimb (F) (see Supplementary information for details). Rate bar 1 for (A) and Rate bar 2 for (B-G). Silhouettes are from PhyloPic.org.

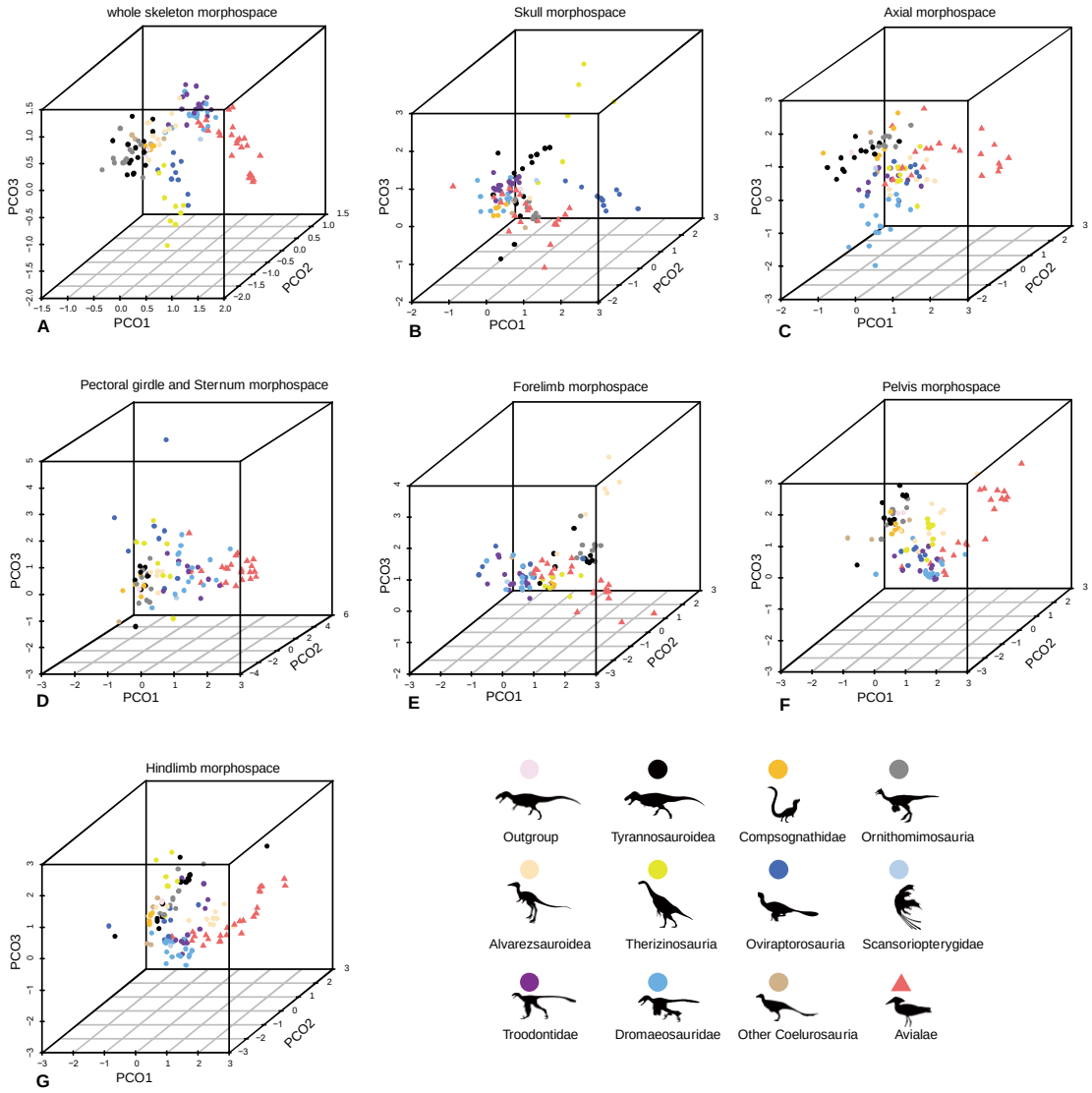


Figure S23. Morphospace comparisons of birds and other coelurosaurian clades based on the results of the disparity analyses of MORD matrices. Non-avian dinosaurs are denoted with colored circles and birds are denoted with red triangles. The comparisons are based on whole skeleton (A), skull (B), axial skeleton (C), pectoral girdle (D), forelimb (E), pelvis (F), and hindlimb (G) morphospaces (see Supplementary information for details). Silhouettes are from PhyloPic.org.

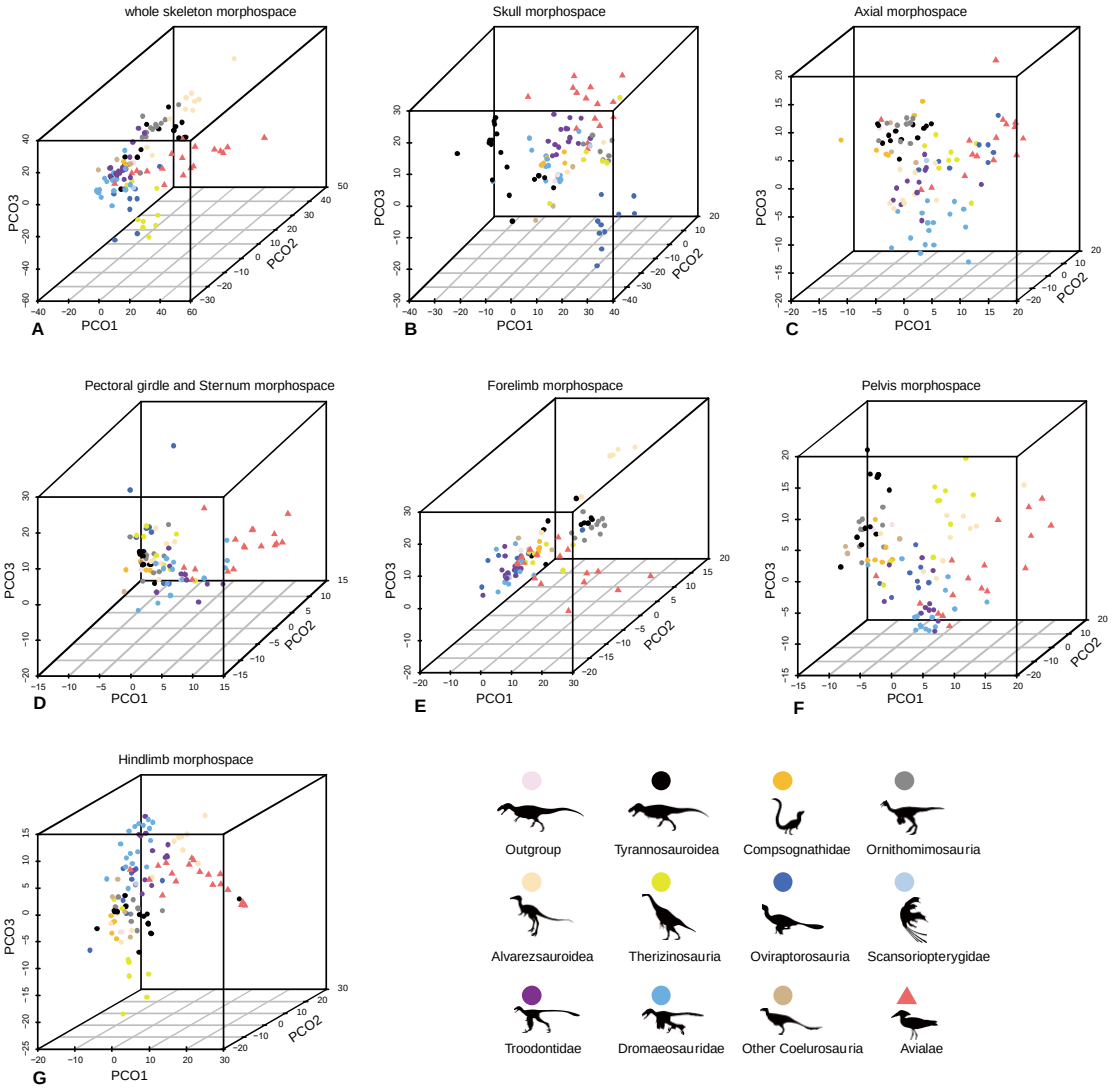


Figure S24. Morphospace comparisons of stem birds and other coelurosaurian clades based on the results of the disparity analyses of GED matrices. Non-avian dinosaurs are denoted with colored circles and birds are denoted with red triangles. The comparisons are based on whole skeleton (A), skull (B), axial skeleton (C), pectoral girdle (D), forelimb (E), pelvis (F), and hindlimb (G) morphospaces (see Supplementary information for details). Silhouettes are from PhyloPic.org.

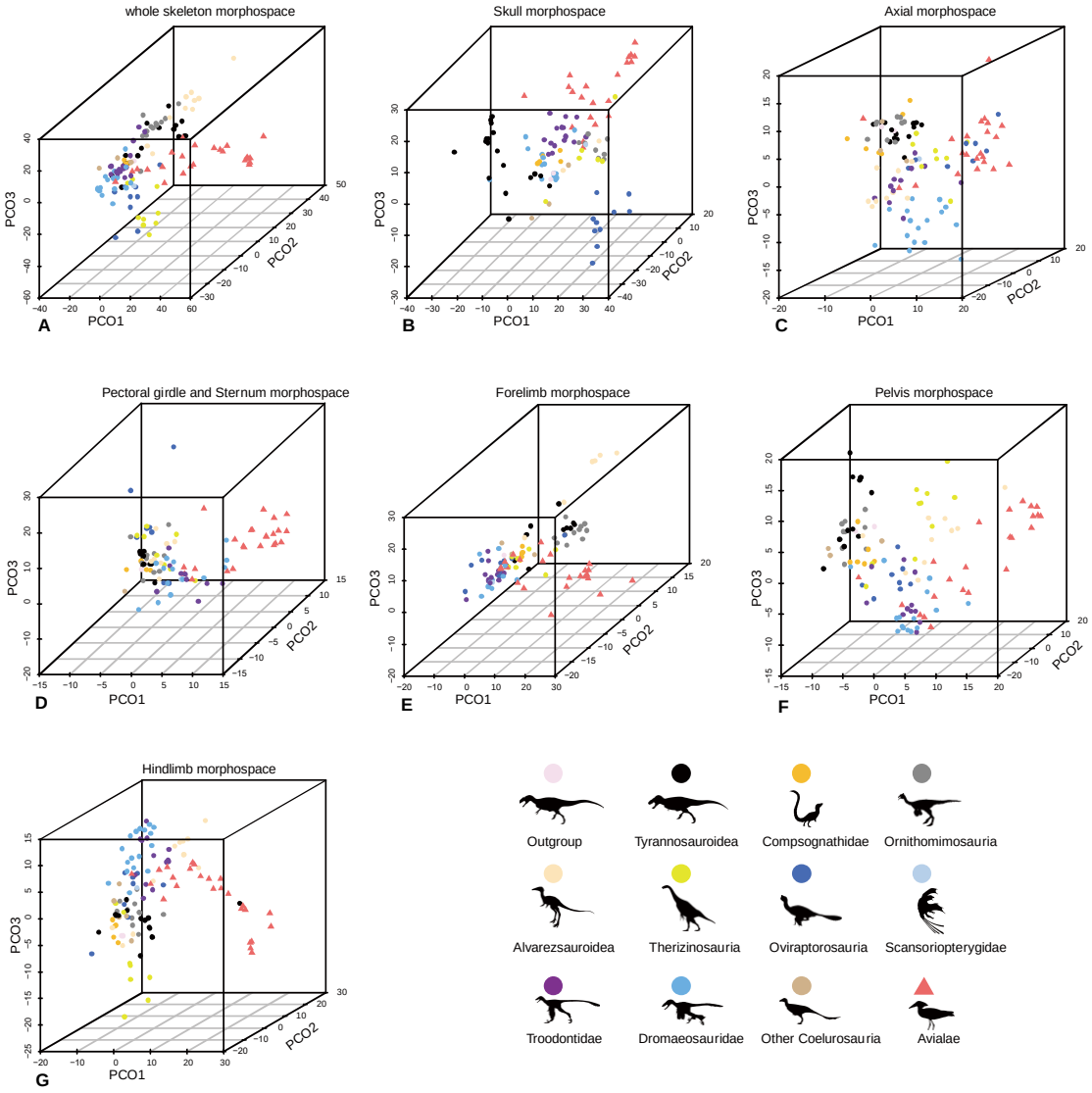


Figure S25. Morphospace comparisons of birds and other coelurosaurian clades based on the results of the disparity analyses of GED matrices. Non-avian dinosaurs are denoted with colored circles and birds are denoted with red triangles. The comparisons are based on whole skeleton (A), skull (B), axial skeleton (C), pectoral girdle (D), forelimb (E), pelvis (F), and hindlimb (G) morphospaces (see Supplementary information for details). Silhouettes are from PhyloPic.org.

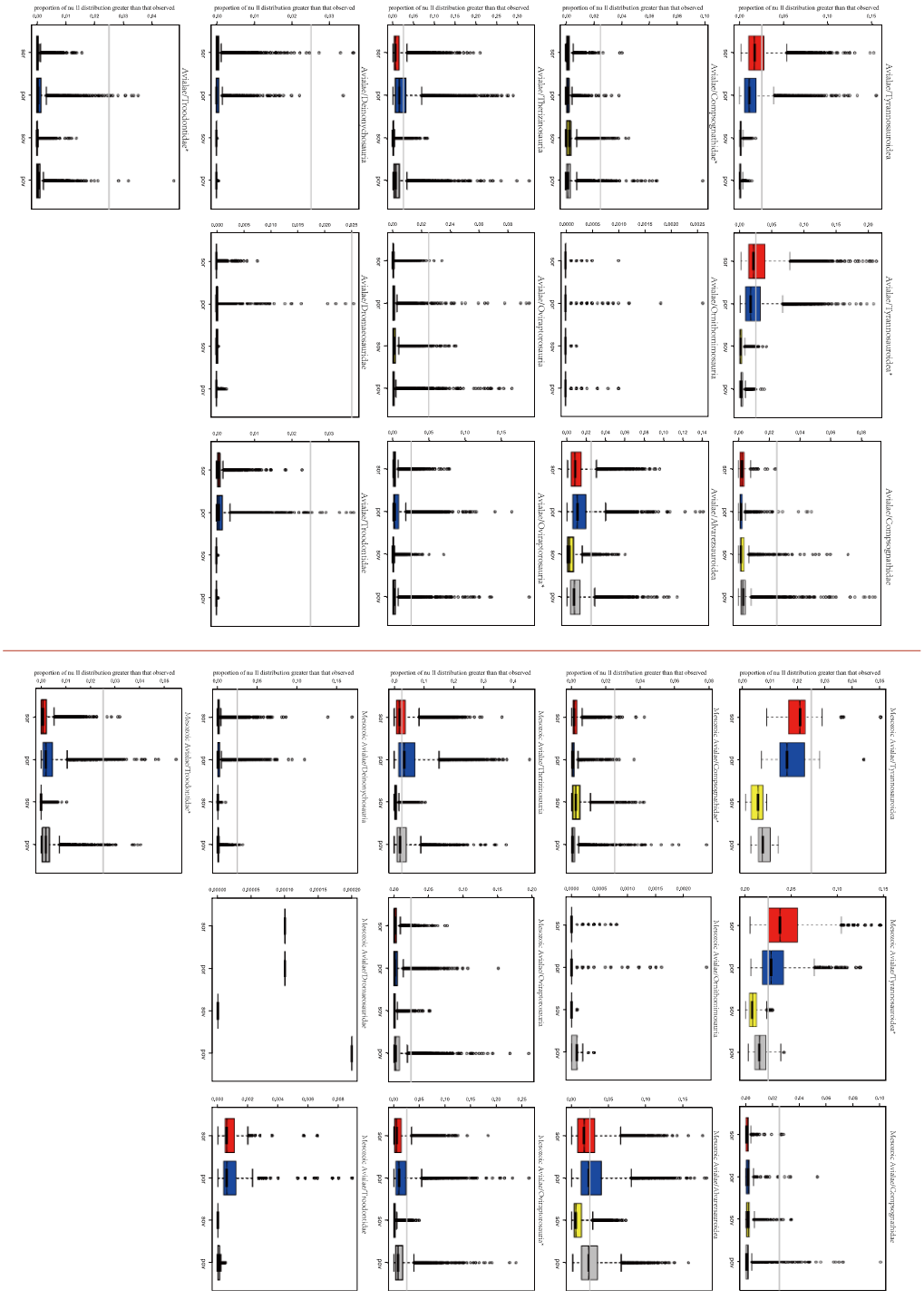


Figure S26. Whole skeleton disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. Each box represents the distribution of the proportion greater than the observed value. Grey line highlights the value of 0.025 or 0.975. Abbreviations: sor: sum of ranges, por: product of

ranges, sov: sum of variance, pov: product of variance. Tyrannosauroidea*:Tyrannosauroidea without *Coelurus* and *Tanycolagreus*, Compsognathidae*:Compsognathidae including *Ornitholestes hermanni*, Oviraptorosauria*:Oviraptorosauria including *Epidexipteryx*, Oviraptorosauria**:Oviraptorosauria including *Epidexipteryx* and *Pedopenna*, Troodontidae*:Troodontidae without *Anchiornis*, *Aurornis*, *Eosinopteryx*, *Caihong Juji* and *Xiaotingia*.

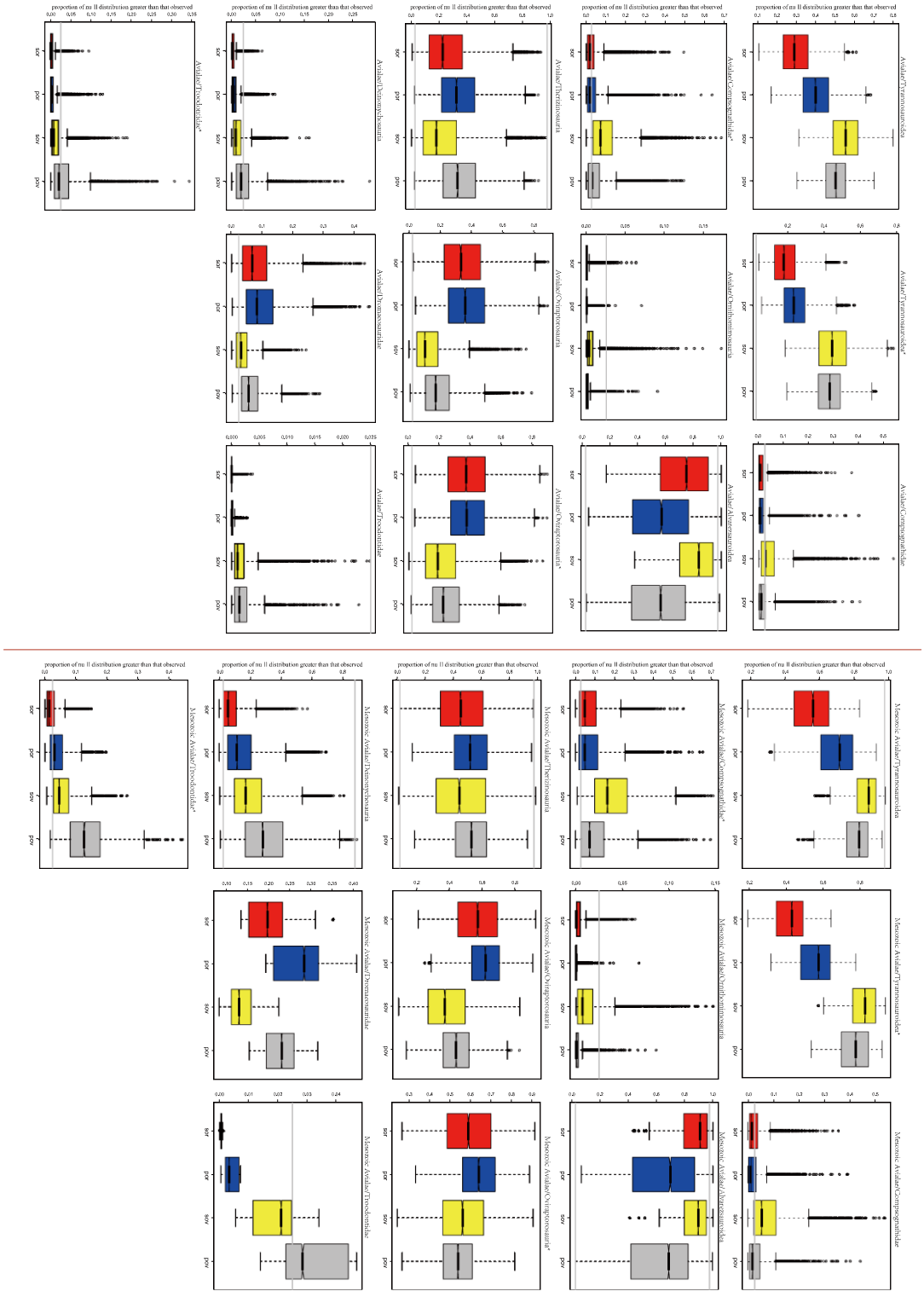


Figure S27. Skull disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

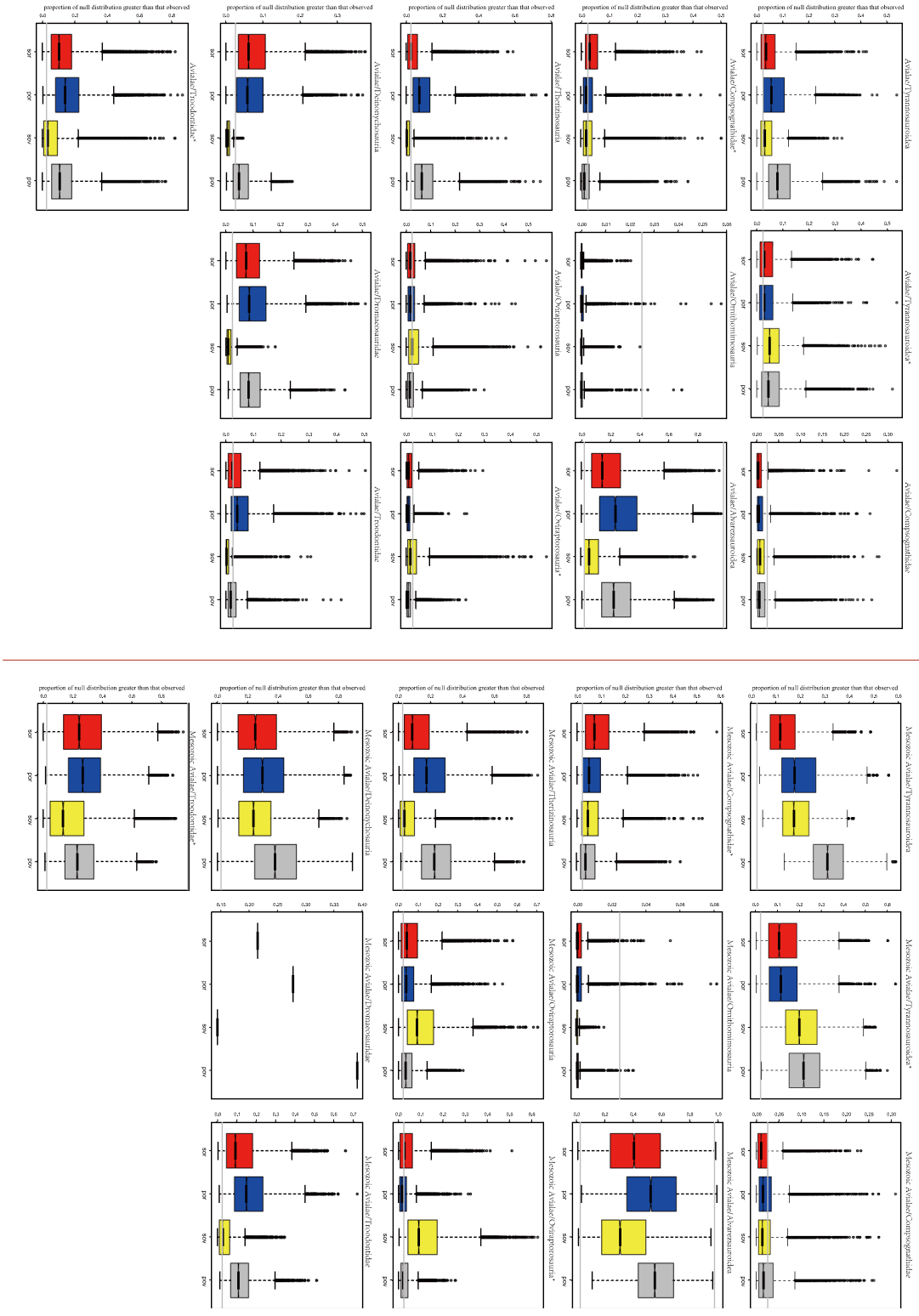


Figure S28. Axial disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

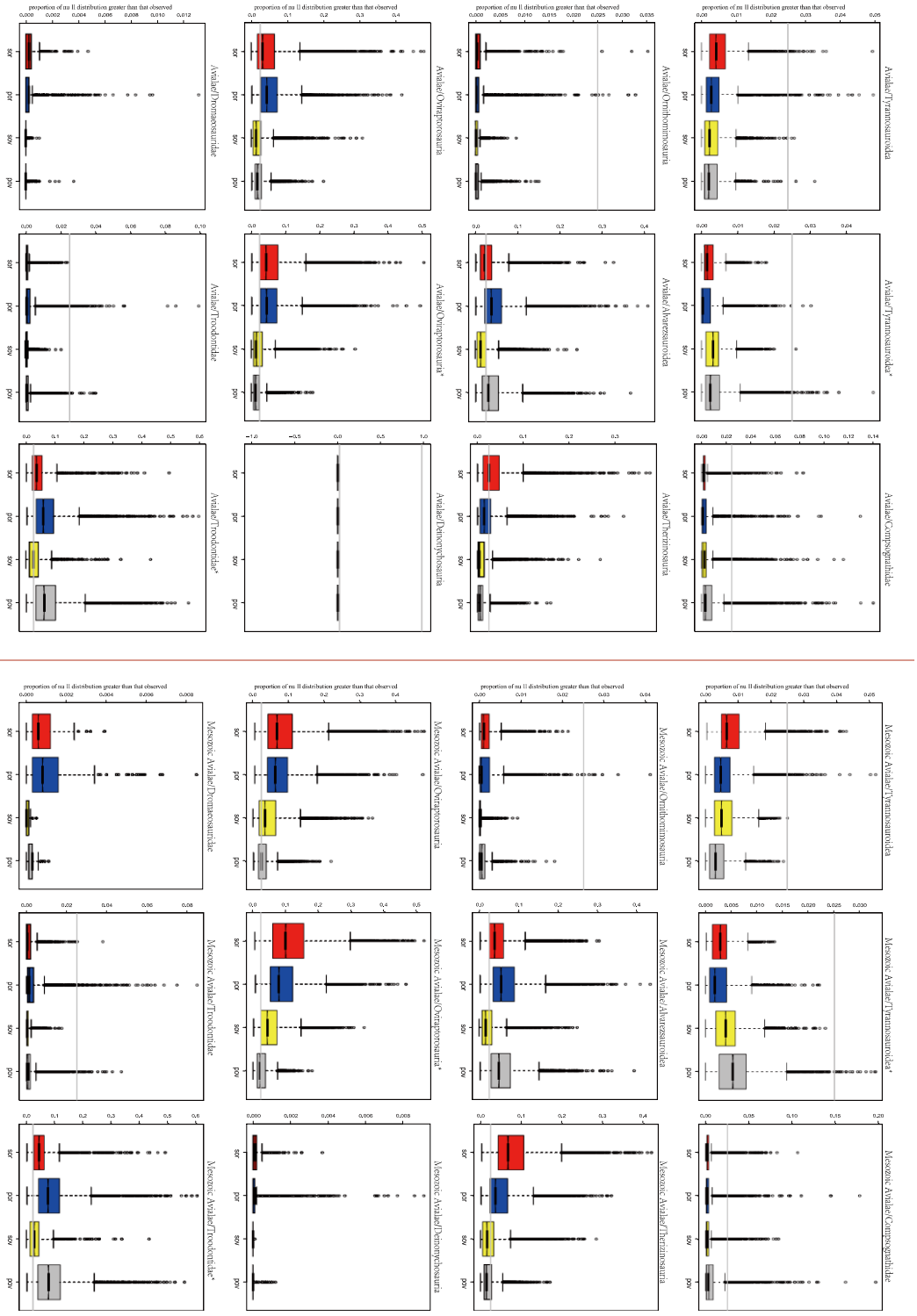


Figure S29. Pectoral girdle disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

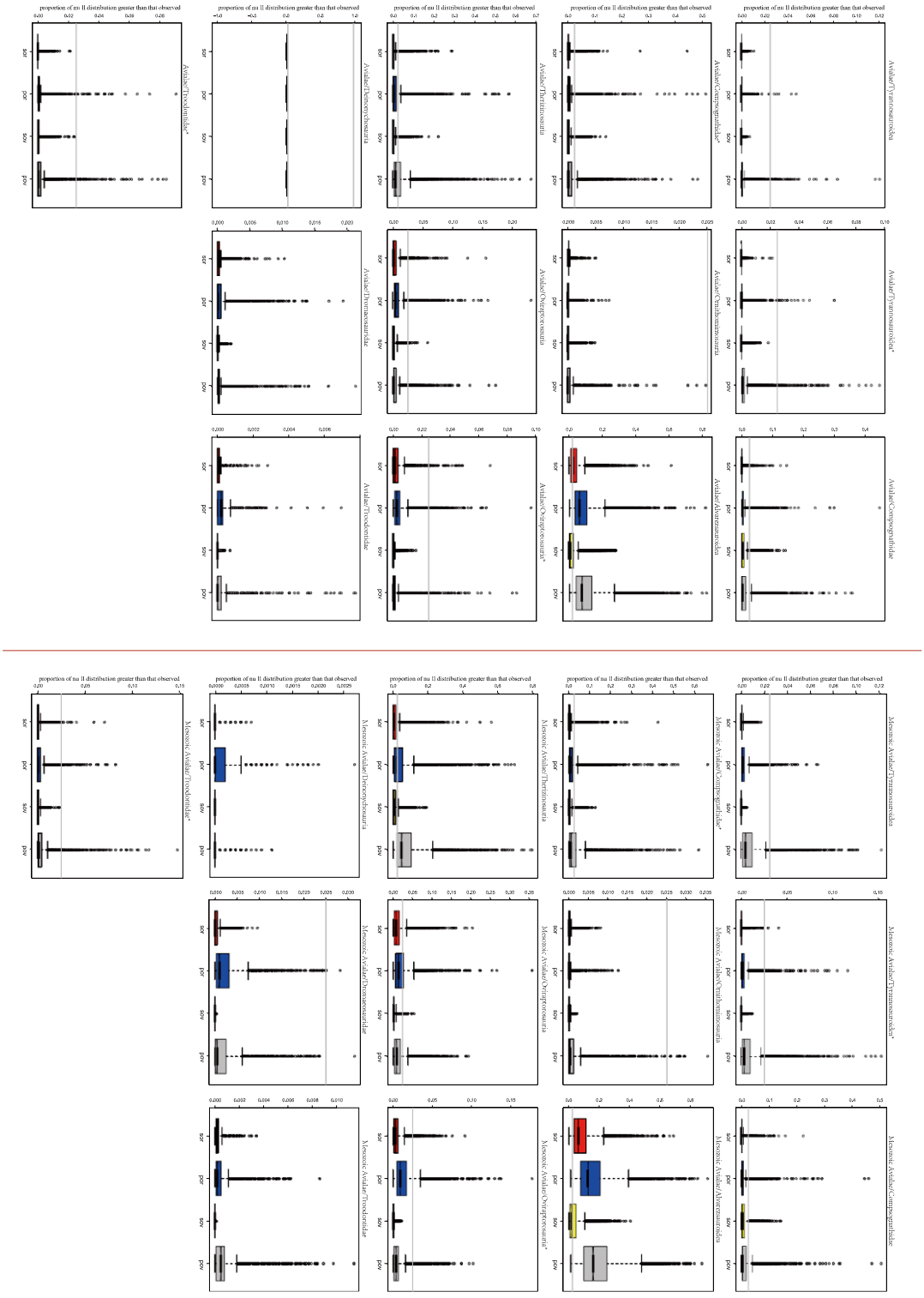


Figure S30. Forelimb disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

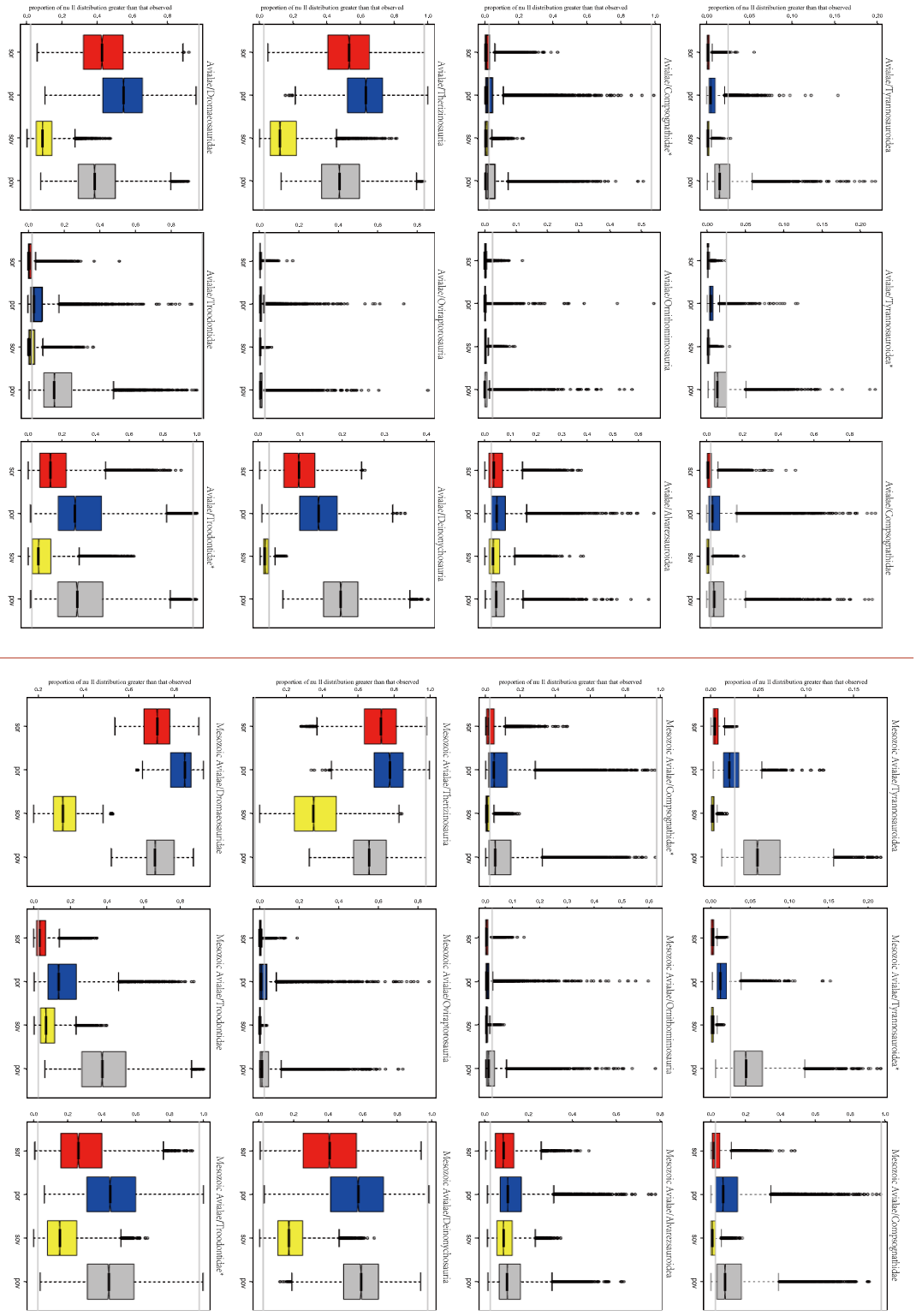


Figure S31. Pelvis disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

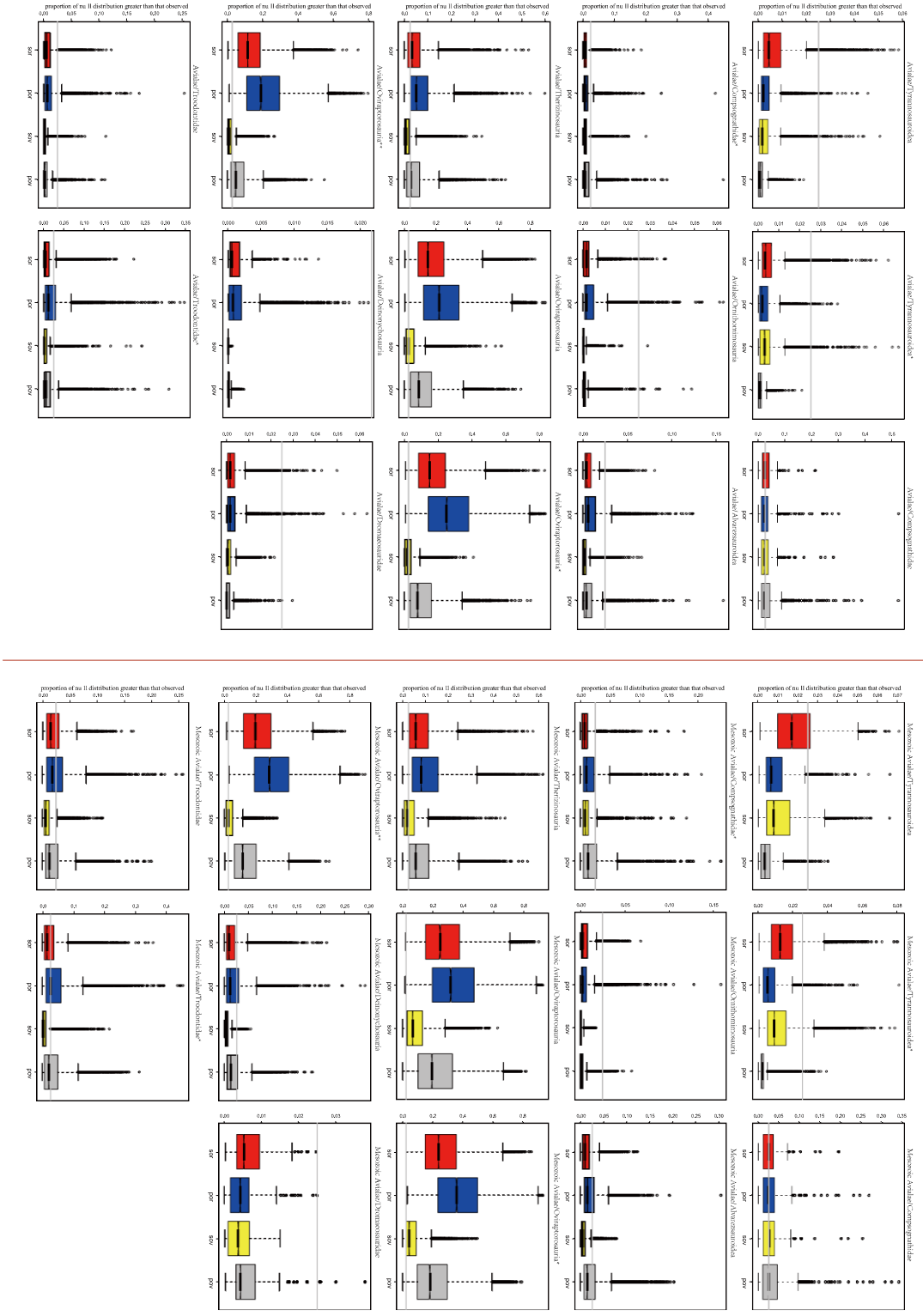


Figure S32. Hindlimb disparity comparison based on permutation tests with sample size corrected (MORD). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

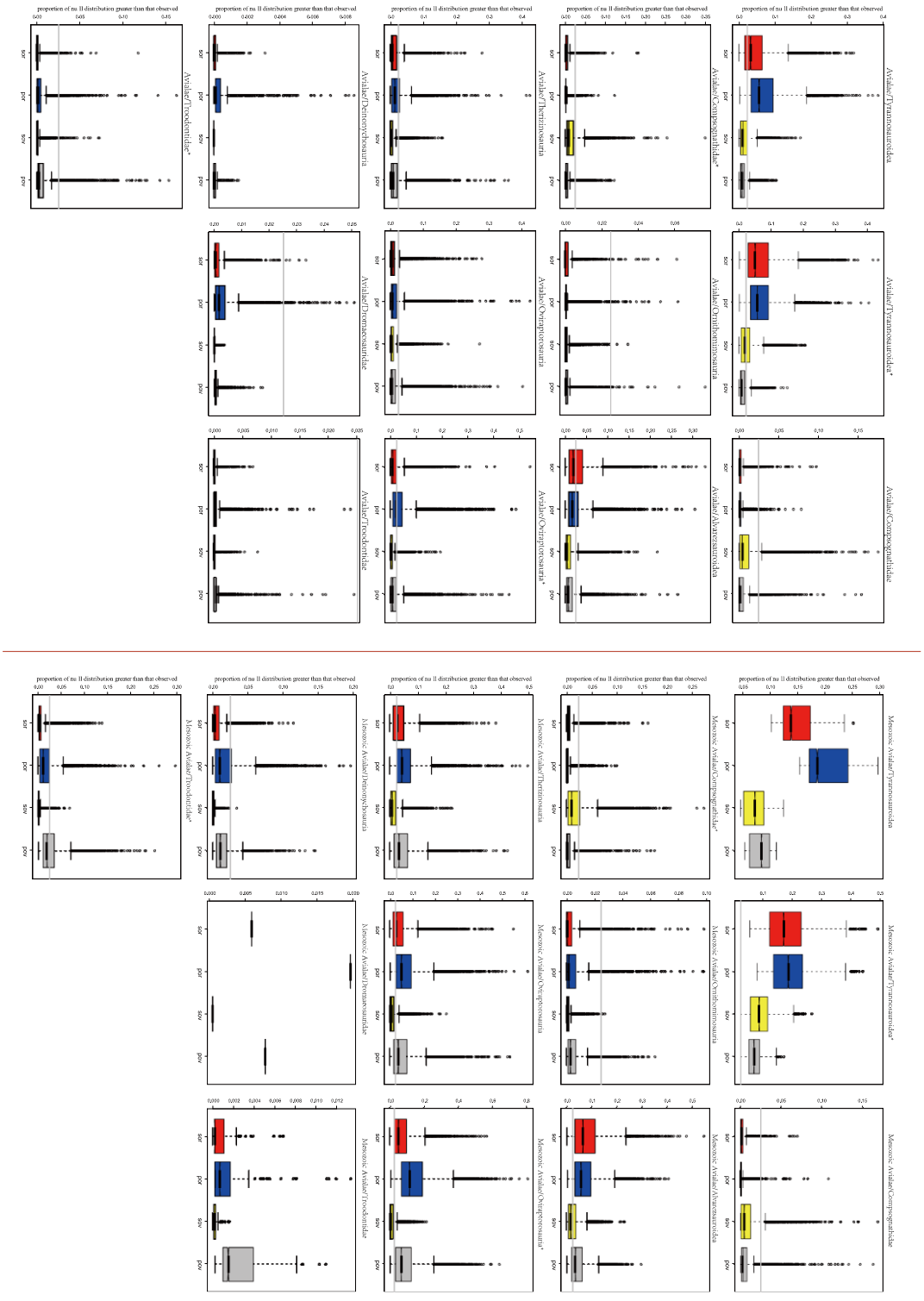


Figure S33. Whole skeleton disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

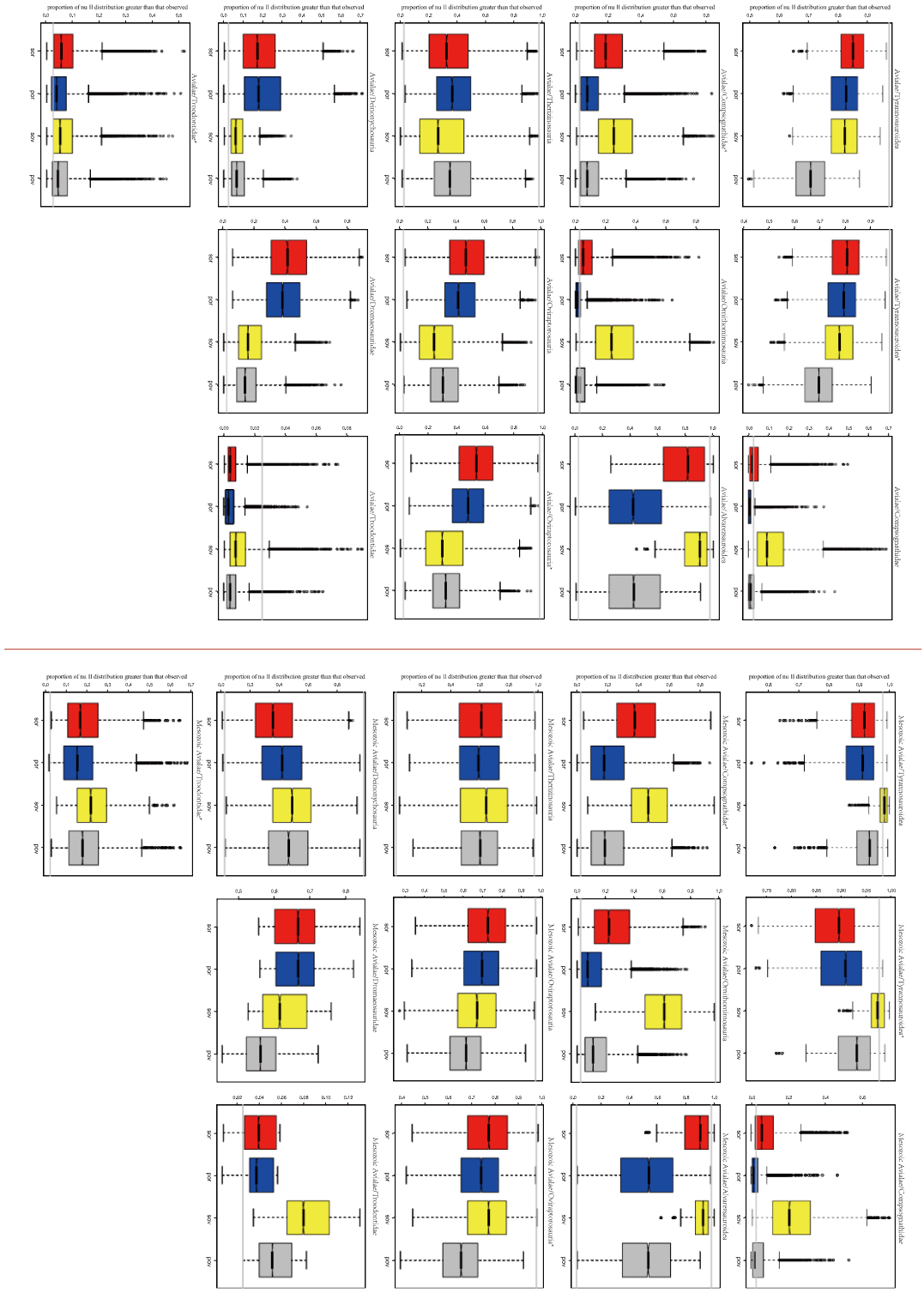


Figure S34. Skull disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

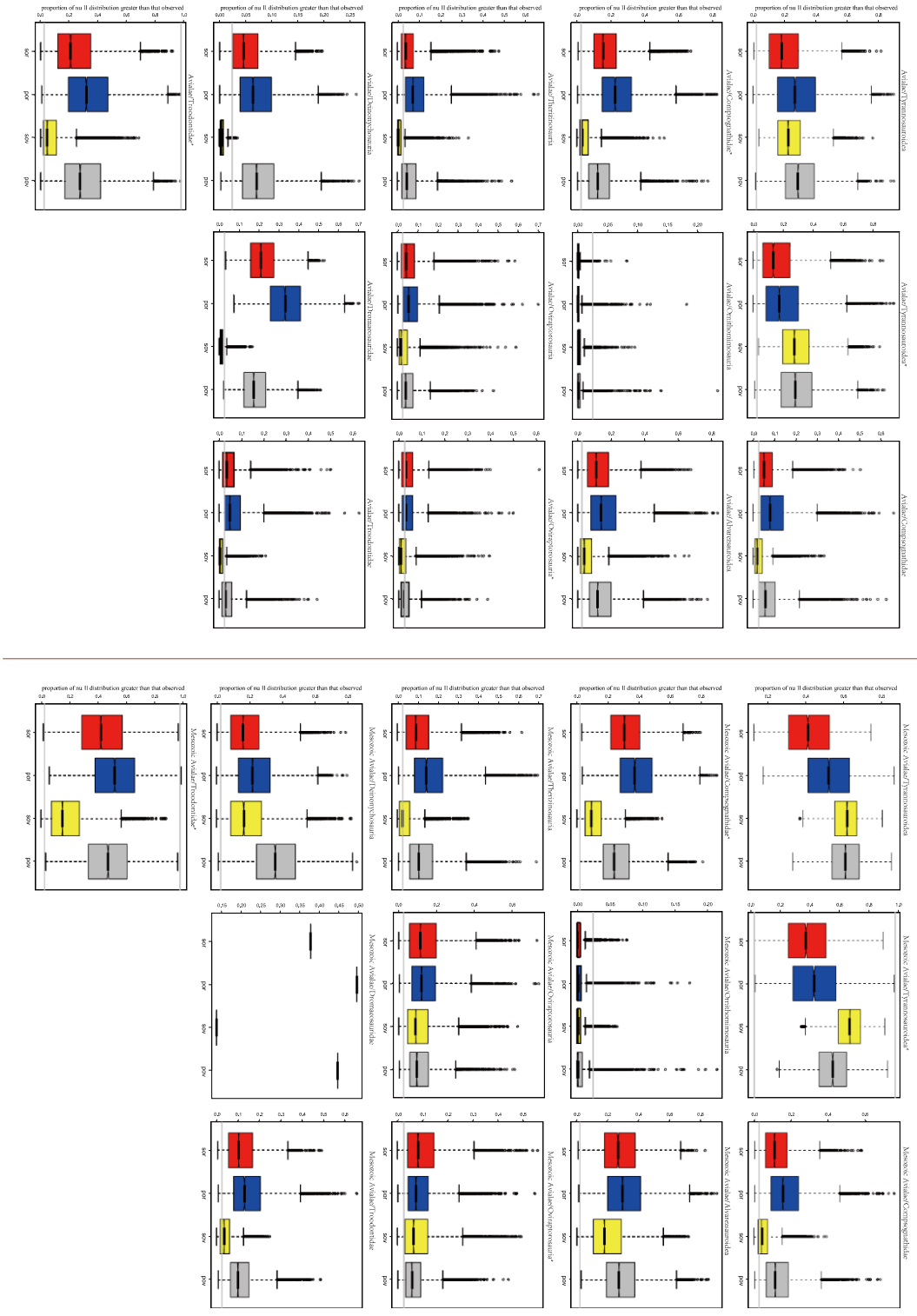


Figure S35. Axial disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

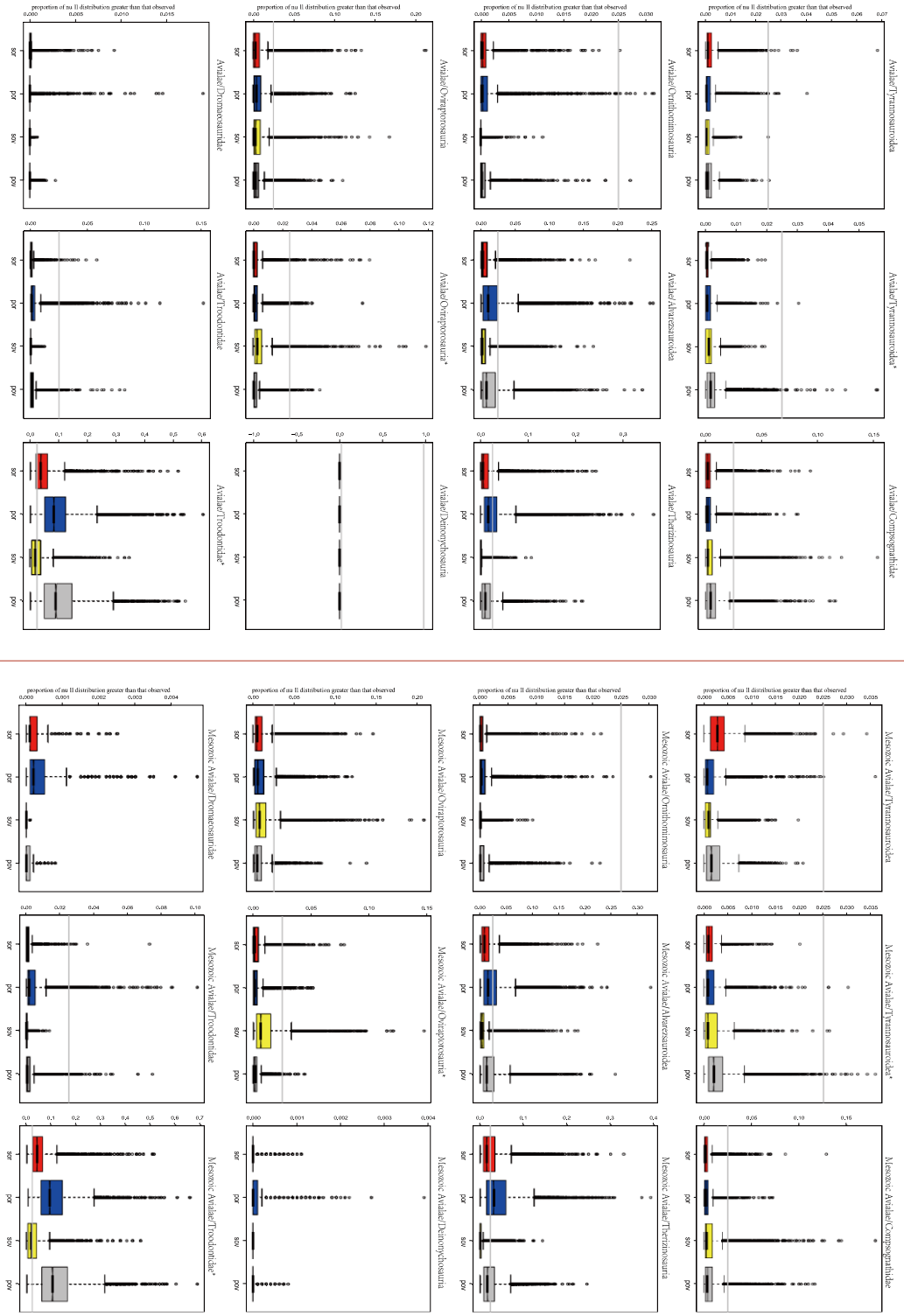


Figure S36. Pectoral girdle disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

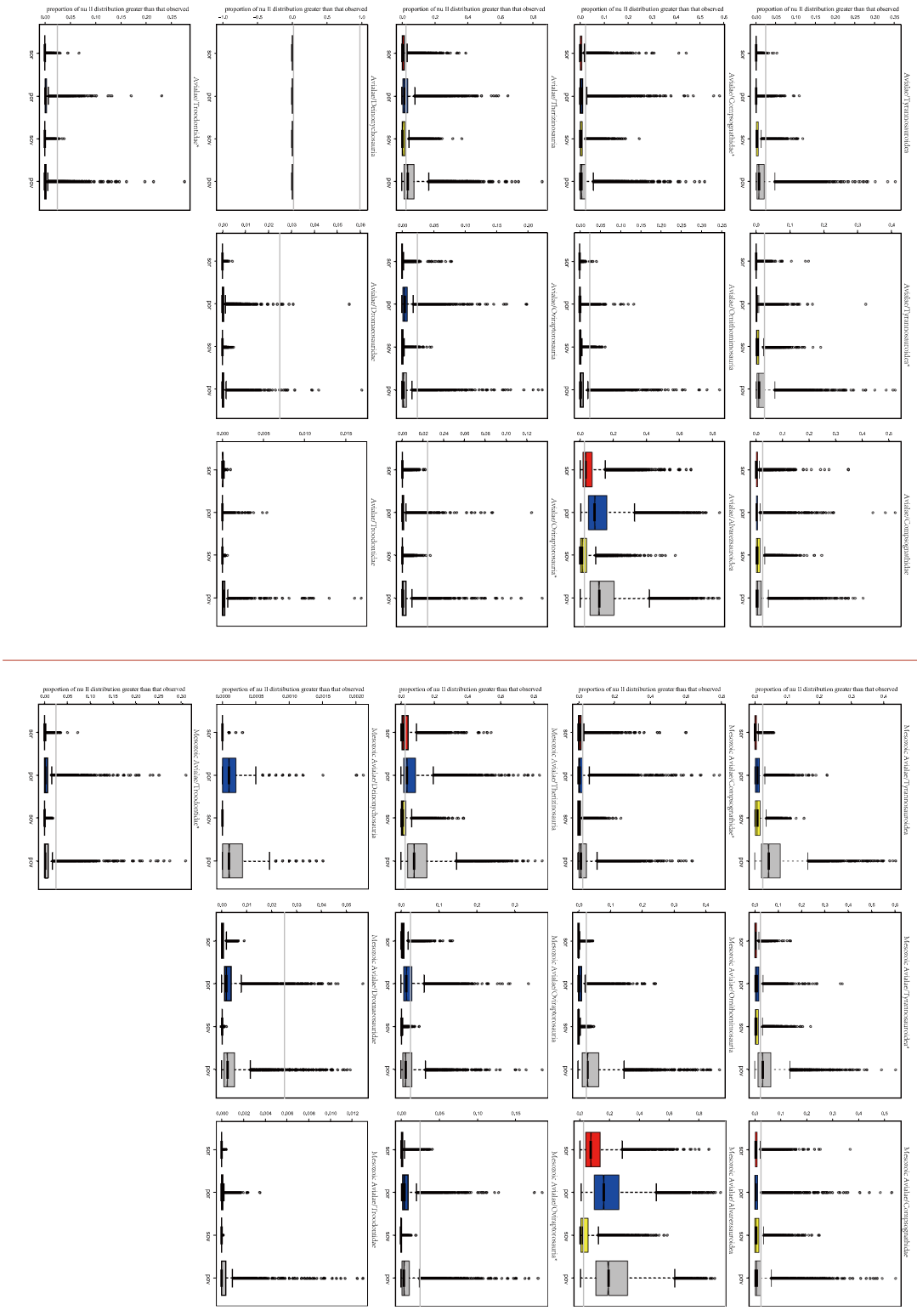


Figure S37. Forelimb disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

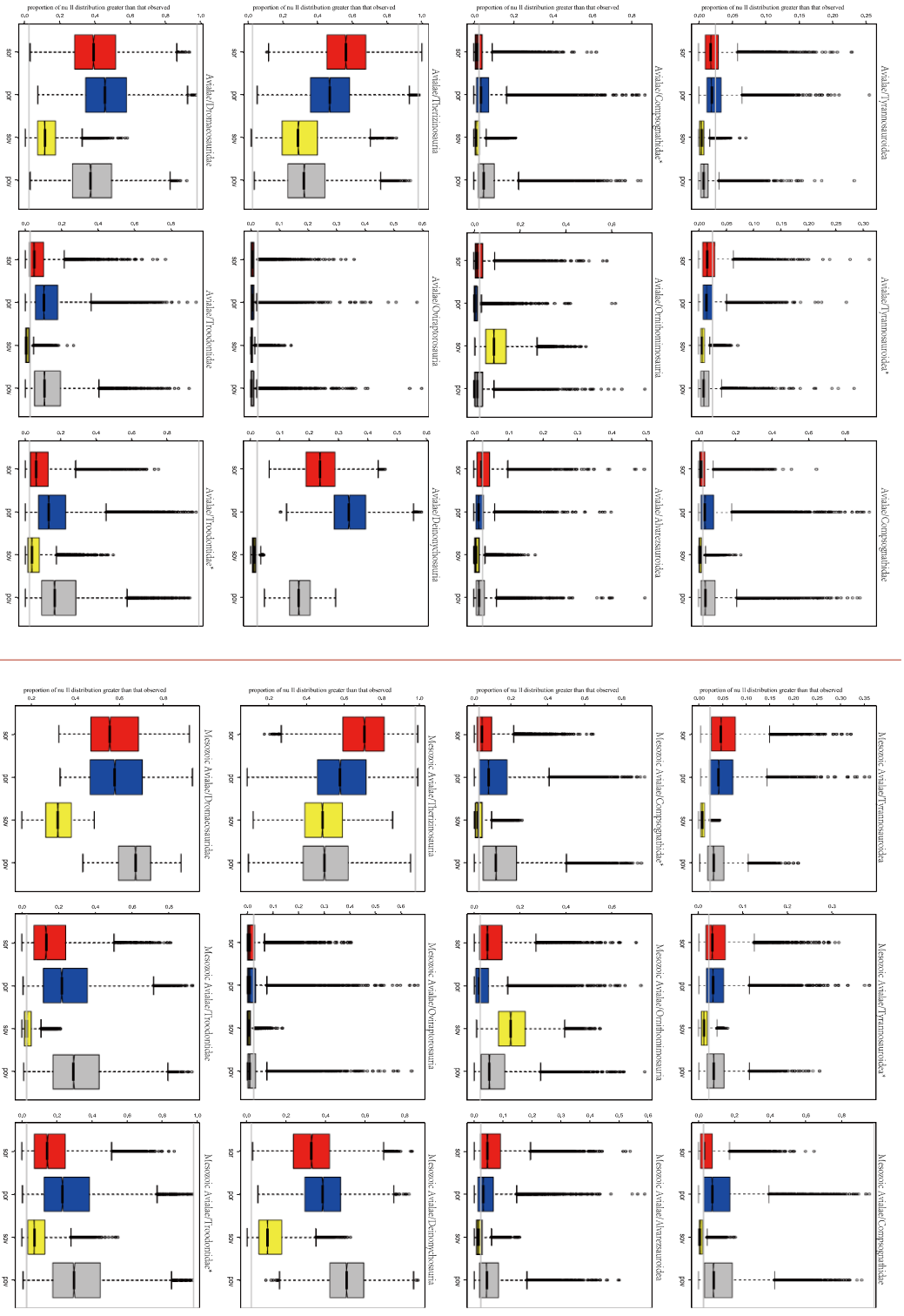


Figure S38. Pelvis disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

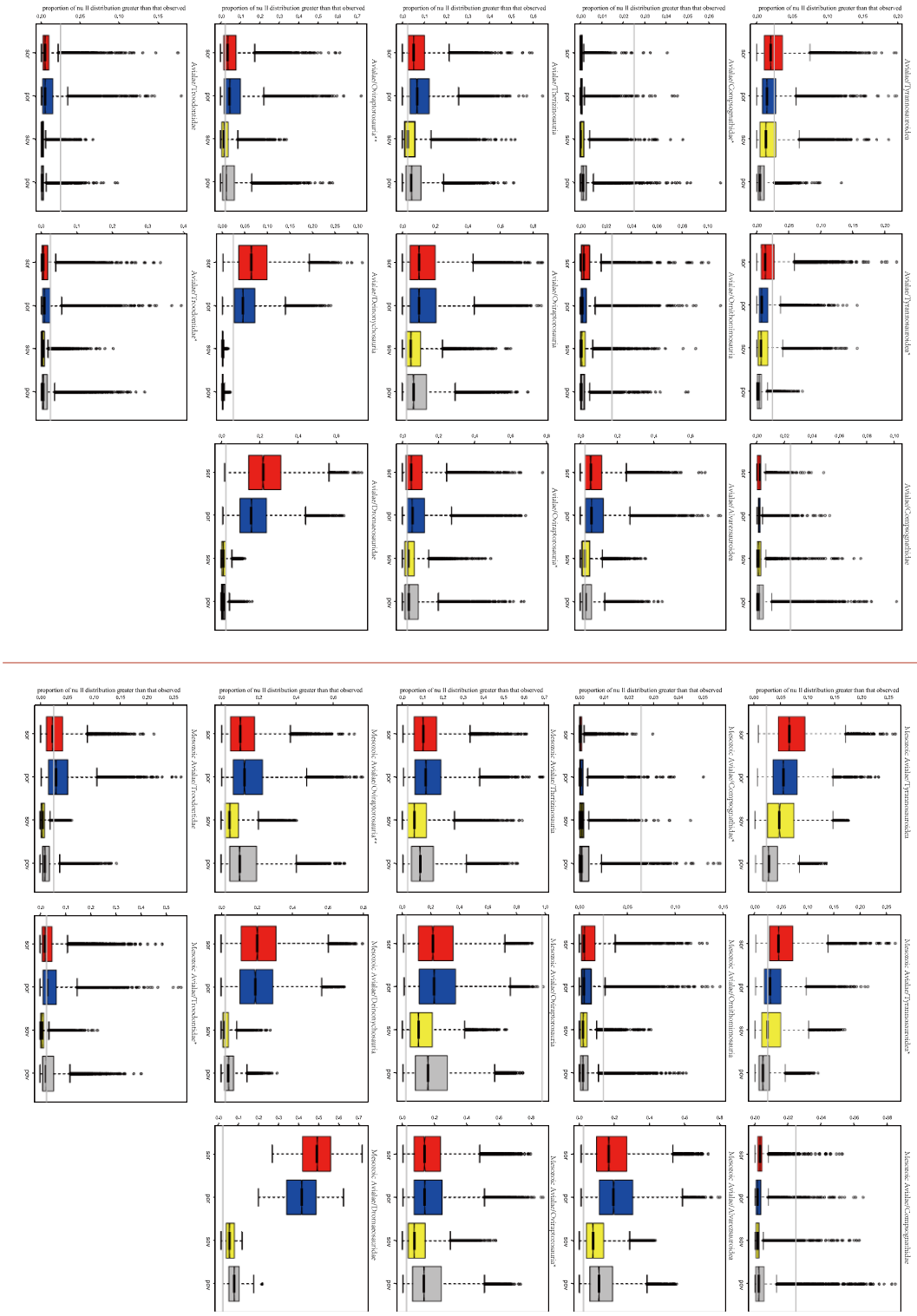


Figure S39. Hindlimb disparity comparison based on permutation tests with sample size corrected (GED). Up: all birds compared with other coelurosaurian clades. Below: Mesozoic birds compared with other coelurosaurian clades. See legend of **Figure S26**.

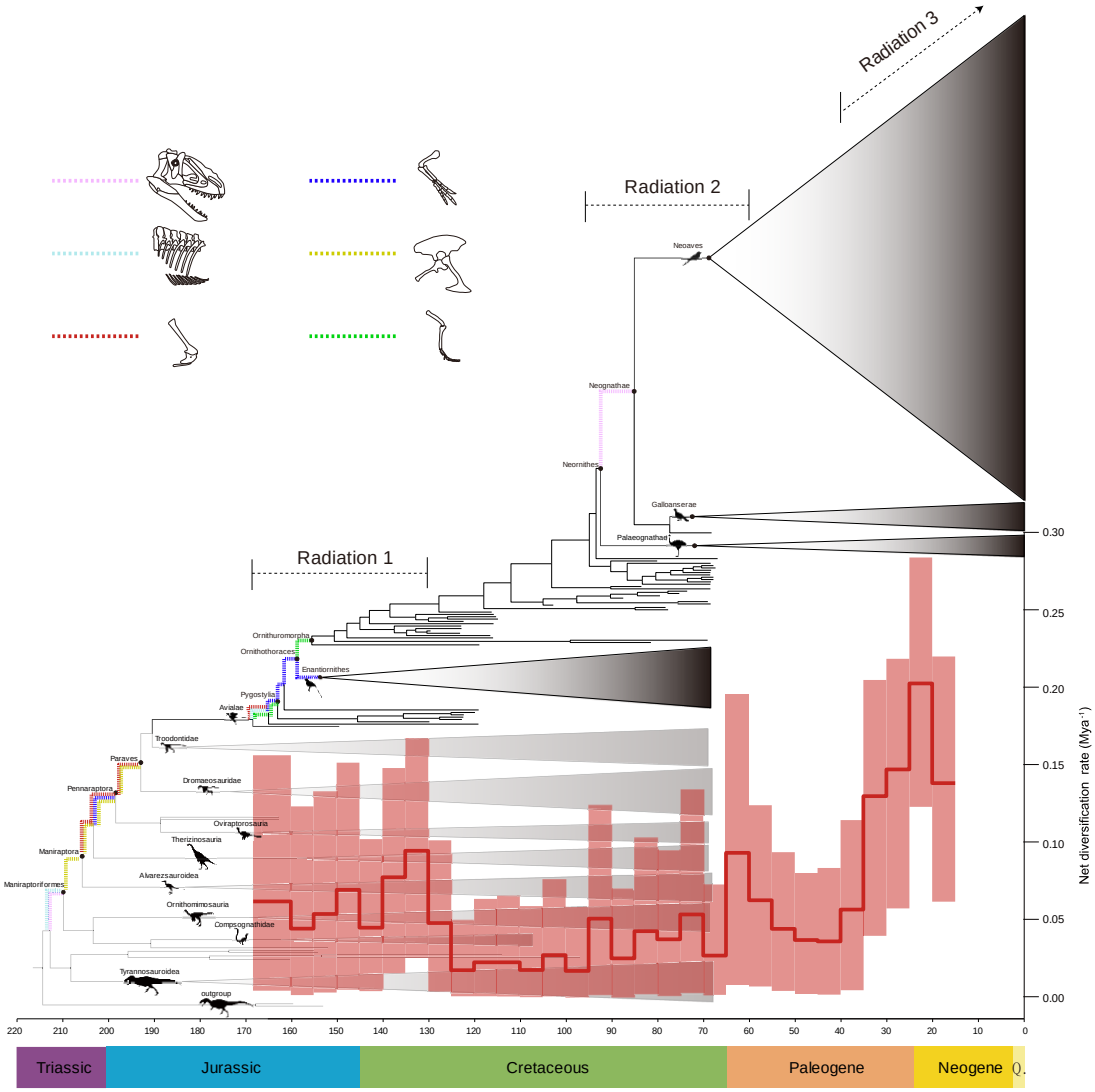


Figure S40. Sketch map of the relationship between morphological evolution and net diversification rate of birds. The tree combined supertree 1 with the non-avian theropods paraphyletic group. The branch length information of non-avian theropods paraphyletic group was derived from the results of morphological rates analyses based on partitioned WN relaxed model. If the morphological rate on a branch is 4 times or more larger than the relative mean (1.0) under both relaxed clock models, it is indicated as a colored dotted line (we only consider branches subtend to birds or in birds). The time spans of the three phases of avian radiation are shown above the tree. Silhouettes are from PhyloPic.org.

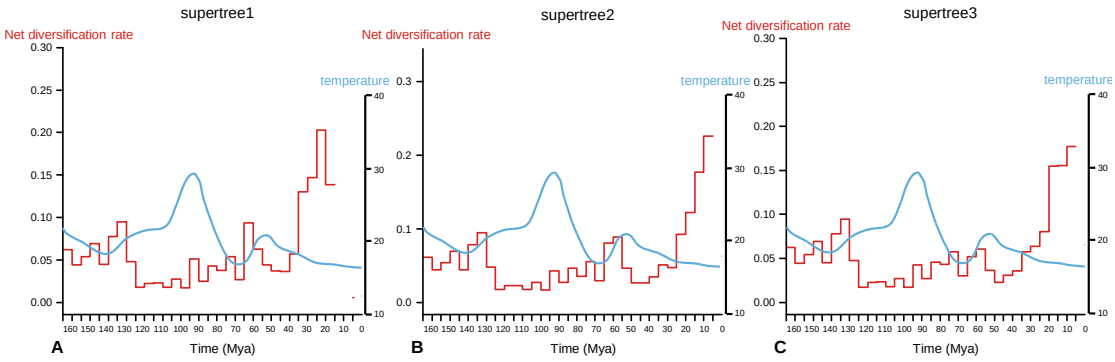


Figure S41. DRTT curve for birds and the temperature curve through Jurassic to the present day. (A)-(C) DRTT curves are derived from Supertree 1-3 (see Supplementary information for details). The temperature curve is derived from ref.(116).

K. Supplemental Tables.

Table S1. Results of permutation tests without sample size corrected.

Number in blanks are proportions greater than the observed values of the test statistic in the null distribution. Red: birds have significantly larger disparity than coelurosaurian clade; yellow: no difference between two groups; white: coelurosaurian clade have significantly larger disparity than birds. Abbreviations: sum of var: sum of variance, product of var: product of variance. Tyrannosauroidea*: Tyrannosauroidea without *Coelurus* and *Tanycolagreus*, Compsognathidae*:Compsognathidae including *Ornitholestes hermanni*, Oviraptorosauria*:Oviraptorosauria including *Epidexipteryx*, Oviraptorosauria**:Oviraptorosauria including *Epidexipteryx* and *Pedopenna*, Troodontidae*:Troodontidae without *Anchiornis*, *Aurornis*, *Eosinopteryx*, *Caihong Juji* and *Xiaotingia*.

MORD					
body	**sum of var**	**product of var**	**body**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0	0.0005	**Mesozoic Avialae/Tyrannosauroidea***	0.005	0.0075
Avialae/Tyrannosauroidea	0.0005	0	**Mesozoic Avialae/Tyrannosauroidea**	0.004	0.009
Avialae/Compsognathidae*	0	0	**Mesozoic Avialae/Compsognathidae***	0	0
Avialae/Compsognathidae	0	0	**Mesozoic Avialae/Compsognathidae**	0	0
Avialae/Ornithomimosauria	0	0	**Mesozoic Avialae/Ornithomimosauria**	0	0
Avialae/Alvarezsauroidea	0	0.001	**Mesozoic Avialae/Alvarezsauroidea**	0.0005	0.0075
Avialae/Therizinosauria	0	0.0005	**Mesozoic Avialae/Therizinosauria**	0	0.008
Avialae/Oviraptorosauria*	0	0	**Mesozoic Avialae/Oviraptorosauria***	0	0.003
Avialae/Oviraptorosauria	0	0	**Mesozoic Avialae/Oviraptorosauria**	0	0
Avialae/Deinonychosauria	0	0	**Mesozoic Avialae/Deinonychosauria**	0	0
Avialae/Dromaeosauridae	0	0	**Mesozoic Avialae/Dromaeosauridae**	0	0
Avialae/Troodontidae	0	0	**Mesozoic Avialae/Troodontidae**	0	0
Avialae/Troodontidae*	0	0	**Mesozoic Avialae/Troodontidae***	0	0.0005
skull	**sum of var**	**product of var**	**skull**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0.452	0.408	**Mesozoic Avialae/Tyrannosauroidea***	0.842	0.7805
Avialae/Tyrannosauroidea	0.564	0.5095	**Mesozoic Avialae/Tyrannosauroidea**	0.882	0.8455
Avialae/Compsognathidae*	0.0185	0.0015	**Mesozoic Avialae/Compsognathidae***	0.096	0.026
Avialae/Compsognathidae	0.001	0	**Mesozoic Avialae/Compsognathidae**	0.0085	0.0005
Avialae/Ornithomimosauria	0	0	**Mesozoic Avialae/Ornithomimosauria**	0.0005	0
Avialae/Alvarezsauroidea	0.9065	0.673	**Mesozoic Avialae/Alvarezsauroidea**	0.953	0.7555

Avialae/Therizinosauria	0.1605	0.3095	Mesozoic Avialae/Therizinosauria	0.462	0.5745
Avialae/Oviraptorosauria*	0.1555	0.1915	Mesozoic Avialae/Oviraptorosauria*	0.555	0.546
Avialae/Oviraptorosauria	0.0655	0.115	Mesozoic Avialae/Oviraptorosauria	0.3855	0.4465
Avialae/Deinonychosauria	0.0065	0.011	Mesozoic Avialae/Deinonychosauria	0.141	0.2265
Avialae/Dromaeosauridae	0.0095	0.0315	Mesozoic Avialae/Dromaeosauridae	0.123	0.2305
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0.016	0.021
Avialae/Troodontidae*	0.0015	0.0085	Mesozoic Avialae/Troodontidae*	0.0385	0.0885
axial	sum of var	product of var	axial	sum of var	product of var
Avialae/Tyrannosauroidea*	0.017	0.0145	Mesozoic Avialae/Tyrannosauroidea*	0.1615	0.166
Avialae/Tyrannosauroidea	0.0095	0.0375	Mesozoic Avialae/Tyrannosauroidea	0.152	0.31
Avialae/Compsognathidae*	0	0.001	Mesozoic Avialae/Compsognathidae*	0.0105	0.0085
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0.002
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0
Avialae/Alvarezsauroidea	0.031	0.1695	Mesozoic Avialae/Alvarezsauroidea	0.3165	0.554
Avialae/Therizinosauria	0.001	0.0335	Mesozoic Avialae/Therizinosauria	0.0225	0.1875
Avialae/Oviraptorosauria*	0.0035	0.0005	Mesozoic Avialae/Oviraptorosauria*	0.0695	0.005
Avialae/Oviraptorosauria	0.0035	0.005	Mesozoic Avialae/Oviraptorosauria	0.0565	0.0105
Avialae/Deinonychosauria	0.004	0.026	Mesozoic Avialae/Deinonychosauria	0.215	0.3785
Avialae/Dromaeosauridae	0.0035	0.049	Mesozoic Avialae/Dromaeosauridae	0.1455	0.393
Avialae/Troodontidae	0.0005	0.004	Mesozoic Avialae/Troodontidae	0.017	0.0815
Avialae/Troodontidae*	0.013	0.033	Mesozoic Avialae/Troodontidae*	0.095	0.157
ps	sum of var	product of var	ps	sum of var	product of var
Avialae/Tyrannosauroidea*	0	0	Mesozoic Avialae/Tyrannosauroidea*	0	0.0005
Avialae/Tyrannosauroidea	0	0	Mesozoic Avialae/Tyrannosauroidea	0.001	0.0005
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0
Avialae/Alvarezsauroidea	0	0.002	Mesozoic Avialae/Alvarezsauroidea	0.001	0.006
Avialae/Therizinosauria	0	0	Mesozoic Avialae/Therizinosauria	0.0005	0.0015
Avialae/Oviraptorosauria*	0.0015	0.001	Mesozoic Avialae/Oviraptorosauria*	0.009	0.0065
Avialae/Oviraptorosauria	0.0005	0.0015	Mesozoic Avialae/Oviraptorosauria	0.0105	0.007
Avialae/Deinonychosauria	0	0	Mesozoic Avialae/Deinonychosauria	0	0
Avialae/Dromaeosauridae	0	0	Mesozoic Avialae/Dromaeosauridae	0	0
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0	0.0005
Avialae/Troodontidae*	0	0.002	Mesozoic Avialae/Troodontidae*	0.0005	0.008
forelimb	sum of var	product of var	forelimb	sum of var	product of var
Avialae/Tyrannosauroidea*	0	0	Mesozoic Avialae/Tyrannosauroidea*	0	0
Avialae/Tyrannosauroidea	0	0	Mesozoic Avialae/Tyrannosauroidea	0	0.001

Avialae/Compsognathidae*	0	0	Mesozoic Avialae/Compsognathidae*	0	0
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0
Avialae/Alvarezsauroidea	0.0005	0.0255	Mesozoic Avialae/Alvarezsauroidea	0.0005	0.084
Avialae/Therizinosauria	0	0.005	Mesozoic Avialae/Therizinosauria	0	0.022
Avialae/Oviraptorosauria*	0	0	Mesozoic Avialae/Oviraptorosauria*	0	0.0005
Avialae/Oviraptorosauria	0	0	Mesozoic Avialae/Oviraptorosauria	0	0.005
Avialae/Deinonychosauria	0	0	Mesozoic Avialae/Deinonychosauria	0	0
Avialae/Dromaeosauridae	0	0	Mesozoic Avialae/Dromaeosauridae	0	0.0005
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0	0
Avialae/Troodontidae*	0	0	Mesozoic Avialae/Troodontidae*	0	0
pl	**sum of var**	**product of var**	**pl**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0	0.004	Mesozoic Avialae/Tyrannosauroidea*	0	0.0275
Avialae/Tyrannosauroidea	0	0.005	Mesozoic Avialae/Tyrannosauroidea	0	0.0355
Avialae/Compsognathidae*	0	0.015	Mesozoic Avialae/Compsognathidae*	0	0.032
Avialae/Compsognathidae	0	0.0125	Mesozoic Avialae/Compsognathidae	0	0.05
Avialae/Ornithomimosauria	0	0.001	Mesozoic Avialae/Ornithomimosauria	0	0.008
Avialae/Alvarezsauroidea	0.0065	0.014	Mesozoic Avialae/Alvarezsauroidea	0.038	0.0735
Avialae/Therizinosauria	0.0735	0.5045	Mesozoic Avialae/Therizinosauria	0.335	0.708
Avialae/Oviraptorosauria	0	0	Mesozoic Avialae/Oviraptorosauria	0	0.0075
Avialae/Deinonychosauria	0.0105	0.1745	Mesozoic Avialae/Deinonychosauria	0.153	0.5885
Avialae/Dromaeosauridae	0.066	0.406	Mesozoic Avialae/Dromaeosauridae	0.3145	0.7285
Avialae/Troodontidae	0.004	0.1255	Mesozoic Avialae/Troodontidae	0.034	0.3585
Avialae/Troodontidae*	0.0175	0.2305	Mesozoic Avialae/Troodontidae*	0.0755	0.4265
hindlimb	**sum of var**	**product of var**	**hindlimb**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0	0	Mesozoic Avialae/Tyrannosauroidea*	0.0025	0.0015
Avialae/Tyrannosauroidea	0	0	Mesozoic Avialae/Tyrannosauroidea	0.0055	0.001
Avialae/Compsognathidae*	0	0	Mesozoic Avialae/Compsognathidae*	0	0
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0.0005
Avialae/Alvarezsauroidea	0.0005	0	Mesozoic Avialae/Alvarezsauroidea	0.0015	0.004
Avialae/Therizinosauria	0.001	0.004	Mesozoic Avialae/Therizinosauria	0.0025	0.0205
Avialae/Oviraptorosauria**	0.0015	0.0265	Mesozoic Avialae/Oviraptorosauria**	0.01	0.0945
Avialae/Oviraptorosauria*	0.0015	0.0535	Mesozoic Avialae/Oviraptorosauria*	0.0175	0.155
Avialae/Oviraptorosauria	0.0055	0.059	Mesozoic Avialae/Oviraptorosauria	0.0335	0.1885
Avialae/Deinonychosauria	0	0	Mesozoic Avialae/Deinonychosauria	0.0015	0.0045
Avialae/Dromaeosauridae	0	0	Mesozoic Avialae/Dromaeosauridae	0.0025	0.0065
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0.003	0.0055

Avialae/Troodontidae*	0	0.0015	Mesozoic Avialae/Troodontidae*	0	0.005
GED					
body	**sum of var**	**product of var**	**body**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0.0125	0.0045	Mesozoic Avialae/Tyrannosauroidea*	0.077	0.056
Avialae/Tyrannosauroidea	0.0045	0.006	Mesozoic Avialae/Tyrannosauroidea	0.068	0.0775
Avialae/Compsognathidae*	0	0	Mesozoic Avialae/Compsognathidae*	0	0
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0.0005
Avialae/Alvarezsauroidea	0	0.0015	Mesozoic Avialae/Alvarezsauroidea	0.0025	0.023
Avialae/Therizinosauria	0	0.002	Mesozoic Avialae/Therizinosauria	0	0.0145
Avialae/Oviraptorosauria*	0	0.004	Mesozoic Avialae/Oviraptorosauria*	0.001	0.055
Avialae/Oviraptorosauria	0	0.001	Mesozoic Avialae/Oviraptorosauria	0	0.0265
Avialae/Deinonychosauria	0	0	Mesozoic Avialae/Deinonychosauria	0	0.0015
Avialae/Dromaeosauridae	0	0	Mesozoic Avialae/Dromaeosauridae	0	0.005
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0.0005	0.002
Avialae/Troodontidae*	0	0.001	Mesozoic Avialae/Troodontidae*	0	0.0065
skull	**sum of var**	**product of var**	**skull**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0.8045	0.722	Mesozoic Avialae/Tyrannosauroidea*	0.9765	0.927
Avialae/Tyrannosauroidea	0.8175	0.722	Mesozoic Avialae/Tyrannosauroidea	0.98	0.9355
Avialae/Compsognathidae*	0.1425	0.018	Mesozoic Avialae/Compsognathidae*	0.452	0.1485
Avialae/Compsognathidae	0.0295	0.0005	Mesozoic Avialae/Compsognathidae	0.1455	0.0025
Avialae/Ornithomimosauria	0.195	0.007	Mesozoic Avialae/Ornithomimosauria	0.6215	0.088
Avialae/Alvarezsauroidea	0.961	0.546	Mesozoic Avialae/Alvarezsauroidea	0.9845	0.6675
Avialae/Therizinosauria	0.2505	0.354	Mesozoic Avialae/Therizinosauria	0.63	0.664
Avialae/Oviraptorosauria*	0.257	0.3005	Mesozoic Avialae/Oviraptorosauria*	0.779	0.709
Avialae/Oviraptorosauria	0.212	0.293	Mesozoic Avialae/Oviraptorosauria	0.716	0.6385
Avialae/Deinonychosauria	0.048	0.0635	Mesozoic Avialae/Deinonychosauria	0.487	0.4285
Avialae/Dromaeosauridae	0.141	0.136	Mesozoic Avialae/Dromaeosauridae	0.6275	0.577
Avialae/Troodontidae	0.006	0.0025	Mesozoic Avialae/Troodontidae	0.075	0.0465
Avialae/Troodontidae*	0.0275	0.0175	Mesozoic Avialae/Troodontidae*	0.2065	0.162
axial	**sum of var**	**product of var**	**axial**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0.23	0.207	Mesozoic Avialae/Tyrannosauroidea*	0.6685	0.556
Avialae/Tyrannosauroidea	0.1745	0.258	Mesozoic Avialae/Tyrannosauroidea	0.623	0.615
Avialae/Compsognathidae*	0.004	0.0475	Mesozoic Avialae/Compsognathidae*	0.03	0.1725
Avialae/Compsognathidae	0.0005	0.0175	Mesozoic Avialae/Compsognathidae	0.0035	0.046
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0.0005	0
Avialae/Alvarezsauroidea	0.011	0.081	Mesozoic Avialae/Alvarezsauroidea	0.1445	0.247

Avialae/Therizinosauria	0	0.013	Mesozoic Avialae/Therizinosauria	0.005	0.069
Avialae/Oviraptorosauria*	0.002	0.006	Mesozoic Avialae/Oviraptorosauria*	0.0345	0.0255
Avialae/Oviraptorosauria	0.005	0.011	Mesozoic Avialae/Oviraptorosauria	0.0505	0.052
Avialae/Deinonychosauria	0.001	0.0655	Mesozoic Avialae/Deinonychosauria	0.1515	0.3365
Avialae/Dromaeosauridae	0.0035	0.1415	Mesozoic Avialae/Dromaeosauridae	0.1535	0.4355
Avialae/Troodontidae	0	0.013	Mesozoic Avialae/Troodontidae	0.0185	0.073
Avialae/Troodontidae*	0.009	0.1995	Mesozoic Avialae/Troodontidae*	0.109	0.432
ps	sum of var	product of var	ps	sum of var	product of var
Avialae/Tyrannosauroidea*	0	0	Mesozoic Avialae/Tyrannosauroidea*	0	0
Avialae/Tyrannosauroidea	0	0	Mesozoic Avialae/Tyrannosauroidea	0.0005	0.0005
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0
Avialae/Alvarezsauroidea	0	0	Mesozoic Avialae/Alvarezsauroidea	0	0.001
Avialae/Therizinosauria	0	0	Mesozoic Avialae/Therizinosauria	0	0.0025
Avialae/Oviraptorosauria*	0	0	Mesozoic Avialae/Oviraptorosauria*	0.0005	0
Avialae/Oviraptorosauria	0	0.0005	Mesozoic Avialae/Oviraptorosauria	0	0.0005
Avialae/Deinonychosauria	0	0	Mesozoic Avialae/Deinonychosauria	0	0
Avialae/Dromaeosauridae	0	0	Mesozoic Avialae/Dromaeosauridae	0	0
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0	0
Avialae/Troodontidae*	0.0005	0.0055	Mesozoic Avialae/Troodontidae*	0	0.015
forelimb	sum of var	product of var	forelimb	sum of var	product of var
Avialae/Tyrannosauroidea*	0	0.0025	Mesozoic Avialae/Tyrannosauroidea*	0.0005	0.0175
Avialae/Tyrannosauroidea	0	0.0005	Mesozoic Avialae/Tyrannosauroidea	0.001	0.03
Avialae/Compsognathidae*	0	0.0015	Mesozoic Avialae/Compsognathidae*	0	0.0045
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0.001
Avialae/Ornithomimosauria	0	0.001	Mesozoic Avialae/Ornithomimosauria	0	0.017
Avialae/Alvarezsauroidea	0.0005	0.0545	Mesozoic Avialae/Alvarezsauroidea	0.0055	0.1615
Avialae/Therizinosauria	0.0005	0.011	Mesozoic Avialae/Therizinosauria	0.0005	0.0465
Avialae/Oviraptorosauria*	0	0	Mesozoic Avialae/Oviraptorosauria*	0	0.0025
Avialae/Oviraptorosauria	0	0	Mesozoic Avialae/Oviraptorosauria	0	0.0075
Avialae/Deinonychosauria	0	0	Mesozoic Avialae/Deinonychosauria	0	0
Avialae/Dromaeosauridae	0	0	Mesozoic Avialae/Dromaeosauridae	0	0.0015
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0	0
Avialae/Troodontidae*	0	0	Mesozoic Avialae/Troodontidae*	0	0.0005
pl	sum of var	product of var	pl	sum of var	product of var
Avialae/Tyrannosauroidea*	0.0005	0.002	Mesozoic Avialae/Tyrannosauroidea*	0.002	0.027
Avialae/Tyrannosauroidea	0	0.0005	Mesozoic Avialae/Tyrannosauroidea	0.004	0.025

Avialae/Compsognathidae*	0.0005	0.023	Mesozoic Avialae/Compsognathidae*	0.003	0.096
Avialae/Compsognathidae	0	0.021	Mesozoic Avialae/Compsognathidae	0	0.0685
Avialae/Ornithomimosauria	0.023	0.0085	Mesozoic Avialae/Ornithomimosauria	0.0905	0.065
Avialae/Alvarezsauroidea	0	0.0065	Mesozoic Avialae/Alvarezsauroidea	0.005	0.028
Avialae/Therizinosauria	0.2155	0.3165	Mesozoic Avialae/Therizinosauria	0.4895	0.532
Avialae/Oviraptorosauria	0	0.0005	Mesozoic Avialae/Oviraptorosauria	0	0.008
Avialae/Deinonychosauria	0.0085	0.1535	Mesozoic Avialae/Deinonychosauria	0.092	0.507
Avialae/Dromaeosauridae	0.0855	0.371	Mesozoic Avialae/Dromaeosauridae	0.3	0.6785
Avialae/Troodontidae	0.0005	0.088	Mesozoic Avialae/Troodontidae	0.014	0.2725
Avialae/Troodontidae*	0.004	0.1025	Mesozoic Avialae/Troodontidae*	0.016	0.2805
hindlimb	**sum of var**	**product of var**	**hindlimb**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0.0015	0	Mesozoic Avialae/Tyrannosauroidea*	0.012	0.007
Avialae/Tyrannosauroidea	0.004	0.002	Mesozoic Avialae/Tyrannosauroidea	0.0335	0.021
Avialae/Compsognathidae*	0	0	Mesozoic Avialae/Compsognathidae*	0	0
Avialae/Compsognathidae	0	0	Mesozoic Avialae/Compsognathidae	0	0
Avialae/Ornithomimosauria	0	0	Mesozoic Avialae/Ornithomimosauria	0	0.001
Avialae/Alvarezsauroidea	0.015	0.019	Mesozoic Avialae/Alvarezsauroidea	0.065	0.087
Avialae/Therizinosauria	0.0045	0.0095	Mesozoic Avialae/Therizinosauria	0.0145	0.047
Avialae/Oviraptorosauria**	0.0015	0.015	Mesozoic Avialae/Oviraptorosauria**	0.02	0.091
Avialae/Oviraptorosauria*	0.006	0.023	Mesozoic Avialae/Oviraptorosauria*	0.0415	0.1245
Avialae/Oviraptorosauria	0.0185	0.0405	Mesozoic Avialae/Oviraptorosauria	0.072	0.1875
Avialae/Deinonychosauria	0.0015	0.0015	Mesozoic Avialae/Deinonychosauria	0.012	0.0265
Avialae/Dromaeosauridae	0.006	0.005	Mesozoic Avialae/Dromaeosauridae	0.049	0.0705
Avialae/Troodontidae	0	0	Mesozoic Avialae/Troodontidae	0.0005	0.004
Avialae/Troodontidae*	0	0.001	Mesozoic Avialae/Troodontidae*	0	0.009

Table S2. Observed values of test statistics (for permutation tests without sample size corrected).

All values are obtained by using disparity metrics of birds/Mesozoic birds minus values of Coelurosauria clades. Red: positive; yellow: negative. Abbreviations: sum of var: sum of variance, product of var: product of variance. Tyrannosauroidea*:Tyrannosauroidea without *Coelurus* and *Tanycolagreus*, Compsognathidae*:Compsognathidae including *Ornitholestes hermanni*, Oviraptorosauria*:Oviraptorosauria including *Epidexipteryx*, Oviraptorosauria**:Oviraptorosauria including *Epidexipteryx* and *Pedopenna*, Troodontidae*:Troodontidae without *Anchiornis*, *Aurornis*, *Eosinopteryx*, *Caihong Juji* and *Xiaotingia*.

MORD					
body	**sum of var**	**product of var**	**body**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	1.836519	0.03030516	Mesozoic Avialae/Tyrannosauroidea*	1.373223	0.02256718
Avialae/Tyrannosauroidea	1.755359	0.02795776	Mesozoic Avialae/Tyrannosauroidea	1.292063	0.02021978
Avialae/Compsognathidae*	4.427376	0.07732588	Mesozoic Avialae/Compsognathidae*	3.96408	0.0695879
Avialae/Compsognathidae	5.449977	0.08945233	Mesozoic Avialae/Compsognathidae	4.986681	0.08171435
Avialae/Ornithomimosauria	4.056086	0.05212995	Mesozoic Avialae/Ornithomimosauria	3.59279	0.04439197
Avialae/Alvarezsauroidea	2.237594	0.03685364	Mesozoic Avialae/Alvarezsauroidea	1.774297	0.02911566
Avialae/Therizinosauria	2.305758	0.03594896	Mesozoic Avialae/Therizinosauria	1.842462	0.02821098
Avialae/Oviraptorosauria*	2.35023	0.03514326	Mesozoic Avialae/Oviraptorosauria*	1.886934	0.02740528
Avialae/Oviraptorosauria	2.633554	0.04187939	Mesozoic Avialae/Oviraptorosauria	2.170258	0.03414141
Avialae/Deinonychosauria	2.153933	0.02530361	Mesozoic Avialae/Deinonychosauria	1.690637	0.01756563
Avialae/Dromaeosauridae	2.453886	0.02941463	Mesozoic Avialae/Dromaeosauridae	1.99059	0.02167665
Avialae/Troodontidae	2.540111	0.03333327	Mesozoic Avialae/Troodontidae	2.076814	0.02559529
Avialae/Troodontidae*	2.705319	0.04176391	Mesozoic Avialae/Troodontidae*	2.242023	0.03402593
skull	**sum of var**	**product of var**	**skull**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	0.1404011	0.007620836	Mesozoic Avialae/Tyrannosauroidea*	-0.9784114	-0.01724771
Avialae/Tyrannosauroidea	-0.1331142	0.002420267	Mesozoic Avialae/Tyrannosauroidea	-1.251927	-0.02244827
Avialae/Compsognathidae*	2.807569	0.0842526	Mesozoic Avialae/Compsognathidae*	1.688757	0.05938406
Avialae/Compsognathidae	4.12998	0.1105412	Mesozoic Avialae/Compsognathidae	3.011167	0.08567261
Avialae/Ornithomimosauria	4.54847	0.106727	Mesozoic Avialae/Ornithomimosauria	3.429657	0.08185845
Avialae/Alvarezsauroidea	-3.285651	0.1435464	Mesozoic Avialae/Alvarezsauroidea	-4.404463	0.1186779
Avialae/Therizinosauria	1.252406	0.05188041	Mesozoic Avialae/Therizinosauria	0.1335933	0.02701186
Avialae/Oviraptorosauria*	1.453167	0.03033098	Mesozoic Avialae/Oviraptorosauria*	-0.1885536	0.005462442
Avialae/Oviraptorosauria	0.9302589	0.03882911	Mesozoic Avialae/Oviraptorosauria	0.3343544	0.01396057
Avialae/Deinonychosauria	2.028048	0.02655671	Mesozoic Avialae/Deinonychosauria	0.9092359	0.001688164
Avialae/Dromaeosauridae	2.283413	0.04019029	Mesozoic Avialae/Dromaeosauridae	1.1646	0.01532175

Avialae/Troodontidae	3.155294	0.05344876	Mesozoic Avialae/Troodontidae	2.036482	0.02858022
Avialae/Troodontidae*	3.079181	0.05365593	Mesozoic Avialae/Troodontidae*	1.960368	0.02878739
axial	**sum of var**	**product of var**	**axial**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	2.603356	0.05963269	Mesozoic Avialae/Tyrannosauroidea*	1.147023	0.03149537
Avialae/Tyrannosauroidea	2.589522	0.04520739	Mesozoic Avialae/Tyrannosauroidea	1.133189	0.01707006
Avialae/Compsognathidae*	4.43466	0.1087619	Mesozoic Avialae/Compsognathidae*	2.978327	0.08062458
Avialae/Compsognathidae	5.90854	0.1342217	Mesozoic Avialae/Compsognathidae	4.452207	0.1060844
Avialae/Ornithomimosauria	6.996289	0.1284743	Mesozoic Avialae/Ornithomimosauria	5.539956	0.100337
Avialae/Alvarezsauroidea	1.925439	0.04385036	Mesozoic Avialae/Alvarezsauroidea	0.4691064	0.01571304
Avialae/Therizinosauria	3.337242	0.0721411	Mesozoic Avialae/Therizinosauria	1.880909	0.04400378
Avialae/Oviraptorosauria*	3.021305	0.100173	Mesozoic Avialae/Oviraptorosauria*	1.564972	0.07203572
Avialae/Oviraptorosauria	3.198422	0.1022111	Mesozoic Avialae/Oviraptorosauria	1.742089	0.07407373
Avialae/Deinonychosauria	2.01498	0.02390381	Mesozoic Avialae/Deinonychosauria	0.5586466	-0.004233513
Avialae/Dromaeosauridae	2.372526	0.03300829	Mesozoic Avialae/Dromaeosauridae	0.9161931	0.004870967
Avialae/Troodontidae	3.184039	0.06679364	Mesozoic Avialae/Troodontidae	1.727706	0.03865631
Avialae/Troodontidae*	2.886125	0.09082582	Mesozoic Avialae/Troodontidae*	1.429792	0.06268849
ps	**sum of var**	**product of var**	**ps**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	11.38247	0.2313726	Mesozoic Avialae/Tyrannosauroidea*	10.5235	0.207338
Avialae/Tyrannosauroidea	8.92147	0.1935494	Mesozoic Avialae/Tyrannosauroidea	8.062501	0.1695148
Avialae/Compsognathidae	11.00156	0.2426113	Mesozoic Avialae/Compsognathidae	10.14259	0.2185767
Avialae/Ornithomimosauria	7.987622	0.1564745	Mesozoic Avialae/Ornithomimosauria	7.128653	0.1324399
Avialae/Alvarezsauroidea	6.007073	0.1455171	Mesozoic Avialae/Alvarezsauroidea	5.148104	0.1214825
Avialae/Therizinosauria	4.876908	0.1461231	Mesozoic Avialae/Therizinosauria	4.017939	0.1220885
Avialae/Oviraptorosauria*	4.12169	0.1318377	Mesozoic Avialae/Oviraptorosauria*	3.262721	0.1078031
Avialae/Oviraptorosauria	4.546355	0.1461231	Mesozoic Avialae/Oviraptorosauria	3.687385	0.1206633
Avialae/Deinonychosauria	6.562313	0.1125468	Mesozoic Avialae/Deinonychosauria	5.703344	0.08851223
Avialae/Dromaeosauridae	6.359819	0.1226607	Mesozoic Avialae/Dromaeosauridae	5.50085	0.09862611
Avialae/Troodontidae	7.891767	0.1486851	Mesozoic Avialae/Troodontidae	7.032798	0.1246505
Avialae/Troodontidae*	9.132515	0.2079392	Mesozoic Avialae/Troodontidae*	8.273545	0.1839046
forelimb	**sum of var**	**product of var**	**forelimb**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	7.538262	0.1683443	Mesozoic Avialae/Tyrannosauroidea*	6.51274	0.136911
Avialae/Tyrannosauroidea	6.630236	0.1360554	Mesozoic Avialae/Tyrannosauroidea	5.604715	0.1046221
Avialae/Compsognathidae*	8.429358	0.185409	Mesozoic Avialae/Compsognathidae*	7.403836	0.1539758
Avialae/Compsognathidae	9.740895	0.2303766	Mesozoic Avialae/Compsognathidae	8.715373	0.1989433
Avialae/Ornithomimosauria	7.670018	0.1597032	Mesozoic Avialae/Ornithomimosauria	6.644496	0.12827
Avialae/Alvarezsauroidea	5.823901	0.1398188	Mesozoic Avialae/Alvarezsauroidea	4.798379	0.1083855
Avialae/Therizinosauria	6.286698	0.1386556	Mesozoic Avialae/Therizinosauria	5.261176	0.1072224
Avialae/Oviraptorosauria*	7.494188	0.162984	Mesozoic Avialae/Oviraptorosauria*	6.468666	0.1315508
Avialae/Oviraptorosauria	7.003404	0.1614896	Mesozoic Avialae/Oviraptorosauria	5.977883	0.1300563
Avialae/Deinonychosauria	8.522149	0.140505	Mesozoic Avialae/Deinonychosauria	7.496627	0.1090718

Avialae/Dromaeosauridae	8.264534	0.1467163	Mesozoic Avialae/Dromaeosauridae	7.239013	0.1152831
Avialae/Troodontidae	9.152749	0.1640593	Mesozoic Avialae/Troodontidae	8.127228	0.1326261
Avialae/Troodontidae*	10.42593	0.2271403	Mesozoic Avialae/Troodontidae*	9.400413	0.1957071
pl	**sum of var**	**product of var**	**pl**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	6.326836	0.09508845	Mesozoic Avialae/Tyrannosauroidea*	5.299934	0.07118622
Avialae/Tyrannosauroidea	5.776974	0.08844386	Mesozoic Avialae/Tyrannosauroidea	4.750072	0.06454163
Avialae/Compsognathidae*	5.17573	0.097099	Mesozoic Avialae/Compsognathidae*	4.148828	0.07319677
Avialae/Compsognathidae	5.887233	0.1089591	Mesozoic Avialae/Compsognathidae	4.860331	0.08505685
Avialae/Ornithomimosauria	5.363363	0.09813872	Mesozoic Avialae/Ornithomimosauria	4.336461	0.07423649
Avialae/Alvarezsauroidea	3.392823	0.08052071	Mesozoic Avialae/Alvarezsauroidea	2.365921	0.05661848
Avialae/Therizinosauria	1.456209	0.02593463	Mesozoic Avialae/Therizinosauria	0.4293074	0.002032397
Avialae/Oviraptorosauria	6.492346	0.1022459	Mesozoic Avialae/Oviraptorosauria	5.465444	0.07834371
Avialae/Deinonychosauria	1.95489	0.01072192	Mesozoic Avialae/Deinonychosauria	0.9279884	-0.01318031
Avialae/Dromaeosauridae	1.576803	0.01518595	Mesozoic Avialae/Dromaeosauridae	0.5499006	-0.008716284
Avialae/Troodontidae	2.760347	0.04368024	Mesozoic Avialae/Troodontidae	1.733445	0.01977801
Avialae/Troodontidae*	2.931797	0.07128852	Mesozoic Avialae/Troodontidae*	1.904895	0.04738628
hindlimb	**sum of var**	**product of var**	**hindlimb**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	5.073081	0.1052723	Mesozoic Avialae/Tyrannosauroidea*	4.067858	0.08663305
Avialae/Tyrannosauroidea	4.404758	0.08803177	Mesozoic Avialae/Tyrannosauroidea	3.399535	0.0693925
Avialae/Compsognathidae*	12.68504	0.1981522	Mesozoic Avialae/Compsognathidae*	11.67981	0.1795129
Avialae/Compsognathidae	13.97075	0.2259598	Mesozoic Avialae/Compsognathidae	12.96553	0.2073205
Avialae/Ornithomimosauria	7.496349	0.1099799	Mesozoic Avialae/Ornithomimosauria	6.491126	0.0913406
Avialae/Alvarezsauroidea	5.357913	0.08231922	Mesozoic Avialae/Alvarezsauroidea	4.35269	0.06367995
Avialae/Therizinosauria	4.802731	0.08844915	Mesozoic Avialae/Therizinosauria	3.797508	0.06980988
Avialae/Oviraptorosauria**	4.195129	0.06441739	Mesozoic Avialae/Oviraptorosauria**	3.189906	0.04577812
Avialae/Oviraptorosauria*	3.937067	0.06221974	Mesozoic Avialae/Oviraptorosauria*	2.931844	0.04358047
Avialae/Oviraptorosauria	3.78564	0.06809676	Mesozoic Avialae/Oviraptorosauria	2.780417	0.0494575
Avialae/Deinonychosauria	3.809091	0.04766798	Mesozoic Avialae/Deinonychosauria	2.803868	0.02902871
Avialae/Dromaeosauridae	4.170164	0.06136195	Mesozoic Avialae/Dromaeosauridae	3.164941	0.04272268
Avialae/Troodontidae	4.198399	0.0741417	Mesozoic Avialae/Troodontidae	3.193176	0.05550243
Avialae/Troodontidae*	5.284466	0.09271711	Mesozoic Avialae/Troodontidae*	4.279243	0.07407784
GED					
body	**sum of var**	**product of var**	**body**	**sum of var**	**product of var**
Avialae/Tyrannosauroidea*	1337.597	24.50169	Mesozoic Avialae/Tyrannosauroidea*	780.9658	15.14252
Avialae/Tyrannosauroidea	1309.857	21.19596	Mesozoic Avialae/Tyrannosauroidea	753.2258	11.83679
Avialae/Compsognathidae*	2861.836	60.47967	Mesozoic Avialae/Compsognathidae*	2305.204	51.12051
Avialae/Compsognathidae	3625.36	71.56681	Mesozoic Avialae/Compsognathidae	3068.728	62.20765
Avialae/Ornithomimosauria	2328.148	30.80402	Mesozoic Avialae/Ornithomimosauria	1771.517	21.44486
Avialae/Alvarezsauroidea	1925.259	30.73379	Mesozoic Avialae/Alvarezsauroidea	1368.628	21.37462
Avialae/Therizinosauria	2338.443	34.04354	Mesozoic Avialae/Therizinosauria	1781.812	24.68438

Avialae/Oviraptorosauria*	2096.546	26.47233	Mesozoic Avialae/Oviraptorosauria*	1539.914	17.11316
Avialae/Oviraptorosauria	2245.491	31.47126	Mesozoic Avialae/Oviraptorosauria	1688.86	22.11209
Avialae/Deinonychosauria	1996.767	20.5397	Mesozoic Avialae/Deinonychosauria	1440.135	11.18054
Avialae/Dromaeosauridae	2123.283	23.88041	Mesozoic Avialae/Dromaeosauridae	1566.652	14.52124
Avialae/Troodontidae	2364.062	27.19105	Mesozoic Avialae/Troodontidae	1807.431	17.83189
Avialae/Troodontidae*	2466.764	31.6609	Mesozoic Avialae/Troodontidae*	1910.132	22.30173
skull	sum of var	product of var	skull	sum of var	product of var
Avialae/Tyrannosauroidea*	-322.6829	-2.436811	Mesozoic Avialae/Tyrannosauroidea*	-794.5921	-10.25142
Avialae/Tyrannosauroidea	-358.5693	-2.746485	Mesozoic Avialae/Tyrannosauroidea	-830.4786	-10.5611
Avialae/Compsognathidae*	525.5285	22.85729	Mesozoic Avialae/Compsognathidae*	53.61928	15.04267
Avialae/Compsognathidae	1003.738	33.90249	Mesozoic Avialae/Compsognathidae	531.8285	26.08788
Avialae/Ornithomimosauria	351.3031	19.39985	Mesozoic Avialae/Ornithomimosauria	-120.6062	11.58524
Avialae/Alvarezsauroidea	-1812.565	45.25066	Mesozoic Avialae/Alvarezsauroidea	-2284.474	37.43605
Avialae/Therizinosauria	312.7159	13.64008	Mesozoic Avialae/Therizinosauria	-159.1934	5.825465
Avialae/Oviraptorosauria*	204.0499	6.880663	Mesozoic Avialae/Oviraptorosauria*	-267.8593	-0.9339505
Avialae/Oviraptorosauria	288.2305	8.457632	Mesozoic Avialae/Oviraptorosauria	-183.6787	0.6430188
Avialae/Deinonychosauria	481.0012	5.78219	Mesozoic Avialae/Deinonychosauria	9.091911	-2.032423
Avialae/Dromaeosauridae	367.1288	7.415392	Mesozoic Avialae/Dromaeosauridae	-104.7804	-0.3992212
Avialae/Troodontidae	942.8145	15.70459	Mesozoic Avialae/Troodontidae	470.9053	7.889981
Avialae/Troodontidae*	771.4293	14.89191	Mesozoic Avialae/Troodontidae*	299.5201	7.0773
axial	sum of var	product of var	axial	sum of var	product of var
Avialae/Tyrannosauroidea*	72.64665	2.198112	Mesozoic Avialae/Tyrannosauroidea*	-45.08351	0.5299609
Avialae/Tyrannosauroidea	87.54501	1.666513	Mesozoic Avialae/Tyrannosauroidea	-30.18515	-0.001638477
Avialae/Compsognathidae*	292.4493	5.463515	Mesozoic Avialae/Compsognathidae*	174.7191	3.795364
Avialae/Compsognathidae	388.0763	7.424561	Mesozoic Avialae/Compsognathidae	270.3461	5.75641
Avialae/Ornithomimosauria	417.6334	7.177035	Mesozoic Avialae/Ornithomimosauria	299.9032	5.508884
Avialae/Alvarezsauroidea	195.0778	3.97444	Mesozoic Avialae/Alvarezsauroidea	77.34769	2.306289
Avialae/Therizinosauria	312.2808	5.772011	Mesozoic Avialae/Therizinosauria	194.5506	4.10386
Avialae/Oviraptorosauria*	251.2863	6.189713	Mesozoic Avialae/Oviraptorosauria*	133.5561	4.521562
Avialae/Oviraptorosauria	254.1029	6.138786	Mesozoic Avialae/Oviraptorosauria	136.3727	4.470635
Avialae/Deinonychosauria	178.3377	1.615356	Mesozoic Avialae/Deinonychosauria	60.60753	-0.05279561
Avialae/Dromaeosauridae	187.363	1.862874	Mesozoic Avialae/Dromaeosauridae	69.6328	0.1947231
Avialae/Troodontidae	264.6011	4.399877	Mesozoic Avialae/Troodontidae	146.871	2.731725
Avialae/Troodontidae*	227.4479	4.606387	Mesozoic Avialae/Troodontidae*	109.7178	2.938235
ps	sum of var	product of var	ps	sum of var	product of var
Avialae/Tyrannosauroidea*	397.1871	7.937311	Mesozoic Avialae/Tyrannosauroidea*	369.2983	7.177872
Avialae/Tyrannosauroidea	320.5816	6.830278	Mesozoic Avialae/Tyrannosauroidea	292.6927	6.070839
Avialae/Compsognathidae	353.4228	8.684512	Mesozoic Avialae/Compsognathidae	325.534	7.925072
Avialae/Ornithomimosauria	292.3038	5.692084	Mesozoic Avialae/Ornithomimosauria	264.4149	4.932644
Avialae/Alvarezsauroidea	239.8983	5.575977	Mesozoic Avialae/Alvarezsauroidea	212.0094	4.816538

Avialae/Therizinosauria	232.6614	4.924471	Mesozoic Avialae/Therizinosauria	204.7725	4.165032
Avialae/Oviraptorosauria*	211.5094	5.972305	Mesozoic Avialae/Oviraptorosauria*	183.6205	5.212865
Avialae/Oviraptorosauria	212.4522	5.849302	Mesozoic Avialae/Oviraptorosauria	184.5633	5.089863
Avialae/Deinonychosauria	250.0384	4.133734	Mesozoic Avialae/Deinonychosauria	222.1495	3.374294
Avialae/Dromaeosauridae	240.7337	4.44145	Mesozoic Avialae/Dromaeosauridae	212.8449	3.682011
Avialae/Troodontidae	297.1602	5.221069	Mesozoic Avialae/Troodontidae	269.2714	4.461629
Avialae/Troodontidae*	290.9133	6.670859	Mesozoic Avialae/Troodontidae*	263.0244	5.91142
forelimb	sum of var	product of var	forelimb	sum of var	product of var
Avialae/Tyrannosauroidea*	438.5092	9.97301	Mesozoic Avialae/Tyrannosauroidea*	346.6358	7.351981
Avialae/Tyrannosauroidea	387.6845	8.135477	Mesozoic Avialae/Tyrannosauroidea	295.8111	5.514448
Avialae/Compsognathidae*	602.2468	13.13104	Mesozoic Avialae/Compsognathidae*	510.3733	10.51001
Avialae/Compsognathidae	684.1589	15.85974	Mesozoic Avialae/Compsognathidae	592.2854	13.23871
Avialae/Ornithomimosauria	443.873	7.936997	Mesozoic Avialae/Ornithomimosauria	351.9995	5.315968
Avialae/Alvarezsauroidea	422.861	9.440815	Mesozoic Avialae/Alvarezsauroidea	330.9876	6.819786
Avialae/Therizinosauria	473.9741	10.21523	Mesozoic Avialae/Therizinosauria	382.1007	7.594205
Avialae/Oviraptorosauria*	624.4997	11.72867	Mesozoic Avialae/Oviraptorosauria*	532.6262	9.107639
Avialae/Oviraptorosauria	589.4998	11.69603	Mesozoic Avialae/Oviraptorosauria	497.6264	9.075004
Avialae/Deinonychosauria	709.715	10.52469	Mesozoic Avialae/Deinonychosauria	617.8416	7.903664
Avialae/Dromaeosauridae	650.7791	10.95898	Mesozoic Avialae/Dromaeosauridae	558.9057	8.337952
Avialae/Troodontidae	791.0786	12.336	Mesozoic Avialae/Troodontidae	699.2052	9.714973
Avialae/Troodontidae*	859.082	14.97975	Mesozoic Avialae/Troodontidae*	767.2086	12.35872
pl	sum of var	product of var	pl	sum of var	product of var
Avialae/Tyrannosauroidea*	384.0114	7.11767	Mesozoic Avialae/Tyrannosauroidea*	313.049	5.387535
Avialae/Tyrannosauroidea	365.959	6.646984	Mesozoic Avialae/Tyrannosauroidea	294.9966	4.916849
Avialae/Compsognathidae*	385.2138	6.78003	Mesozoic Avialae/Compsognathidae*	314.2514	5.049895
Avialae/Compsognathidae	465.6019	7.973474	Mesozoic Avialae/Compsognathidae	394.6395	6.243339
Avialae/Ornithomimosauria	216.2413	6.539257	Mesozoic Avialae/Ornithomimosauria	145.2789	4.809122
Avialae/Alvarezsauroidea	330.486	7.058233	Mesozoic Avialae/Alvarezsauroidea	259.5236	5.328097
Avialae/Therizinosauria	73.51285	2.656089	Mesozoic Avialae/Therizinosauria	2.550458	0.9259541
Avialae/Oviraptorosauria	422.1245	8.183621	Mesozoic Avialae/Oviraptorosauria	351.1621	6.453485
Avialae/Deinonychosauria	166.7555	1.229171	Mesozoic Avialae/Deinonychosauria	95.79309	-0.500964
Avialae/Dromaeosauridae	110.8298	1.128875	Mesozoic Avialae/Dromaeosauridae	39.86744	-0.60126
Avialae/Troodontidae	269.9346	3.630753	Mesozoic Avialae/Troodontidae	198.9722	1.900618
Avialae/Troodontidae*	310.1277	6.025876	Mesozoic Avialae/Troodontidae*	239.1653	4.295741
hindlimb	sum of var	product of var	hindlimb	sum of var	product of var
Avialae/Tyrannosauroidea*	416.2658	8.297683	Mesozoic Avialae/Tyrannosauroidea*	312.4903	6.298378
Avialae/Tyrannosauroidea	341.0269	6.903335	Mesozoic Avialae/Tyrannosauroidea	237.2514	4.90403
Avialae/Compsognathidae*	1283.006	18.2412	Mesozoic Avialae/Compsognathidae*	1179.231	16.24189
Avialae/Compsognathidae	1354	19.62419	Mesozoic Avialae/Compsognathidae	1250.225	17.62488
Avialae/Ornithomimosauria	588.3462	9.72513	Mesozoic Avialae/Ornithomimosauria	484.5707	7.725824

Avialae/Alvarezsauroidea	312.9624	5.571554	**Mesozoic Avialae/Alvarezsauroidea**	209.1869	3.572248
Avialae/Therizinosauria	417.8653	8.323004	**Mesozoic Avialae/Therizinosauria**	314.0899	6.323699
Avialae/Oviraptorosauria**	406.621	6.287302	**Mesozoic Avialae/Oviraptorosauria****	302.8455	4.287996
Avialae/Oviraptorosauria*	379.0731	6.462454	**Mesozoic Avialae/Oviraptorosauria***	275.2977	4.463148
Avialae/Oviraptorosauria	338.5053	6.499493	**Mesozoic Avialae/Oviraptorosauria**	234.7298	4.500188
Avialae/Deinonychosauria	359.602	4.573775	**Mesozoic Avialae/Deinonychosauria**	255.8265	2.574469
Avialae/Dromaeosauridae	322.8993	4.847911	**Mesozoic Avialae/Dromaeosauridae**	219.1238	2.848606
Avialae/Troodontidae	486.7357	7.703678	**Mesozoic Avialae/Troodontidae**	382.9602	5.704373
Avialae/Troodontidae*	549.2862	8.805567	**Mesozoic Avialae/Troodontidae***	445.5107	6.806261

Table S3. P-values of PERMANOVA.

Red: significant; yellow: insignificant. Tyrannosauroidea*:Tyrannosauroidea without *Coelurus* and *Tanycolagreus*, Compsognathidae*:Compsognathidae including *Ornitholestes hermanni*, Oviraptorosauria*:Oviraptorosauria including *Epidexipteryx*, Oviraptorosauria**:Oviraptorosauria including *Epidexipteryx* and *Pedopenna*, Troodontidae*:Troodontidae without *Anchiornis*, *Aurornis*, *Eosinopteryx*, *Caihong Juji* and *Xiaotingia*.

MORD							
body	F value	P value	permutation	**body**	F value	P value	permutation
Avialae/Tyrannosauroidea*	27.553	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	24.279	0.0003332	3000
Avialae/Tyrannosauroidea	28.081	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	23.973	0.0003332	3000
Avialae/Compsognathidae*	11.434	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	10.84	0.0003332	3000
Avialae/Compsognathidae	10.404	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	10.117	0.0006664	3000
Avialae/Ornithomimosauria	29.067	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	28.542	0.0003332	3000
Avialae/Alvarezsauroidea	16.649	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	15.544	0.0003332	3000
Avialae/Therizinosauria	17.655	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	18.782	0.0003332	3000
Avialae/Oviraptorosauria*	15.719	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	14.96	0.0003332	3000
Avialae/Oviraptorosauria	16.221	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	16.045	0.0003332	3000
Avialae/Deinonychosauria	27.2	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	19.314	0.0003332	3000
Avialae/Dromaeosauridae	22.008	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	17.295	0.0003332	3000
Avialae/Troodontidae*	15.825	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	14.42	0.0003332	3000
Avialae/Troodontidae	19.207	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	16.202	0.0003332	3000
Avialae/non-ave	28.379	0.0003332	3000	**Mesozoic Avialae/non-ave**	18.015	0.0003332	3000

skull	F value	P value	permutation	skull	F value	P value	permutation
Avialae/Tyrannosauroidea*	26.719	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	24.123	0.0003332	3000
Avialae/Tyrannosauroidea	25.451	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	22.142	0.0003332	3000
Avialae/Compsognathidae*	7.3552	0.0006664	3000	**Mesozoic Avialae/Compsognathidae***	7.2008	0.0003332	3000
Avialae/Compsognathidae	6.776	0.001333	3000	**Mesozoic Avialae/Compsognathidae**	7.1135	0.0003332	3000
Avialae/Ornithomimosauria	11.305	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	14.041	0.0003332	3000
Avialae/Alvarezsauroidea	2.4808	0.05098	3000	**Mesozoic Avialae/Alvarezsauroidea**	2.6924	0.02799	3000
Avialae/Therizinosauria	18.23	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	22.403	0.0003332	3000
Avialae/Oviraptorosauria*	21.67	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	27.344	0.0003332	3000
Avialae/Oviraptorosauria	24.337	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	33.15	0.0003332	3000
Avialae/Deinonychosauria	16.638	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	9.3626	0.0003332	3000
Avialae/Dromaeosauridae	15.23	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	11.312	0.0003332	3000
Avialae/Troodontidae*	11.92	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	10.716	0.0003332	3000
Avialae/Troodontidae	15.583	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	12.677	0.0003332	3000
Avialae/non-ave	13.611	0.0003332	3000	**Mesozoic Avialae/non-ave**	7.3141	0.0003332	3000
axial	F value	P value	permutation	**axial**	F value	P value	permutation
Avialae/Tyrannosauroidea*	13.676	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	11.606	0.0003332	3000
Avialae/Tyrannosauroidea	14.185	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	11.458	0.0003332	3000
Avialae/Compsognathidae*	8.689	0.0006664	3000	**Mesozoic Avialae/Compsognathidae***	12.792	0.0003332	3000
Avialae/Compsognathidae	8.2235	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	12.912	0.0003332	3000
Avialae/Ornithomimosauria	10.777	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	10.929	0.0003332	3000

Avialae/Alvarezsauroidea	10.43	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	14.609	0.0003332	3000
Avialae/Therizinosauria	8.4845	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	11.857	0.0003332	3000
Avialae/Oviraptorosauria*	4.9841	0.002999	3000	**Mesozoic Avialae/Oviraptorosauria***	3.7576	0.005332	3000
Avialae/Oviraptorosauria	4.5837	0.005998	3000	**Mesozoic Avialae/Oviraptorosauria**	3.3413	0.01166	3000
Avialae/Deinonychosauria	18.458	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	19.218	0.0003332	3000
Avialae/Dromaeosauridae	17.217	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	18.602	0.0003332	3000
Avialae/Troodontidae*	4.9952	0.002666	3000	**Mesozoic Avialae/Troodontidae***	6.0238	0.0009997	3000
Avialae/Troodontidae	9.1459	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	9.3191	0.0003332	3000
Avialae/non-ave	17.453	0.0003332	3000	**Mesozoic Avialae/non-ave**	10.887	0.0003332	3000
ps	F value	P value	permutation	**ps**	F value	P value	permutation
Avialae/Tyrannosauroidea*	20.305	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	19.663	0.0003332	3000
Avialae/Tyrannosauroidea	23.356	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	22.251	0.0003332	3000
Avialae/Compsognathidae	14.67	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	14.94	0.0003332	3000
Avialae/Ornithomimosauria	28.832	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	26.486	0.0003332	3000
Avialae/Alvarezsauroidea	15.356	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	14.719	0.0003332	3000
Avialae/Therizinosauria	17.161	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	17.152	0.0003332	3000
Avialae/Oviraptorosauria*	10.845	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	10.918	0.0003332	3000
Avialae/Oviraptorosauria	20.135	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	20.446	0.0003332	3000
Avialae/Deinonychosauria	23.099	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	18.695	0.0003332	3000
Avialae/Dromaeosauridae	16.692	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	14.024	0.0003332	3000
Avialae/Troodontidae*	6.14	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	5.539	0.0003332	3000

Avialae/Troodontidae	11.057	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	9.864	0.0003332	3000
Avialae/non-ave	31.533	0.0003332	3000	**Mesozoic Avialae/non-ave**	22.377	0.0003332	3000
forelimb	F value	P value	permutation	**forelimb**	F value	P value	permutation
Avialae/Tyrannosauroidea*	17.951	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	16.751	0.0003332	3000
Avialae/Tyrannosauroidea	19.541	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	17.71	0.0003332	3000
Avialae/Compsognathidae*	12.036	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	10.213	0.0003332	3000
Avialae/Compsognathidae	10.954	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	9.3629	0.0009997	3000
Avialae/Ornithomimosauria	28.226	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	26.132	0.0003332	3000
Avialae/Alvarezsauroidea	11.806	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	11.205	0.0003332	3000
Avialae/Therizinosauria	9.0851	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	8.1116	0.0003332	3000
Avialae/Oviraptorosauria*	15.284	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	11.192	0.0003332	3000
Avialae/Oviraptorosauria	13.928	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	10.147	0.0006664	3000
Avialae/Deinonychosauria	36.547	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	28.655	0.0003332	3000
Avialae/Dromaeosauridae	18.6	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	14.806	0.0003332	3000
Avialae/Troodontidae*	13.647	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	11.756	0.0006664	3000
Avialae/Troodontidae	20.854	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	17.023	0.0003332	3000
Avialae/non-ave	38.588	0.0003332	3000	**Mesozoic Avialae/non-ave**	25.419	0.0003332	3000
pl	F value	P value	permutation	**pl**	F value	P value	permutation
Avialae/Tyrannosauroidea*	19.383	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	17.822	0.0003332	3000
Avialae/Tyrannosauroidea	21.384	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	18.985	0.0003332	3000
Avialae/Compsognathidae*	10.416	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	9.3695	0.0006664	3000

Avialae/Compsognathidae	9.5466	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	8.6948	0.0003332	3000
Avialae/Ornithomimosauria	16.638	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	14.967	0.0003332	3000
Avialae/Alvarezsauroidea	10.503	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	11.338	0.0003332	3000
Avialae/Therizinosauria	16.292	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	18.231	0.0003332	3000
Avialae/Oviraptorosauria	9.4067	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	9.3032	0.0003332	3000
Avialae/Deinonychosauria	16.099	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	12.589	0.0003332	3000
Avialae/Dromaeosauridae	12.062	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	9.7379	0.0003332	3000
Avialae/Troodontidae*	6.0615	0.002999	3000	**Mesozoic Avialae/Troodontidae***	6.4236	0.004332	3000
Avialae/Troodontidae	8.0946	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	7.3455	0.001333	3000
Avialae/non-ave	20.227	0.0003332	3000	**Mesozoic Avialae/non-ave**	13.267	0.0003332	3000
hindlimb	F value	P value	permutation	**hindlimb**	F value	P value	permutation
Avialae/Tyrannosauroidea*	24.792	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	20.83	0.0003332	3000
Avialae/Tyrannosauroidea	26.154	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	21.278	0.0003332	3000
Avialae/Compsognathidae*	12.844	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	11.622	0.0003332	3000
Avialae/Compsognathidae	10.81	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	10.042	0.0003332	3000
Avialae/Ornithomimosauria	30.606	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	29.6	0.0003332	3000
Avialae/Alvarezsauroidea	17.13	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	14.283	0.0003332	3000
Avialae/Therizinosauria	22.191	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	22.631	0.0003332	3000
Avialae/Oviraptorosauria**	18.631	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria****	15.458	0.0003332	3000
Avialae/Oviraptorosauria*	17.08	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	14.693	0.0003332	3000
Avialae/Oviraptorosauria	16.265	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	14.149	0.0003332	3000

Avialae/Deinonychosauria	39.397	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	28.658	0.0003332	3000
Avialae/Dromaeosauridae	28.586	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	22.254	0.0003332	3000
Avialae/Troodontidae*	28.567	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	27.775	0.0003332	3000
Avialae/Troodontidae	24.358	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	20.485	0.0003332	3000
Avialae/non-ave	44.21	0.0003332	3000	**Mesozoic Avialae/non-ave**	27.491	0.0003332	3000
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body	F value	P value	permutation	**body**	F value	P value	permutation
Avialae/Tyrannosauroidea*	33.94	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	29.969	0.0003332	3000
Avialae/Tyrannosauroidea	35.114	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	30.049	0.0003332	3000
Avialae/Compsognathidae*	14.355	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	14.129	0.0003332	3000
Avialae/Compsognathidae	13.097	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	13.353	0.0003332	3000
Avialae/Ornithomimosauria	35.94	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	37.906	0.0003332	3000
Avialae/Alvarezsauroidea	17.78	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	27.424	0.0003332	3000
Avialae/Therizinosauria	19.709	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	23.031	0.0003332	3000
Avialae/Oviraptorosauria*	18.736	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	18.993	0.0003332	3000
Avialae/Oviraptorosauria	19.033	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	20.047	0.0003332	3000
Avialae/Deinonychosauria	37.355	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	27.216	0.0003332	3000
Avialae/Dromaeosauridae	27.545	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	22.645	0.0003332	3000
Avialae/Troodontidae*	17.538	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	16.747	0.0003332	3000
Avialae/Troodontidae	22.726	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	20.098	0.0003332	3000
Avialae/non-ave	39.436	0.0003332	3000	**Mesozoic Avialae/non-ave**	23.571	0.0003332	3000
skull	F value	P value	permutation	**skull**	F value	P value	permutation

Avialae/Tyrannosauroidea*	31.105	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	24.666	0.0003332	3000
Avialae/Tyrannosauroidea	29.959	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	23.009	0.0003332	3000
Avialae/Compsognathidae*	5.2224	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	5.963	0.0003332	3000
Avialae/Compsognathidae	8.565	0.001666	3000	**Mesozoic Avialae/Compsognathidae**	9.7338	0.0006664	3000
Avialae/Ornithomimosauria	14.646	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	18.987	0.0003332	3000
Avialae/Alvarezsauroidea	3.4335	0.02832	3000	**Mesozoic Avialae/Alvarezsauroidea**	3.4148	0.02399	3000
Avialae/Therizinosauria	24.13	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	36.739	0.0003332	3000
Avialae/Oviraptorosauria*	30.339	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	42.981	0.0003332	3000
Avialae/Oviraptorosauria	34.206	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	54.362	0.0003332	3000
Avialae/Deinonychosauria	24.102	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	11.498	0.0003332	3000
Avialae/Dromaeosauridae	21.587	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	15.898	0.0003332	3000
Avialae/Troodontidae*	16.446	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	14.261	0.0003332	3000
Avialae/Troodontidae	22.126	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	17.668	0.0003332	3000
Avialae/non-ave	14.806	0.0003332	3000	**Mesozoic Avialae/non-ave**	6.059	0.0009997	3000
axial	F value	P value	permutation	**axial**	F value	P value	permutation
Avialae/Tyrannosauroidea*	16.894	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	15.476	0.0003332	3000
Avialae/Tyrannosauroidea	16.939	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	14.618	0.0003332	3000
Avialae/Compsognathidae*	6.1587	0.0006664	3000	**Mesozoic Avialae/Compsognathidae***	7.6022	0.0003332	3000
Avialae/Compsognathidae	11.489	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	14.514	0.0003332	3000
Avialae/Ornithomimosauria	13.35	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	14.698	0.0003332	3000
Avialae/Alvarezsauroidea	10.604	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	13.389	0.0003332	3000

Avialae/Therizinosauria	7.8551	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	13.107	0.0003332	3000
Avialae/Oviraptorosauria*	4.4121	0.007997	3000	**Mesozoic Avialae/Oviraptorosauria***	3.171	0.03865	3000
Avialae/Oviraptorosauria	3.926	0.01566	3000	**Mesozoic Avialae/Oviraptorosauria**	2.9004	0.05098	3000
Avialae/Deinonychosauria	20.973	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	20.58	0.0003332	3000
Avialae/Dromaeosauridae	18.161	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	21.174	0.0003332	3000
Avialae/Troodontidae*	4.7216	0.006998	3000	**Mesozoic Avialae/Troodontidae***	6.3675	0.001999	3000
Avialae/Troodontidae	9.5984	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	10.534	0.0003332	3000
Avialae/non-ave	21.505	0.0003332	3000	**Mesozoic Avialae/non-ave**	13.485	0.0003332	3000
ps	F value	P value	permutation	**ps**	F value	P value	permutation
Avialae/Tyrannosauroidea*	19.312	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	20.012	0.0003332	3000
Avialae/Tyrannosauroidea	21.857	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	22.357	0.0003332	3000
Avialae/Compsognathidae	12.621	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	13.743	0.0003332	3000
Avialae/Ornithomimosauria	12.972	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	13.358	0.0003332	3000
Avialae/Alvarezsauroidea	12.972	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	13.358	0.0003332	3000
Avialae/Therizinosauria	14.722	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	15.96	0.0003332	3000
Avialae/Oviraptorosauria*	15.987	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	16.284	0.0003332	3000
Avialae/Oviraptorosauria	17.409	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	18.52	0.0003332	3000
Avialae/Deinonychosauria	20.651	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	17.751	0.0003332	3000
Avialae/Dromaeosauridae	15.123	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	13.313	0.0003332	3000
Avialae/Troodontidae*	4.3851	0.001333	3000	**Mesozoic Avialae/Troodontidae***	4.0841	0.002999	3000
Avialae/Troodontidae	8.2125	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	7.836	0.0006664	3000

Avialae/non-ave	34.291	0.0003332	3000	**Mesozoic Avialae/non-ave**	25.239	0.0003332	3000
forelimb	F value	P value	permutation	**forelimb**	F value	P value	permutation
Avialae/Tyrannosauroidea*	17.921	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	16.828	0.0003332	3000
Avialae/Tyrannosauroidea	19.87	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	17.981	0.0003332	3000
Avialae/Compsognathidae*	12.749	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	10.659	0.0003332	3000
Avialae/Compsognathidae	11.673	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	9.8399	0.0003332	3000
Avialae/Ornithomimosauria	29.247	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	28.2	0.0003332	3000
Avialae/Alvarezsauroidea	10.329	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	10.353	0.0003332	3000
Avialae/Therizinosauria	9.445	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	8.0835	0.001666	3000
Avialae/Oviraptorosauria*	13.669	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	10.361	0.001333	3000
Avialae/Oviraptorosauria	12.873	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	9.7139	0.0006664	3000
Avialae/Deinonychosauria	34.194	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	26.898	0.0003332	3000
Avialae/Dromaeosauridae	17.683	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	14.006	0.0006664	3000
Avialae/Troodontidae*	10.596	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	8.8603	0.002666	3000
Avialae/Troodontidae	17.723	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	14.435	0.0006664	3000
Avialae/non-ave	45.522	0.0003332	3000	**Mesozoic Avialae/non-ave**	28.704	0.0003332	3000
pl	F value	P value	permutation	**pl**	F value	P value	permutation
Avialae/Tyrannosauroidea*	28.133	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	24.246	0.0003332	3000
Avialae/Tyrannosauroidea	30.13	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	25.39	0.0003332	3000
Avialae/Compsognathidae*	6.8035	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	11.379	0.0003332	3000
Avialae/Compsognathidae	12.017	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	10.552	0.0006664	3000

Avialae/Ornithomimosauria	23.716	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	20.707	0.0003332	3000
Avialae/Alvarezsauroidea	9.3777	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	9.9275	0.0003332	3000
Avialae/Therizinosauria	20.1	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	21.677	0.0003332	3000
Avialae/Oviraptorosauria	11.819	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	11.327	0.0003332	3000
Avialae/Deinonychosauria	22.282	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	16.339	0.0003332	3000
Avialae/Dromaeosauridae	15.994	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	12.062	0.0003332	3000
Avialae/Troodontidae*	7.9648	0.0009997	3000	**Mesozoic Avialae/Troodontidae***	7.9463	0.0009997	3000
Avialae/Troodontidae	11.219	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	9.7545	0.0003332	3000
Avialae/non-ave	24.153	0.0003332	3000	**Mesozoic Avialae/non-ave**	13.994	0.0003332	3000
hindlimb	F value	P value	permutation	**hindlimb**	F value	P value	permutation
Avialae/Tyrannosauroidea*	31.805	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea***	26.584	0.0003332	3000
Avialae/Tyrannosauroidea	32.338	0.0003332	3000	**Mesozoic Avialae/Tyrannosauroidea**	25.519	0.0003332	3000
Avialae/Compsognathidae*	6.5365	0.0003332	3000	**Mesozoic Avialae/Compsognathidae***	11.988	0.0006664	3000
Avialae/Compsognathidae	11.715	0.0003332	3000	**Mesozoic Avialae/Compsognathidae**	10.732	0.0003332	3000
Avialae/Ornithomimosauria	37.115	0.0003332	3000	**Mesozoic Avialae/Ornithomimosauria**	35.829	0.0003332	3000
Avialae/Alvarezsauroidea	21.536	0.0003332	3000	**Mesozoic Avialae/Alvarezsauroidea**	17.63	0.0003332	3000
Avialae/Therizinosauria	23.212	0.0003332	3000	**Mesozoic Avialae/Therizinosauria**	23.614	0.0003332	3000
Avialae/Oviraptorosauria**	20.852	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria****	17.061	0.0003332	3000
Avialae/Oviraptorosauria*	19.222	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria***	16.218	0.0003332	3000
Avialae/Oviraptorosauria	18.399	0.0003332	3000	**Mesozoic Avialae/Oviraptorosauria**	15.677	0.0003332	3000
Avialae/Deinonychosauria	49.644	0.0003332	3000	**Mesozoic Avialae/Deinonychosauria**	35.818	0.0003332	3000

Avialae/Dromaeosauridae	34.318	0.0003332	3000	**Mesozoic Avialae/Dromaeosauridae**	26.127	0.0003332	3000
Avialae/Troodontidae*	29.095	0.0003332	3000	**Mesozoic Avialae/Troodontidae***	28.174	0.0003332	3000
Avialae/Troodontidae	27.416	0.0003332	3000	**Mesozoic Avialae/Troodontidae**	23.442	0.0003332	3000
Avialae/non-ave	57.245	0.0003332	3000	**Mesozoic Avialae/non-ave**	33.525	0.0003332	3000

Table S4. Results of comparative distance test.

We performed three analyses for each distance matrix (MORD and GED). For test 1 and test 2, the distances of other coelurosarian groups and their closely-related clades are from following pairs: Compsognathidae and Ornithomimosauria, Therizinosauria and Alvarezsauroidea, Troodontidae and Dromaeosauridae, Ornithomimosauria and Alvarezsauroidea, Compsognathidae and Tyrannosauroidea, Ornithomimosauria and Tyrannosauroidea, Oviraptorosauria and Therizinosauria, Oviraptorosauria and Dromaeosauridae, Oviraptorosauria and Troodontidae, Therizinosauria and Dromaeosauridae, Therizinosauria and Troodontidae, Alvarezsauroidea and Dromaeosauridae, and Alvarezsauroidea and Troodontidae. For test 3, the distances of other coelurosarian groups and their closely-related clades are from following pairs: Compsognathidae and Ornithomimosauria, Therizinosauria and Alvarezsauroidea, Ornithomimosauria and Alvarezsauroidea, Compsognathidae and Tyrannosauroidea, Ornithomimosauria and Tyrannosauroidea, Oviraptorosauria and Therizinosauria, Oviraptorosauria and Deinonychosauria, Alvarezsauroidea and Deinonychosauria, and Therizinosauria and Deinonychosauria.

MORD								
Anatomical regions		whole skeleton	skull	axial	ps	forelimb	pl	hindlimb
1. distance between Mesozoic birds and Dromaeosauridae VS distance of other coelurosarian groups and their closely-related clades	P-value	7.58E-08	2.20E-16	2.20E-16	3.41E-06	2.20E-16	0.000421	2.20E-16
	median distance of Mesozoic birds and Dromaeosauridae	5.61E-01	4.96E-01	6.47E-01	5.87E-01	5.68E-01	0.523599	5.38E-01
	median distance of other coelurosarian groups and their closely-related clades	0.526244	0.5932	0.538664	0.523599	0.426425	0.490883	0.459537
2. **distance between Mesozoic birds and Troodontidae VS distance of other coelurosarian groups and their closely-related**	P-value	1.52E-05	2.20E-16	0.000106	0.000103	4.78E-16	0.03344	2.20E-16
	median distance of Mesozoic birds and Troodontidae	5.55E-01	4.80E-01	0.583821	0.578163	5.73E-01	0.523599	6.08E-01
	median distance of other coelurosarian groups and their	0.526244	0.5932	0.538664	0.523599	0.426425	0.490883	0.459537

clades	closely-related clades							
3. **distance between Mesozoic birds and Deinonychosauria VS distance of other coelurosarian groups and their closely-related clades**	P-value	0.0008167	2.20E-16	2.20E-16	0.001811	2.20E-16	0.03545	2.20E-16
	median distance of Mesozoic birds and Deinonychosauria	0.559533	0.490883	0.630771	0.582941	0.568563	0.523599	0.563943
	median distance of other coelurosarian groups and their closely-related clades	0.541939	0.639694	0.55298	0.549467	0.454817	0.503213	0.467872
GED								
Anatomical regions		whole skeleton	skull	axial	ps	forelimb	pl	hindlimb
1. **distance between Mesozoic birds and Dromaeosauridae VS distance of other coelurosarian groups and their closely-related clades**	P-value	4.75E-11	2.20E-16	2.20E-16	1.03E-10	2.20E-16	0.000261	2.20E-16
	median distance of Mesozoic birds and Dromaeosauridae	1.05E+01	5.62E+00	4.50E+00	3.55E+00	4.54E+00	4.081147	4.56E+00
	median distance of other coelurosarian groups and their closely-related clades	9.571805	8.053235	3.559026	2.909055	3.210207	3.503564	3.788597
2. **distance between Mesozoic birds and Troodontidae VS distance of other coelurosarian groups and their closely-related clades**	P-value	5.39E-07	2.20E-16	0.002078	2.11E-06	2.80E-14	0.106	2.20E-16
	median distance of Mesozoic birds and Troodontidae	1.04E+01	5.49E+00	3.920702	3.362609	4.47E+00	3.714821	4.98E+00
	median distance of other coelurosarian groups and their closely-related clades	9.571805	8.053235	3.559026	2.909055	3.210207	3.503564	3.788597

3. distance between Mesozoic birds and Deinonychosauria VS distance of other coelurosarian groups and their closely-related clades	P-value	1.41E-05	2.20E-16	1.37E-14	3.92E-07	2.20E-16	0.05194	2.20E-16
	median distance of Mesozoic birds and Deinonychosauria	10.4686	5.556627	4.388069	3.464947	4.505736	3.86538	4.784599
	median distance of other coelurosarian groups and their closely-related clades	10.05837	8.848325	3.66572	3.081926	3.496565	3.618484	3.970441

Table S5. Details of supertree construction.

Source tree	**Molecular time tree in ref.(4) (Bayesian tree)**	**Molecular time tree 1 in ref.(3) (Bayesian tree)**	**Molecular time tree 2 in ref.(3) (Bayesian tree)**
Supertree number	Assemble to supertree 1	Assemble to supertree 2	Assemble to supertree 3
Source tree topology	**strict consensus tree in ref.(13)**		
Supertree number	all supertrees		
Dating	ILN√		
	WN		
Node constraints	NO		
Tip calibrations	uniform distributions		
Polytomy resolved	Yes		
Sampled fossil taxa	*Archaeopteryx, Jeholornis, Sapeornis, Confuciusornis_sanctus, Changchengornis, Eoconfuciusornis, Jinzhouornis, Confuciusornis_dui, Boluochia, Concornis, Eoalulavis, Cathayornis, Eocathayornis, Eoenantiornis, Gobipteryx, Longipteryx, Longirostravis, Neuquenornis, Pengornis, Eopengornis, Protopteryx, Rapaxavis, Shanweiniao, Vescornis, Vorona, Schizooura, Jianchangornis, Archaeorhynchus, Songlingornis, Apsaravis, Yanornis, Patagopteryx, Yixianornis, Gansus, Ichthyornis, Hesperornis, Parahesperornis, Enaliornis, Baptornis_advenus, Baptornis_varneri, Fortunguavis, Qiliania, Shenqiornis, Sulcavis, Bohaiornis, Parabohaiornis, Longusunguis, Zhouornis, Piscivoravis, Hongshanornis, Longicrusavis, Archaeornithura, Parahongshanornis, Tianyuornis*		
Source tree topology	**Adams consensus tree in ref.(28)**		
Supertree number	all supertrees		
Dating	ILN√		
	WN		
Node constraints	Aves(tree root)=fixed(168.4654) [refer to the dating results of strict consensus tree in ref.(13)]		

Tip calibrations	Tips that are identical with those in the tree of ref.(13) were given fixed ages as the posterior medians estimated already.		
Polytomy resolved	Yes		
Sampled fossil taxa	Enantiornithine_A, Enantiornithine_B, Hesperornithiform_A, Hesperornithiform_B, Ornithurine_D, *Cimolopteryx_rara, Cimolopteryx_minima, Cimolopteryx_maxima, Cimolopteryx_petra, Ceramornis_major, Iaceornis_marshii,* Ornithurine_A, Ornithurine_B, Ornithurine_C, Ornithurine_E, Ornithurine_F, *Palintropus_retusus*		
Source tree topology	**Parsimony tree in ref.(29)**		
Supertree number	supertree1	supertree2	supertree3
Dating	ILN√	ILN	ILN
	WN	WN√	WN√
Node constraints	node_*Ichthyornis*+Neornithes(tree root)=fixed(103.22483) [refer to the dating results of strict consensus tree in ref.(13)] crown_Anseriformes=fixed(69.3289) [refer to the node age of crown Anseriformes of the molecular time tree in ref.(4)] crown_Galliformes=fixed(57.729207) [refer to the node age of crown Galliformes of the molecular time tree in ref.(4)] Galloanserae=fixed(72.500508) [refer to the node age of Galloanserae of the molecular time tree in ref.(4)] Neornithes=fixed(92.515403) [refer to the root age of the molecular time tree in ref.(4)]	node_*Ichthyornis*+Neornithes(tree root)=fixed(103.22483) [refer to the dating results of strict consensus tree in ref.(13)] Neornithes=fixed(72.907242) [refer to the root age of the molecular time tree1 in ref.(3)]	node_*Ichthyornis*+Neornithes(tree root)=fixed(103.22483) [refer to the dating results of strict consensus tree in ref.(13)] crown_Anseriformes=fixed(69.683984) [refer to the node age of crown Anseriformes of the molecular time tree2 in ref.(3)] crown_Galliformes=fixed(60.934716) [refer to the node age of crown Anseriformes of the molecular time tree2 in ref.(3)] Galloanserae=fixed(72.34804) [refer to the node age of crown Anseriformes of the molecular time tree2 in ref.(3)] Neornithes=fixed(78.353363) [refer to the root age of the molecular time tree2 in ref.(3)]
Tip calibrations	Tips that are identical with those in the tree of ref.(13) were given fixed ages as the posterior medians estimated already.		
Polytomy resolved	No		

Sampled fossil taxa	*Gallinuloides_wyomingensis, Pelagornis_chilensis, Protodontopteryx_ruthae, Conflicto_antarcticus, Anatalavis_oxfordi, Wilaru_tedfordi, Presbyornis_pervetus, Asteriornis_maastrichtensis, Lithornis_promiscuus, Lithornis_plebius, Paracathartes_howardae, Vegavis*		
Source tree topology	**Bayesian maximum clade credibility consensus tree in ref.(31)**		
Supertree number	supertree1	supertree2	supertree3
Dating	ILN√	ILN√	ILN
	WN	WN	WN√
Node constraints	Neornithes=fixed(92.515403) [refer to the root age of the molecular time tree in ref.(4)] Galloanserae=fixed(72.500508) [refer to the node age of crown Anseriformes of the molecular time tree in ref.(4)] Anseriformes=fixed(72.0561757) [refer to the dating results of parsimony tree in ref.(29)] Galliformes=fixed(64.122604)[refer to the dating results of parsimony tree in ref.(29)]	Neornithes=fixed(72.907242) [refer to the root age of the molecular time tree1 in ref.(3)] Galloanserae=fixed(70.162369) [refer to the dating results of parsimony tree in ref.(29)] Anseriformes=fixed(68.834254) [refer to the dating results of parsimony tree in ref.(29)] Galliformes=fixed(57.942579) [refer to the dating results of parsimony tree in ref.(29)]	Neornithes=fixed(78.353363) [refer to the node age of Neornithes of the molecular time tree2 in ref.(3)] Galloanserae=fixed(72.34804)[refer to the node age of crown Anseriformes of the molecular time tree2 in ref.(3)] Anseriformes=fixed(71.6635676) [refer to the dating results of parsimony tree in ref.(29)] Galliformes=fixed(66.224786) [refer to the dating results of parsimony tree in ref.(29)]
Tip calibrations	Tips that are identical with those in the tree of ref.(29) were given fixed ages as the posterior medians estimated already.	Tips that are identical with those in the tree of ref.(29) were given fixed ages as the posterior medians estimated already.	Tips that are identical with those in the tree of ref.(29) were given fixed ages as the posterior medians estimated already.
Polytomy resolved	No		
Sampled fossil taxa	*Gastornis_giganteus, Gastornis_parisensis, Barawertornis_tedfordi, Ilbandornis_lawsoni, Ilbandornis_woodburnei, Dromornis_murrayi, Dromornis_planei, Dromornis_stirtoni, Genyornis_newtoni, Megavitiornis_altirostris, Sylviornis_neocaledoniae*		
Source tree topology	**Phylogeny in ref.(30)**		
Supertree number	supertree1	supertree2	supertree3
Dating	ILN√	ILN√	ILN√
	WN	WN	WN

Node constraints	Passeriformes=fixed(52.995798) [refer to the node age of Passeriformes of the molecular time tree in ref.(4)] Passeriformes+Psittaciformes=fixed(60.592768)[refer to the node age of Passeriformes+Psittaciformes of the molecular time tree in ref.(4)] Coraciiformes=fixed(53.957599) [refer to the node age of Coraciiformes of the molecular time tree in ref.(4)] Neoaves=fixed(68.878066) [refer to the node age of Neoaves of the molecular time tree in ref.(4)]	Passeriformes=fixed(49.702671) [refer to the node age of Passeriformes of the molecular time tree1 in ref.(3)] Passeriformes+Psittaciformes=fixed(56.259866) [refer to the node age of Passeriformes+Psittaciformes of the molecular time tree1 in ref.(3)] Coraciiformes=fixed(53.206289) [refer to the node age of Coraciiformes of the molecular time tree1 in ref.(3)] Neoaves=fixed(67.419484)[refer to the node age of Neoaves of the molecular time tree1 in ref.(3)]	Passeriformes=fixed(51.874177)[refer to the node age of Passeriformes of the molecular time tree2 in ref.(3)] Passeriformes+Psittaciformes=fixed(58.27697)[refer to the node age of Passeriformes+Psittaciformes of the molecular time tree2 in ref.(3)] Coraciiformes=fixed(53.331673)[refer to the node age of Coraciiformes of the molecular time tree2 in ref.(3)] Neoaves=fixed(69.469754)[refer to the node age of Neoaves of the molecular time tree2 in ref.(3)]
Tip calibrations	uniform distributions		
Polytomy resolved	Yes		
Sampled fossil taxa	*Plesiocathartes_kelleri, Eocoracias_brachyptera, Palaeotodus_itardiensis, Pseudasturides_macrocephalus, Cyrilavis_colburnorum, Zygodactylus_luberonensis, Eozygodactylus_americanus*		

Table S6. Estimates of relative extinction rates.

Supertree 1 relative extinction rate										
prior	default		net diversification rate exp(10)		net diversification rate exp(50)		relative extinction rate beta(1,4)		relative extinction rate beta(2,2)	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[9.5499E-6, 0.8331]	0.3208	[3.8169E-6, 0.7698]	0.2508	[1.9364E-4, 0.9197]	0.5224	[3.559E-7, 0.4588]	0.1204	[0.0429, 0.8135]	0.416
2	[6.5391E-8, 0.7824]	0.2687	[1.4403E-6, 0.7418]	0.228	[3.0649E-5, 0.8577]	0.3766	[2.5511E-6, 0.4363]	0.1138	[0.0401, 0.7776]	0.3798
3	[4.194E-6, 0.7744]	0.2504	[4.7357E-7, 0.7286]	0.2117	[1.1738E-5, 0.862]	0.3735	[4.9159E-7, 0.4238]	0.1056	[0.0272, 0.7658]	0.369
4	[6.0658E-3, 0.8525]	0.4089	[3.4918E-3, 0.8197]	0.3593	[0.0849, 0.9603]	0.5767	[7.5104E-4, 0.5326]	0.1884	[0.0723, 0.8256]	0.4449
5	[3.0989E-6, 0.6956]	0.1814	[1.9532E-7, 0.6756]	0.1663	[4.0427E-6, 0.7747]	0.2421	[5.4219E-8, 0.3918]	0.0895	[0.0139, 0.7093]	0.3128
6	[1.2588E-6, 0.6702]	0.1499	[3.7489E-7, 0.6322]	0.1312	[1.5472E-5, 0.7798]	0.2315	[3.1067E-7, 0.3686]	0.076	[7.8014E-3, 0.6953]	0.2953
7	[2.1712E-6, 0.6549]	0.1269	[6.3114E-6, 0.6252]	0.1122	[2.3942E-6, 0.7576]	0.1945	[7.3453E-7, 0.3531]	0.0646	[7.6263E-3, 0.696]	0.2845
8	[0.015, 0.8599]	0.4197	[0.0127, 0.8491]	0.3977	[0.0555, 0.9103]	0.4892	[3.5832E-3, 0.5455]	0.2023	[0.0849, 0.8324]	0.4547
9	[0.0761, 0.8934]	0.4861	[0.0635, 0.8819]	0.463	[0.1022, 0.9211]	0.5371	[0.0182, 0.5986]	0.2626	[0.1441, 0.8586]	0.4955
10	[0.1694, 0.9707]	0.6145	[0.132, 0.9618]	0.586	[0.2099, 0.9748]	0.6754	[0.0269, 0.672]	0.3169	[0.1945, 0.9096]	0.5813
11	[0.9091, 0.9995]	0.9769	[0.9216, 0.9992]	0.9771	[0.5077, 0.9993]	0.9687	[0.8849, 0.9875]	0.9464	[0.8751, 0.998]	0.9622
12	[5.192E-6, 0.9263]	0.3825	[1.9612E-5, 0.9116]	0.3694	[1.089E-5, 0.9503]	0.4273	[4.0883E-7, 0.505]	0.1353	[0.0516, 0.9071]	0.4648
13	[0.0666, 0.9711]	0.5507	[0.0521, 0.951]	0.5095	[0.1146, 0.9963]	0.6769	[1.6808E-3, 0.6001]	0.2239	[0.1348, 0.9308]	0.5387
14	[3.7679E-6, 0.8096]	0.2769	[1.3365E-6, 0.7943]	0.2637	[3.2302E-7, 0.8504]	0.3145	[2.9908E-7, 0.4614]	0.1209	[0.0384, 0.7994]	0.3931
15	[0.0335, 0.9778]	0.6575	[2.2052E-4, 0.9297]	0.5577	[0.2529, 0.9984]	0.8667	[3.9742E-6, 0.6016]	0.162	[0.1365, 0.9477]	0.6058
16	[3.3685E-3, 0.824]	0.346	[1.6459E-3, 0.8033]	0.3204	[0.0157, 0.8866]	0.4297	[1.1731E-3, 0.5053]	0.1716	[0.0544, 0.7987]	0.4163

17	[0.038, 0.888]	0.4527	[0.0282, 0.8613]	0.4055	[0.107, 0.9663]	0.5935	[0.0135, 0.596]	0.2428	[0.1073, 0.8513]	0.4765
18	[0.1506, 0.9822]	0.6538	[0.1438, 0.9807]	0.6358	[0.2333, 0.9954]	0.7463	[0.0212, 0.7203]	0.3481	[0.2096, 0.9361]	0.6135
19	[0.3382, 0.9973]	0.8123	[0.2649, 0.9974]	0.7821	[0.5756, 0.9988]	0.8856	[5.2684E-3, 0.8]	0.4478	[0.3367, 0.9716]	0.7369
20	[0.8654, 0.9981]	0.9541	[0.8604, 0.9966]	0.9529	[0.8857, 0.9984]	0.9639	[0.7706, 0.9709]	0.8906	[0.8159, 0.9907]	0.9274
21	[0.0843, 0.9657]	0.6415	[4.1781E-3, 0.8855]	0.5308	[0.4876, 0.9971]	0.8371	[3.5686E-4, 0.6831]	0.2792	[0.1881, 0.9243]	0.612
22	[0.0222, 0.8984]	0.4857	[8.4339E-3, 0.8754]	0.4375	[0.11, 0.9796]	0.6421	[3.0903E-3, 0.5925]	0.2207	[0.0978, 0.8695]	0.5035
23	[0.053, 0.9102]	0.4983	[0.048, 0.898]	0.4843	[0.1001, 0.9568]	0.5687	[0.0147, 0.6408]	0.2865	[0.1082, 0.8524]	0.492
24	[7.1788E-4, 0.8908]	0.4691	[1.0386E-3, 0.8854]	0.4598	[2.1287E-3, 0.9058]	0.5257	[1.1603E-4, 0.6001]	0.1939	[0.0755, 0.8703]	0.4899
25	[0.0491, 0.9507]	0.6116	[0.0609, 0.9652]	0.6078	[0.0902, 0.9829]	0.6563	[7.7158E-4, 0.6758]	0.2712	[0.1331, 0.9119]	0.5661
26	[3.3745E-6, 0.8956]	0.479	[6.7166E-6, 0.8874]	0.4493	[5.5781E-7, 0.9109]	0.5563	[2.5022E-6, 0.5919]	0.1288	[0.0606, 0.8778]	0.5131
27	[1.0805E-4, 0.7272]	0.1928	[1.0787E-4, 0.7071]	0.1779	[1.7325E-4, 0.7751]	0.2647	[1.5747E-4, 0.4733]	0.0962	[0.0117, 0.7096]	0.3383
28	[1.2643E-4, 0.6672]	0.1128	[1.6592E-4, 0.6495]	0.1051	[1.4421E-4, 0.7013]	0.1313	[7.893E-5, 0.3772]	0.0506	[4.7028E-3, 0.6947]	0.3047
29	[3.489E-7, 0.3626]	0.016	[3.6175E-7, 0.3621]	0.0158	[2.5137E-7, 0.39]	0.0164	[1.4093E-7, 0.2001]	0.0101	[3.8476E-5, 0.5651]	0.1925
30	[7.6421E-8, 0.3766]	0.0128	[1.683E-8, 0.3678]	0.0124	[5.0992E-7, 0.3824]	0.0135	[5.9469E-7, 0.1739]	7.59E-03	[2.8228E-5, 0.5991]	0.1955
31	[4.6196E-5, 0.4811]	0.0372	[2.5501E-5, 0.4785]	0.0394	[1.8611E-5, 0.4811]	0.0378	[1.3604E-5, 0.2254]	0.0223	[1.0386E-3, 0.6289]	0.2128
32	[8.3111E-5, 0.6229]	0.1182	[3.8533E-4, 0.6277]	0.1211	[2.8431E-4, 0.6217]	0.1203	[1.0371E-4, 0.327]	0.0609	[5.8787E-3, 0.6794]	0.2731
33	[8.0025E-5, 0.5793]	0.0905	[7.822E-5, 0.5875]	0.0913	[1.2147E-4, 0.5839]	0.092	[1.2501E-4, 0.2998]	0.0467	[8.4198E-4, 0.6637]	0.2561
prior	**relative extinction rate beta(4,1)**		**relative fossilization rate beta(1,4)**		**relative fossilization rate beta(2,2)**		**relative fossilization rate beta(4,1)**			
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median		

1	[0.3825, 0.9878]	0.7503	[7.6243E-5, 0.8695]	0.419	[7.9112E-6, 0.7814]	0.2657	[1.586E-6, 0.6172]	0.1302	
2	[0.3385, 0.98]	0.7055	[8.9555E-7, 0.8264]	0.3459	[4.192E-7, 0.7438]	0.2402	[1.1728E-7, 0.6176]	0.1329	
3	[0.3271, 0.9787]	0.6973	[1.5851E-6, 0.8254]	0.3375	[1.2298E-6, 0.721]	0.2087	[1.5096E-6, 0.5459]	0.1016	
4	[0.3707, 0.9831]	0.7252	[0.0899, 0.9273]	0.528	[9.7445E-3, 0.8076]	0.3459	[3.7096E-4, 0.6164]	0.1573	
5	[0.263, 0.963]	0.6346	[1.257E-5, 0.747]	0.2469	[1.6231E-6, 0.6281]	0.1448	[2.3257E-6, 0.4262]	0.0645	
6	[0.2512, 0.9574]	0.6236	[1.1554E-7, 0.738]	0.2257	[2.5147E-7, 0.522]	0.093	[1.4881E-7, 0.2503]	0.0313	
7	[0.247, 0.9525]	0.6213	[5.5081E-7, 0.7285]	0.193	[6.2045E-8, 0.4322]	0.0596	[5.2791E-7, 0.1566]	0.0178	
8	[0.369, 0.9795]	0.7226	[0.1144, 0.9261]	0.5351	[0.0191, 0.8036]	0.3358	[5.5501E-4, 0.5759]	0.1471	
9	[0.3965, 0.9799]	0.7326	[0.2676, 0.9625]	0.6404	[0.1357, 0.9119]	0.5304	[0.0342, 0.839]	0.4073	
10	[0.4874, 0.9962]	0.7957	[0.3849, 0.9915]	0.7514	[0.2319, 0.9734]	0.6494	[0.0985, 0.9262]	0.5187	
11	[0.707, 0.9992]	0.9549	[0.9092, 0.9997]	0.9787	[0.9238, 0.9989]	0.9753	[0.9112, 0.9991]	0.9697	
12	[0.4267, 0.9982]	0.8164	[2.6531E-4, 0.9448]	0.4881	[7.8221E-6, 0.8384]	0.3094	[1.5026E-6, 0.7009]	0.178	
13	[0.4905, 0.9968]	0.8372	[0.1667, 0.9922]	0.6759	[0.0425, 0.9102]	0.4753	[8.0331E-4, 0.7753]	0.2795	
14	[0.3579, 0.9889]	0.7287	[3.2338E-5, 0.8513]	0.3629	[1.44E-5, 0.7632]	0.2493	[1.0467E-5, 0.6545]	0.1517	
15	[0.5837, 0.9982]	0.8764	[0.1507, 0.9971]	0.7672	[1.2109E-6, 0.8719]	0.329	[9.3648E-8, 0.534]	0.0682	
16	[0.3614, 0.9846]	0.7245	[0.0466, 0.8937]	0.4615	[5.1309E-3, 0.7827]	0.3111	[2.9258E-4, 0.663]	0.1848	
17	[0.3962, 0.9829]	0.745	[0.1741, 0.9585]	0.5951	[0.0542, 0.8609]	0.4293	[7.6715E-3, 0.7446]	0.2767	
18	[0.5019, 0.9991]	0.8167	[0.3305, 0.9963]	0.7619	[0.1605, 0.9661]	0.6157	[0.044, 0.8929]	0.4592	
19	[0.6264, 0.9966]	0.8714	[0.4846, 0.9976]	0.849	[0.0635, 0.9773]	0.6487	[3.1809E-5, 0.6813]	0.0804	
20	[0.8675, 0.9985]	0.9553	[0.8779, 0.9984]	0.96	[0.8508, 0.9983]	0.9489	[0.8501, 0.9988]	0.9464	

21	[0.4923, 0.9847]	0.7822	[0.1955, 0.9866]	0.6873	[1.0507E-3, 0.8452]	0.3684	[1.1096E-5, 0.4118]	0.047	
22	[0.4088, 0.9834]	0.7563	[0.1124, 0.9438]	0.5739	[7.963E-3, 0.811]	0.3255	[2.885E-4, 0.5329]	0.1038	
23	[0.3539, 0.9755]	0.7061	[0.1799, 0.946]	0.5975	[0.0814, 0.8985]	0.4861	[6.9677E-3, 0.7911]	0.3103	
24	[0.3866, 0.9861]	0.7436	[0.0502, 0.9349]	0.5456	[8.302E-4, 0.8373]	0.3023	[2.137E-5, 0.5182]	0.0604	
25	[0.4453, 0.9876]	0.7815	[0.1576, 0.9767]	0.6674	[9.5565E-3, 0.8858]	0.442	[2.0719E-4, 0.5732]	0.0692	
26	[0.4171, 0.9895]	0.7577	[5.4506E-5, 0.9115]	0.5489	[4.2572E-7, 0.7062]	0.059	[1.4723E-8, 0.0655]	5.56E-03	
27	[0.2439, 0.9063]	0.5886	[2.0841E-3, 0.7718]	0.2813	[1.9791E-4, 0.3784]	0.0407	[3.2134E-5, 0.0431]	6.36E-03	
28	[0.2507, 0.9247]	0.6034	[6.6103E-4, 0.7018]	0.182	[7.6941E-5, 0.2016]	0.0163	[3.3442E-5, 0.0176]	2.63E-03	
29	[0.1586, 0.8348]	0.4913	[3.1148E-7, 0.4007]	0.0272	[1.0075E-7, 0.0244]	1.24E-03	[1.9483E-7, 1.8234E-3]	1.93E-04	
30	[0.1951, 0.9297]	0.5699	[1.7617E-7, 0.4146]	0.022	[4.3107E-8, 0.0156]	7.60E-04	[3.4096E-7, 1.1634E-3]	1.03E-04	
31	[0.2209, 0.9529]	0.605	[5.6338E-5, 0.537]	0.0665	[1.7049E-5, 0.0919]	5.79E-03	[1.4276E-5, 9.2494E-3]	9.45E-04	
32	[0.2582, 0.9623]	0.6332	[9.0046E-4, 0.6739]	0.1725	[4.159E-4, 0.5385]	0.0738	[5.0675E-5, 0.4518]	0.0296	
33	[0.2423, 0.9548]	0.6263	[3.9465E-4, 0.6274]	0.1319	[9.0163E-5, 0.4818]	0.0442	[3.8449E-5, 0.4066]	0.0172	

Supertree 2 relative extinction rate

prior	**default**		**net diversification rate exp(10)**		**net diversification rate exp(50)**		**relative extinction rate beta(1,4)**		**relative extinction rate beta(2,2)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[1.0015E-6, 0.8335]	0.3243	[1.5013E-5, 0.7735]	0.2486	[7.6036E-5, 0.9186]	0.5209	[1.0155E-6, 0.4588]	0.1209	[0.0398, 0.8103]	0.418

2	[4.5527E-7, 0.7819]	0.2665	[7.2084E-7, 0.7374]	0.2263	[2.2689E-6, 0.8597]	0.3778	[4.1868E-7, 0.4367]	0.1142	[0.0362, 0.7762]	0.3782
3	[1.5112E-7, 0.771]	0.2475	[1.2315E-6, 0.7334]	0.2125	[9.9799E-6, 0.8633]	0.376	[3.8343E-7, 0.4274]	0.1083	[0.0313, 0.7692]	0.3693
4	[8.4869E-3, 0.8501]	0.406	[3.7948E-3, 0.8227]	0.3581	[0.0887, 0.9636]	0.5711	[1.2157E-3, 0.5349]	0.1884	[0.072, 0.8245]	0.4459
5	[8.7492E-8, 0.6954]	0.1807	[1.3581E-7, 0.6685]	0.164	[1.7566E-5, 0.7691]	0.2407	[1.0095E-6, 0.3898]	0.0901	[0.0173, 0.7149]	0.3123
6	[1.936E-6, 0.6712]	0.1485	[4.5416E-6, 0.6441]	0.1335	[2.9002E-6, 0.785]	0.2331	[1.179E-6, 0.3741]	0.0771	[9.872E-3, 0.6968]	0.2943
7	[1.4731E-6, 0.6582]	0.1263	[9.4013E-7, 0.6274]	0.1124	[1.1795E-6, 0.7612]	0.1937	[4.5153E-7, 0.3534]	0.0644	[8.0721E-3, 0.6919]	0.2843
8	[0.0198, 0.8622]	0.4171	[0.0168, 0.8576]	0.4005	[0.0495, 0.9045]	0.4848	[3.9685E-3, 0.5464]	0.2023	[0.0883, 0.8353]	0.4532
9	[0.0729, 0.8966]	0.489	[0.0638, 0.8871]	0.464	[0.1029, 0.9203]	0.5296	[0.0153, 0.593]	0.2614	[0.1396, 0.8571]	0.4964
10	[0.1527, 0.9657]	0.613	[0.1357, 0.9623]	0.5781	[0.2188, 0.9825]	0.6764	[0.025, 0.6719]	0.3181	[0.1965, 0.9123]	0.5856
11	[0.9148, 0.9994]	0.9772	[0.9243, 0.9995]	0.9775	[0.6897, 0.9994]	0.9738	[0.8839, 0.9876]	0.9463	[0.8889, 0.9973]	0.9625
12	[3.1851E-6, 0.9253]	0.3858	[4.6099E-6, 0.9126]	0.3682	[2.7587E-5, 0.9514]	0.427	[1.7927E-6, 0.5097]	0.1346	[0.051, 0.8982]	0.4646
13	[0.0695, 0.9722]	0.5498	[0.0459, 0.9444]	0.5087	[0.1124, 0.9964]	0.6602	[3.3925E-3, 0.5977]	0.2221	[0.1248, 0.9195]	0.5384
14	[4.0501E-7, 0.8122]	0.2778	[3.6376E-6, 0.7952]	0.2666	[1.6369E-6, 0.8507]	0.32	[2.8822E-6, 0.4595]	0.1203	[0.0383, 0.8013]	0.3961
15	[1.4299E-5, 0.9301]	0.5221	[1.2481E-5, 0.9049]	0.4381	[0.0697, 0.9966]	0.7304	[2.0841E-6, 0.5486]	0.1411	[0.089, 0.9108]	0.5301
16	[3.6986E-3, 0.8309]	0.3529	[1.9645E-3, 0.8058]	0.3234	[0.0142, 0.8877]	0.4327	[1.3156E-3, 0.5119]	0.1728	[0.0671, 0.8157]	0.4206
17	[0.0382, 0.8971]	0.4754	[0.0227, 0.8639]	0.4253	[0.126, 0.9701]	0.6193	[9.522E-3, 0.6044]	0.2521	[0.1117, 0.8561]	0.4939
18	[0.1368, 0.9824]	0.6276	[0.1135, 0.9669]	0.6012	[0.1941, 0.9943]	0.7074	[0.0248, 0.7083]	0.3258	[0.1859, 0.9258]	0.5919
19	[0.4044, 0.9946]	0.8383	[0.3345, 0.9979]	0.8119	[0.6305, 0.9985]	0.9023	[7.285E-3, 0.8214]	0.4813	[0.3808, 0.9805]	0.7673
20	[0.8875, 0.9987]	0.9643	[0.8884, 0.9986]	0.9638	[0.9023, 0.9987]	0.9709	[0.8086, 0.9806]	0.911	[0.8454, 0.9944]	0.9427
21	[0.3096, 0.9959]	0.7851	[0.1309, 0.9845]	0.6775	[0.6976, 0.9975]	0.9093	[1.5809E-3, 0.7462]	0.361	[0.3255, 0.9633]	0.7229

22	[9.9723E-3, 0.8805]	0.4631	[4.3214E-3, 0.8398]	0.3908	[0.1812, 0.9867]	0.7128	[2.5573E-3, 0.5725]	0.2094	[0.0942, 0.8616]	0.4852
23	[0.0593, 0.9152]	0.5157	[0.0487, 0.9057]	0.494	[0.1169, 0.9606]	0.6107	[0.0129, 0.6352]	0.2752	[0.116, 0.8625]	0.4987
24	[0.027, 0.9025]	0.482	[0.0303, 0.8973]	0.4779	[0.0439, 0.9212]	0.5204	[2.6863E-3, 0.6051]	0.2393	[0.0886, 0.8532]	0.4778
25	[0.0247, 0.9104]	0.5107	[0.0151, 0.9028]	0.5008	[0.0377, 0.9297]	0.5409	[1.5647E-3, 0.6127]	0.2252	[0.0981, 0.8788]	0.4976
26	[2.2147E-6, 0.7838]	0.2068	[9.8967E-7, 0.7844]	0.2042	[7.3838E-8, 0.8134]	0.2369	[5.1144E-6, 0.4511]	0.0867	[0.0128, 0.7701]	0.3634
27	[1.4136E-3, 0.8489]	0.3592	[1.2864E-3, 0.8383]	0.3408	[1.9133E-3, 0.8712]	0.4163	[5.1545E-4, 0.5474]	0.1619	[0.0335, 0.8019]	0.4128
28	[4.8791E-4, 0.8065]	0.2769	[1.0333E-3, 0.8029]	0.2692	[6.2085E-4, 0.8335]	0.3138	[2.3463E-4, 0.4863]	0.1275	[0.0246, 0.771]	0.3691
29	[2.9161E-7, 0.7036]	0.109	[1.576E-7, 0.7039]	0.105	[1.6349E-6, 0.745]	0.1377	[9.6673E-7, 0.4008]	0.0479	[2.0116E-3, 0.7081]	0.303
30	[7.619E-7, 0.6758]	0.074	[6.2237E-8, 0.6548]	0.0686	[6.9439E-7, 0.7091]	0.0865	[3.5945E-8, 0.3538]	0.0304	[6.0374E-4, 0.7045]	0.2932
31	[3.4567E-5, 0.5335]	0.063	[1.0885E-4, 0.534]	0.0607	[1.4686E-4, 0.5642]	0.066	[5.3136E-5, 0.3008]	0.036	[1.7629E-3, 0.6195]	0.2325
32	[2.4057E-5, 0.4772]	0.0325	[6.5422E-5, 0.4904]	0.0349	[4.6357E-5, 0.4993]	0.0363	[4.1986E-5, 0.2451]	0.0174	[6.954E-4, 0.6109]	0.2242
33	[8.6923E-5, 0.379]	0.0314	[3.6255E-5, 0.389]	0.0309	[3.1634E-5, 0.3979]	0.0319	[4.8503E-5, 0.2044]	0.0203	[1.5719E-3, 0.5769]	0.1817

prior	**relative extinction rate beta(4,1)**		**relative fossilization rate beta(1,4)**		**relative fossilization rate beta(2,2)**		**relative fossilization rate beta(4,1)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[0.3877, 0.9915]	0.7515	[6.2735E-5, 0.8703]	0.4196	[7.7223E-6, 0.7784]	0.2644	[2.588E-6, 0.6153]	0.1279
2	[0.3444, 0.9859]	0.7059	[9.7698E-5, 0.8263]	0.3471	[4.0858E-6, 0.7435]	0.2396	[6.5532E-8, 0.6189]	0.1358
3	[0.3367, 0.9841]	0.6999	[4.5205E-5, 0.8257]	0.3401	[5.2465E-6, 0.7169]	0.2057	[1.1879E-7, 0.5423]	0.0988
4	[0.3731, 0.9836]	0.7259	[0.0957, 0.9316]	0.5267	[7.1945E-3, 0.8062]	0.347	[2.573E-4, 0.6118]	0.1574
5	[0.2663, 0.9612]	0.6331	[7.4832E-7, 0.7522]	0.2475	[5.338E-8, 0.6304]	0.1467	[8.3991E-7, 0.4253]	0.0635

6	[0.2505, 0.9556]	0.6214	[1.6561E-5, 0.7427]	0.2262	[7.9479E-7, 0.527]	0.0933	[1.3401E-6, 0.253]	0.0311	
7	[0.2468, 0.956]	0.6239	[1.5279E-6, 0.7255]	0.1945	[2.8448E-7, 0.4307]	0.0598	[2.2119E-7, 0.1545]	0.0177	
8	[0.371, 0.9875]	0.721	[0.1238, 0.9361]	0.5366	[0.0175, 0.8053]	0.3347	[6.6465E-4, 0.5827]	0.1483	
9	[0.3965, 0.978]	0.7309	[0.263, 0.9588]	0.6395	[0.1442, 0.9231]	0.5343	[0.0447, 0.847]	0.4082	
10	[0.481, 0.9921]	0.7937	[0.3791, 0.9898]	0.7493	[0.2348, 0.9864]	0.645	[0.0961, 0.9226]	0.5196	
11	[0.7276, 0.9998]	0.9582	[0.9157, 0.9992]	0.9786	[0.9254, 0.9996]	0.9753	[0.9112, 0.9988]	0.9683	
12	[0.4346, 0.9995]	0.8164	[1.5745E-5, 0.9453]	0.4901	[2.5152E-5, 0.8384]	0.3085	[5.8431E-7, 0.702]	0.1769	
13	[0.4853, 0.9985]	0.8341	[0.1741, 0.9937]	0.6756	[0.0472, 0.9186]	0.4731	[1.2797E-3, 0.774]	0.2754	
14	[0.3552, 0.9865]	0.7272	[5.1771E-5, 0.8535]	0.365	[6.2977E-6, 0.7672]	0.2475	[7.886E-7, 0.6547]	0.1497	
15	[0.5079, 0.9969]	0.8444	[0.0659, 0.9848]	0.6577	[2.6822E-6, 0.8372]	0.278	[2.9572E-6, 0.56]	0.0852	
16	[0.3633, 0.9825]	0.7278	[0.05, 0.8972]	0.4726	[6.4837E-3, 0.7922]	0.3164	[8.0407E-4, 0.6693]	0.1878	
17	[0.4083, 0.9863]	0.755	[0.1891, 0.968]	0.6177	[0.0594, 0.8746]	0.4456	[7.0733E-3, 0.7552]	0.2846	
18	[0.4895, 0.9955]	0.8094	[0.3, 0.9917]	0.7376	[0.1474, 0.9599]	0.6002	[0.0445, 0.8934]	0.4566	
19	[0.6563, 0.9966]	0.8841	[0.5485, 0.9989]	0.8738	[0.0835, 0.9835]	0.7046	[7.8852E-5, 0.7492]	0.0972	
20	[0.8844, 0.9971]	0.9636	[0.9021, 0.9991]	0.9688	[0.8693, 0.9981]	0.9569	[0.8666, 0.9989]	0.9524	
21	[0.6393, 0.9953]	0.8676	[0.4154, 0.9957]	0.81	[0.0507, 0.9583]	0.5985	[5.4476E-5, 0.6323]	0.0877	
22	[0.408, 0.9902]	0.7434	[0.1114, 0.9419]	0.5606	[7.139E-3, 0.7646]	0.2797	[3.5064E-4, 0.4523]	0.0884	
23	[0.368, 0.9796]	0.7211	[0.1738, 0.9449]	0.6029	[0.0512, 0.8721]	0.4615	[2.7628E-3, 0.7519]	0.2509	
24	[0.3648, 0.983]	0.7227	[0.1356, 0.949]	0.5734	[0.0293, 0.8707]	0.4262	[1.6322E-3, 0.7161]	0.1932	
25	[0.3967, 0.9866]	0.7477	[0.121, 0.9523]	0.5893	[0.0113, 0.8593]	0.3933	[2.4574E-4, 0.6312]	0.1177	

26	[0.2922, 0.9673]	0.669	[1.0499E-5, 0.8246]	0.2912	[8.0796E-8, 0.6203]	0.0905	[1.1688E-8, 0.1998]	0.0189	
27	[0.2959, 0.9747]	0.6682	[0.0263, 0.8899]	0.4631	[1.9991E-3, 0.7252]	0.1864	[7.6481E-5, 0.2512]	0.0349	
28	[0.2846, 0.9695]	0.6501	[0.0168, 0.8558]	0.3938	[7.7907E-4, 0.6561]	0.1423	[8.969E-5, 0.1882]	0.028	
29	[0.2049, 0.9108]	0.5707	[8.4112E-8, 0.7566]	0.1808	[1.1292E-7, 0.263]	0.0163	[2.7315E-7, 0.0258]	2.45E-03	
30	[0.2406, 0.9436]	0.6078	[7.9434E-7, 0.7114]	0.1247	[9.672E-8, 0.1269]	6.99E-03	[2.0619E-7, 0.0107]	1.09E-03	
31	[0.1849, 0.884]	0.5395	[3.1685E-4, 0.6039]	0.1073	[9.892E-5, 0.1255]	0.0109	[4.1313E-5, 0.0115]	1.85E-03	
32	[0.1976, 0.9135]	0.5662	[7.7889E-5, 0.5642]	0.0605	[4.0193E-5, 0.0449]	2.98E-03	[1.465E-5, 3.1615E-3]	4.35E-04	
33	[0.187, 0.9314]	0.5627	[1.6165E-4, 0.4379]	0.0524	[2.2171E-5, 0.0989]	5.57E-03	[1.7415E-5, 0.0122]	9.34E-04	

Supertree 3 relative extinction rate

prior	**default**		**net diversification rate exp(10)**		**net diversification rate exp(50)**		**relative extinction rate beta(1,4)**		**relative extinction rate beta(2,2)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[8.6021E-6, 0.8304]	0.3217	[2.8719E-6, 0.7721]	0.2481	[2.6698E-4, 0.9199]	0.5204	[3.3676E-6, 0.4624]	0.1205	[0.0452, 0.817]	0.4182
2	[3.2826E-6, 0.7851]	0.2676	[9.5248E-7, 0.7417]	0.2284	[2.1359E-5, 0.8591]	0.3785	[4.8758E-7, 0.4396]	0.1144	[0.0344, 0.7746]	0.3774
3	[1.6628E-6, 0.7748]	0.2494	[2.1571E-6, 0.7318]	0.2114	[1.7464E-6, 0.8584]	0.3754	[7.0385E-7, 0.4254]	0.1063	[0.0291, 0.7678]	0.368
4	[0.0111, 0.8582]	0.4112	[2.405E-3, 0.8195]	0.3585	[0.0858, 0.9616]	0.5786	[1.0887E-3, 0.5351]	0.1886	[0.082, 0.8327]	0.4447
5	[6.6615E-6, 0.6948]	0.179	[3.8251E-7, 0.6712]	0.1644	[5.4045E-6, 0.7687]	0.2411	[6.2184E-6, 0.3925]	0.09	[0.0169, 0.714]	0.3129
6	[3.8282E-6, 0.6703]	0.1505	[1.0698E-6, 0.6415]	0.1341	[1.9393E-7, 0.7847]	0.2344	[7.0015E-7, 0.3717]	0.0768	[7.4071E-3, 0.6983]	0.2952
7	[2.9794E-6, 0.6518]	0.1257	[3.2272E-7, 0.6288]	0.1131	[1.7894E-6, 0.7564]	0.1935	[4.1032E-7, 0.3505]	0.0639	[7.1636E-3, 0.6917]	0.2831
8	[0.0171, 0.8599]	0.4187	[0.0175, 0.856]	0.3971	[0.0556, 0.9182]	0.4848	[3.1021E-3, 0.55]	0.2011	[0.0898, 0.8349]	0.4525

9	[0.0771, 0.8967]	0.4853	[0.0665, 0.8884]	0.4662	[0.1139, 0.9309]	0.5277	[0.0153, 0.5936]	0.2613	[0.1376, 0.8526]	0.4963
10	[0.1597, 0.9739]	0.6115	[0.143, 0.9632]	0.5798	[0.2137, 0.9844]	0.6736	[0.0269, 0.6711]	0.3179	[0.1979, 0.9117]	0.5811
11	[0.9133, 0.9996]	0.9764	[0.925, 0.999]	0.9769	[0.6352, 0.999]	0.9723	[0.8851, 0.9871]	0.9461	[0.8874, 0.9985]	0.9623
12	[4.0113E-6, 0.9275]	0.3892	[3.6094E-6, 0.9113]	0.3697	[9.0114E-6, 0.9505]	0.4286	[2.1518E-5, 0.5026]	0.1351	[0.0479, 0.899]	0.4642
13	[0.0657, 0.9705]	0.5508	[0.0453, 0.9451]	0.5099	[0.111, 0.9964]	0.6666	[4.1213E-3, 0.6004]	0.2245	[0.1276, 0.9203]	0.5387
14	[2.8922E-6, 0.8072]	0.2791	[4.4568E-6, 0.7968]	0.2656	[3.9154E-5, 0.8473]	0.3191	[1.9593E-6, 0.4647]	0.1209	[0.0334, 0.7939]	0.3946
15	[1.071E-5, 0.9292]	0.5172	[6.1954E-6, 0.9065]	0.442	[0.0718, 0.9955]	0.7375	[1.9863E-6, 0.5493]	0.1433	[0.0926, 0.9125]	0.5288
16	[3.5171E-3, 0.8292]	0.3468	[1.4824E-3, 0.8096]	0.3232	[0.0111, 0.8831]	0.4299	[1.4222E-3, 0.5122]	0.1759	[0.061, 0.8079]	0.4221
17	[0.0493, 0.9086]	0.4741	[0.0287, 0.8715]	0.4295	[0.1294, 0.9754]	0.6219	[0.0104, 0.6095]	0.2588	[0.1206, 0.8692]	0.4982
18	[0.22, 0.9957]	0.7341	[0.1873, 0.9917]	0.7096	[0.3325, 0.9968]	0.8162	[0.0415, 0.7789]	0.4101	[0.2476, 0.9488]	0.6641
19	[0.409, 0.9953]	0.8317	[0.3173, 0.9954]	0.8003	[0.635, 0.9999]	0.8995	[0.0173, 0.8306]	0.5039	[0.3767, 0.974]	0.7559
20	[0.8307, 0.9982]	0.9435	[0.8249, 0.9961]	0.9416	[0.8624, 0.9983]	0.9562	[0.7173, 0.9686]	0.8674	[0.7797, 0.9915]	0.914
21	[0.5575, 0.9981]	0.8517	[0.4566, 0.9975]	0.8187	[0.7272, 0.9977]	0.9137	[0.0938, 0.8882]	0.5772	[0.4924, 0.9789]	0.7919
22	[0.045, 0.926]	0.5309	[0.0132, 0.8864]	0.473	[0.1594, 0.9862]	0.7018	[2.1671E-3, 0.6052]	0.2306	[0.1213, 0.8923]	0.5278
23	[0.071, 0.9259]	0.5315	[0.0561, 0.9137]	0.5091	[0.1239, 0.9607]	0.6028	[0.016, 0.6427]	0.284	[0.1273, 0.8695]	0.5029
24	[4.433E-3, 0.8553]	0.3922	[9.0559E-3, 0.8492]	0.3861	[0.0104, 0.8723]	0.4249	[1.1088E-3, 0.5529]	0.1939	[0.0651, 0.8286]	0.4326
25	[6.8807E-3, 0.8787]	0.4451	[6.6365E-3, 0.8764]	0.44	[0.012, 0.8973]	0.4915	[1.2245E-3, 0.5913]	0.2053	[0.0671, 0.8421]	0.4616
26	[6.4567E-7, 0.8155]	0.2409	[2.0779E-6, 0.8112]	0.2326	[1.7578E-6, 0.8411]	0.2806	[2.0031E-6, 0.4753]	0.0913	[0.0135, 0.7839]	0.3873
27	[7.3959E-4, 0.8169]	0.2932	[5.6674E-4, 0.805]	0.2808	[8.2131E-4, 0.8429]	0.3496	[9.0553E-5, 0.5261]	0.1432	[0.0267, 0.7739]	0.3736
28	[8.6709E-4, 0.8359]	0.3122	[1.6147E-4, 0.8278]	0.2996	[4.0455E-4, 0.8593]	0.3677	[8.389E-5, 0.5237]	0.1296	[0.0178, 0.7843]	0.395

29	[2.5458E-7, 0.7815]	0.1493	[6.8485E-8, 0.7749]	0.143	[2.0285E-7, 0.8137]	0.1892	[3.5905E-7, 0.4231]	0.0481	[3.1407E-3, 0.7597]	0.3608
30	[6.8799E-7, 0.4896]	0.0351	[1.4116E-7, 0.4808]	0.0333	[1.7303E-7, 0.5293]	0.0388	[2.3906E-7, 0.2782]	0.0208	[3.2158E-4, 0.6072]	0.2225
31	[1.448E-4, 0.6925]	0.0966	[8.7654E-5, 0.7121]	0.0944	[8.2149E-5, 0.7225]	0.0998	[5.7628E-5, 0.3376]	0.0355	[1.589E-3, 0.7178]	0.3043
32	[4.8049E-5, 0.3823]	0.0265	[7.6949E-5, 0.3804]	0.0264	[4.8835E-5, 0.4062]	0.0271	[4.2703E-5, 0.2106]	0.0169	[5.7786E-4, 0.5573]	0.1833
33	[7.6099E-5, 0.3807]	0.0296	[7.2853E-5, 0.3813]	0.0311	[5.9356E-5, 0.3955]	0.0311	[2.8578E-5, 0.1969]	0.0194	[7.0205E-4, 0.5658]	0.1742
prior	**relative extinction rate beta(4,1)**		**relative fossilization rate beta(1,4)**		**relative fossilization rate beta(2,2)**		**relative fossilization rate beta(4,1)**			
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median		
1	[0.3876, 0.9905]	0.7524	[2.5833E-5, 0.8733]	0.4188	[1.7262E-6, 0.779]	0.2665	[1.5771E-7, 0.6136]	0.1295		
2	[0.3406, 0.9808]	0.7048	[2.0312E-5, 0.8271]	0.3458	[8.0711E-7, 0.7444]	0.2391	[2.514E-6, 0.6148]	0.1332		
3	[0.3351, 0.9826]	0.698	[2.1697E-6, 0.8231]	0.3389	[1.5824E-6, 0.7151]	0.2065	[5.9451E-7, 0.5422]	0.1001		
4	[0.3647, 0.9819]	0.7256	[0.0924, 0.9278]	0.5248	[7.8175E-3, 0.8076]	0.3437	[4.0384E-4, 0.6139]	0.1591		
5	[0.2585, 0.9568]	0.6345	[5.4272E-6, 0.7486]	0.2461	[2.2458E-6, 0.6276]	0.1447	[1.3433E-7, 0.424]	0.0647		
6	[0.2498, 0.9577]	0.6224	[1.8752E-6, 0.7359]	0.2241	[1.9229E-7, 0.5246]	0.0925	[1.6703E-7, 0.2515]	0.0305		
7	[0.2481, 0.9519]	0.6254	[2.9764E-6, 0.7274]	0.1945	[2.4166E-6, 0.4333]	0.0602	[7.1444E-7, 0.1565]	0.0176		
8	[0.3744, 0.9804]	0.7212	[0.1216, 0.9299]	0.5311	[0.023, 0.8086]	0.3384	[1.515E-3, 0.5799]	0.1491		
9	[0.4019, 0.9845]	0.733	[0.2653, 0.9642]	0.6339	[0.1411, 0.9245]	0.5346	[0.0392, 0.8402]	0.4044		
10	[0.4823, 0.9918]	0.7898	[0.3768, 0.989]	0.7467	[0.2302, 0.9703]	0.6462	[0.1017, 0.9307]	0.5203		
11	[0.7213, 0.9995]	0.9564	[0.9171, 0.9993]	0.979	[0.9251, 0.9995]	0.9752	[0.9112, 0.9987]	0.9687		
12	[0.4292, 0.9984]	0.8179	[4.9425E-5, 0.9463]	0.4944	[1.9365E-5, 0.837]	0.3127	[4.5962E-6, 0.7006]	0.1757		

13	[0.4877, 0.9974]	0.8367	[0.175, 0.9954]	0.6739	[0.0312, 0.9036]	0.4733	[1.8384E-3, 0.7752]	0.2755	
14	[0.3574, 0.9867]	0.7281	[1.5553E-5, 0.8537]	0.3638	[1.9187E-6, 0.7679]	0.2503	[5.0068E-8, 0.6581]	0.1518	
15	[0.5185, 0.9991]	0.847	[0.0744, 0.9923]	0.6613	[3.1236E-7, 0.8374]	0.2796	[7.1599E-7, 0.5735]	0.0861	
16	[0.3603, 0.9817]	0.7267	[0.0505, 0.8966]	0.4665	[5.3618E-3, 0.7876]	0.3157	[2.4921E-4, 0.6705]	0.1892	
17	[0.4129, 0.9866]	0.7579	[0.1946, 0.9715]	0.6187	[0.0587, 0.8746]	0.4485	[6.0822E-3, 0.7592]	0.2957	
18	[0.5506, 0.9965]	0.8448	[0.3985, 0.9967]	0.8084	[0.2018, 0.9866]	0.677	[0.08, 0.9397]	0.5441	
19	[0.6516, 0.998]	0.8803	[0.539, 0.9989]	0.8656	[0.0701, 0.9763]	0.6838	[5.0988E-5, 0.7381]	0.0901	
20	[0.8429, 0.9985]	0.9496	[0.8506, 0.9988]	0.9521	[0.8064, 0.9975]	0.9346	[0.821, 0.9975]	0.9375	
21	[0.7015, 0.997]	0.8908	[0.6046, 0.9986]	0.867	[0.2478, 0.9951]	0.764	[6.3462E-5, 0.8325]	0.1626	
22	[0.4414, 0.9869]	0.7741	[0.1575, 0.9655]	0.6188	[0.0112, 0.8147]	0.3273	[6.2296E-4, 0.4935]	0.0975	
23	[0.3745, 0.9879]	0.729	[0.2094, 0.961]	0.6188	[0.0858, 0.9086]	0.4886	[5.5349E-3, 0.7595]	0.274	
24	[0.3352, 0.9816]	0.6977	[0.0767, 0.9148]	0.4991	[0.0115, 0.8181]	0.3368	[4.624E-4, 0.6165]	0.1401	
25	[0.3434, 0.9819]	0.705	[0.0912, 0.9386]	0.5363	[8.9543E-3, 0.8358]	0.3401	[9.3709E-5, 0.5573]	0.0927	
26	[0.3072, 0.9722]	0.6859	[2.1943E-7, 0.8454]	0.3343	[4.8976E-6, 0.6485]	0.0881	[4.9412E-8, 0.179]	0.0163	
27	[0.2403, 0.9472]	0.6151	[0.014, 0.8526]	0.3968	[8.6329E-4, 0.6775]	0.1479	[1.6635E-4, 0.202]	0.0295	
28	[0.2893, 0.9659]	0.6593	[0.0126, 0.8716]	0.427	[3.5151E-4, 0.635]	0.1086	[6.3161E-5, 0.1165]	0.0167	
29	[0.2833, 0.9644]	0.6592	[2.1255E-6, 0.8239]	0.2401	[4.0164E-7, 0.2915]	0.0157	[2.4056E-7, 0.0232]	2.26E-03	
30	[0.145, 0.8414]	0.4946	[1.2809E-6, 0.5528]	0.0597	[1.6051E-7, 0.0725]	4.27E-03	[6.5946E-8, 6.9746E-3]	6.71E-04	
31	[0.2801, 0.9625]	0.652	[4.026E-4, 0.739]	0.1563	[3.0922E-5, 0.1336]	9.70E-03	[2.9001E-5,	1.54E-03	

							9.4879E-3]		
32	[0.1727, 0.8976]	0.5319	[7.2717E-5, 0.4402]	0.0467	[1.7612E-5, 0.0455]	3.28E-03	[1.1021E-5, 3.7006E-3]	5.52E-04	
33	[0.1694, 0.9243]	0.548	[1.4638E-4, 0.4268]	0.0499	[6.0857E-5, 0.0935]	5.62E-03	[2.1688E-5, 0.011]	8.59E-04	

Table S7. Estimates of relative fossilization rates.

Supertree 1 relative fossilization rate										
prior	**default**		**net diversification rate exp(10)**		**net diversification rate exp(50)**		**relative extinction rate beta(1,4)**		**relative extinction rate beta(2,2)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[7.5715E-6, 0.7043]	0.2037	[3.9767E-6, 0.7098]	0.213	[2.4276E-6, 0.669]	0.1788	[4.3064E-6, 0.8042]	0.3129	[1.0055E-6, 0.6083]	0.173
2	[1.8282E-6, 0.7451]	0.253	[1.2272E-5, 0.7478]	0.2509	[4.1834E-6, 0.7428]	0.2483	[1.6128E-5, 0.8223]	0.341	[9.5708E-6, 0.665]	0.213
3	[1.9938E-6, 0.7072]	0.2131	[7.5764E-6, 0.7115]	0.2139	[1.7421E-6, 0.6939]	0.202	[1.0671E-5, 0.7934]	0.3081	[7.7252E-6, 0.6008]	0.1722
4	[4.8804E-3, 0.6752]	0.2271	[3.5816E-3, 0.6829]	0.2308	[4.2934E-3, 0.6349]	0.2075	[0.0326, 0.8165]	0.3837	[4.6282E-3, 0.5648]	0.2004
5	[2.209E-6, 0.6807]	0.1893	[2.3669E-5, 0.6791]	0.1893	[1.0432E-5, 0.6745]	0.1869	[6.324E-6, 0.7571]	0.2684	[4.2975E-6, 0.5373]	0.14
6	[1.226E-6, 0.6105]	0.1296	[2.0232E-6, 0.6186]	0.1332	[1.7784E-6, 0.5901]	0.1193	[4.1044E-6, 0.7025]	0.2061	[9.43E-6, 0.4103]	0.084
7	[1.0703E-5, 0.5637]	0.0953	[1.1856E-6, 0.5732]	0.0953	[2.7251E-6, 0.5453]	0.0893	[3.9679E-6, 0.6565]	0.1635	[1.2172E-6, 0.3283]	0.0567
8	[0.0233, 0.6633]	0.2243	[0.0169, 0.6559]	0.212	[0.0302, 0.6701]	0.2532	[0.0789, 0.8227]	0.4032	[0.03, 0.566]	0.2204
9	[0.2698, 0.939]	0.6155	[0.2468, 0.9332]	0.5963	[0.3475, 0.9646]	0.6728	[0.4582, 0.9873]	0.7749	[0.2964, 0.9167]	0.6277
10	[0.1534, 0.873]	0.4963	[0.1263, 0.8649]	0.4692	[0.226, 0.9169]	0.5629	[0.3716, 0.9831]	0.72	[0.2014, 0.8785]	0.5334
11	[8.8538E-4, 0.0544]	0.0128	[7.4139E-4, 0.0381]	0.0102	[1.2449E-3, 0.4369]	0.0263	[1.2234E-3, 0.0418]	0.0141	[1.0508E-3, 0.0643]	0.0152
12	[1.452E-6, 0.6918]	0.1649	[3.4316E-6, 0.692]	0.1667	[2.0016E-6, 0.6809]	0.1463	[6.6167E-6, 0.8208]	0.324	[1.3621E-6, 0.6173]	0.1491
13	[5.0063E-4, 0.6861]	0.1998	[6.6542E-4, 0.6968]	0.2143	[1.5647E-4, 0.6353]	0.143	[0.0521, 0.8878]	0.4607	[6.984E-4, 0.6353]	0.2087
14	[3.0927E-6, 0.7151]	0.2021	[2.3887E-5, 0.7173]	0.2063	[4.8569E-6, 0.7066]	0.1883	[3.6416E-5, 0.8149]	0.3236	[3.5472E-6, 0.6406]	0.1781
15	[1.1961E-6, 0.352]	0.0351	[1.0756E-6, 0.4114]	0.0421	[2.5892E-6, 0.1783]	0.0163	[1.186E-5, 0.6558]	0.1608	[1.3044E-6, 0.271]	0.04
16	[6.2742E-3, 0.7469]	0.2798	[6.5137E-3, 0.7483]	0.2834	[2.6819E-3, 0.7231]	0.2479	[0.0653, 0.8906]	0.4697	[7.2648E-3, 0.6748]	0.2603

17	[0.0316, 0.7371]	0.3039	[0.0376, 0.7537]	0.3158	[0.016, 0.683]	0.2483	[0.1835, 0.922]	0.5552	[0.0459, 0.6831]	0.3124
18	[0.0153, 0.6184]	0.1967	[0.012, 0.6081]	0.1922	[0.0106, 0.5629]	0.1717	[0.1055, 0.8695]	0.4617	[0.028, 0.593]	0.2312
19	[1.2215E-4, 0.1134]	0.0188	[8.8317E-5, 0.125]	0.0192	[4.3486E-5, 0.0776]	0.0162	[1.7037E-4, 0.4345]	0.0704	[5.3974E-5, 0.1253]	0.0254
20	[0.0208, 0.194]	0.0892	[0.0213, 0.1898]	0.087	[0.0197, 0.1889]	0.084	[0.0413, 0.2625]	0.1319	[0.0259, 0.2184]	0.1015
21	[9.1289E-5, 0.2112]	0.028	[9.3685E-5, 0.311]	0.0368	[1.85E-4, 0.0688]	0.0158	[3.3554E-4, 0.5303]	0.1044	[2.2429E-4, 0.1335]	0.0273
22	[5.2511E-3, 0.4858]	0.1028	[5.6011E-3, 0.5236]	0.1122	[4.7146E-3, 0.3551]	0.0733	[0.0255, 0.7144]	0.2682	[5.8774E-3, 0.3261]	0.089
23	[0.0355, 0.6202]	0.2145	[0.0364, 0.6195]	0.2183	[0.0291, 0.5657]	0.1856	[0.1212, 0.813]	0.4127	[0.0358, 0.4902]	0.1942
24	[8.2086E-5, 0.4118]	0.0563	[1.7629E-4, 0.4131]	0.0575	[3.901E-5, 0.363]	0.0491	[1.4972E-3, 0.6178]	0.1585	[2.6409E-5, 0.2232]	0.0445
25	[8.1819E-4, 0.3042]	0.0463	[6.0308E-4, 0.2979]	0.0473	[7.7336E-4, 0.2523]	0.0409	[4.4028E-3, 0.5659]	0.1328	[1.0693E-3, 0.1794]	0.0403
26	[1.3608E-6, 0.2597]	0.0117	[1.356E-6, 0.2708]	0.0126	[1.0197E-6, 0.202]	9.37E-03	[1.8267E-6, 0.4548]	0.0423	[1.3012E-6, 0.065]	7.64E-03
27	[2.155E-4, 0.4379]	0.0375	[2.7753E-4, 0.4427]	0.0406	[2.8121E-4, 0.3786]	0.0274	[6.8989E-4, 0.5315]	0.0795	[2.2171E-4, 0.1145]	0.0158
28	[1.8477E-4, 0.4429]	0.0305	[8.2364E-5, 0.442]	0.0328	[4.2362E-5, 0.4276]	0.0266	[3.0554E-4, 0.5148]	0.0677	[7.3239E-5, 0.0786]	8.51E-03
29	[1.2835E-6, 0.4018]	0.0154	[1.0902E-6, 0.4047]	0.0158	[1.2619E-6, 0.3892]	0.0154	[1.424E-6, 0.4268]	0.0248	[1.6447E-6, 0.0196]	1.23E-03
30	[1.0454E-6, 0.3844]	0.0129	[1.1567E-6, 0.3905]	0.0129	[1.0768E-6, 0.3808]	0.0124	[1.1736E-6, 0.4189]	0.0218	[1.0299E-6, 0.0152]	8.53E-04
31	[9.8936E-5, 0.4744]	0.0404	[5.8319E-5, 0.4552]	0.0379	[9.9559E-5, 0.4807]	0.0412	[1.7406E-4, 0.5452]	0.067	[7.8587E-5, 0.0833]	7.10E-03
32	[1.289E-3, 0.6767]	0.1504	[7.747E-4, 0.6728]	0.148	[7.7138E-4, 0.6719]	0.1526	[2.2362E-3, 0.719]	0.2001	[4.9421E-4, 0.5586]	0.0868
33	[4.0252E-4, 0.6375]	0.1177	[4.3279E-4, 0.6377]	0.1157	[4.5356E-4, 0.649]	0.1212	[1.0918E-3, 0.6922]	0.1627	[3.5078E-4, 0.5145]	0.0613
prior	**relative extinction rate beta(4,1)**		**relative fossilization rate beta(1,4)**		**relative fossilization rate beta(2,2)**		**relative fossilization rate beta(4,1)**			
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median		

1	[1.502E-6, 0.3654]	0.0694	[2.9527E-6, 0.3835]	0.0936	[0.023, 0.7225]	0.3331	[0.2958, 0.9673]	0.6593	
2	[1.9702E-6, 0.4571]	0.1025	[6.5054E-6, 0.4142]	0.1094	[0.0349, 0.7578]	0.3635	[0.3224, 0.97]	0.6845	
3	[2.643E-6, 0.359]	0.074	[3.4151E-6, 0.3803]	0.0943	[0.0237, 0.7286]	0.3386	[0.3059, 0.9711]	0.6631	
4	[1.3329E-3, 0.3036]	0.0847	[1.9071E-3, 0.3862]	0.1238	[0.0455, 0.7052]	0.3226	[0.2671, 0.9458]	0.6231	
5	[3.9204E-6, 0.2881]	0.059	[3.2343E-6, 0.3702]	0.0899	[0.0257, 0.7151]	0.3197	[0.2775, 0.9611]	0.6478	
6	[1.3741E-6, 0.1658]	0.0304	[1.6147E-6, 0.3128]	0.0626	[8.0205E-3, 0.673]	0.2764	[0.2562, 0.9603]	0.6279	
7	[1.2966E-6, 0.1226]	0.0214	[2.3561E-6, 0.2768]	0.0454	[5.9708E-3, 0.6625]	0.2515	[0.2406, 0.9565]	0.6204	
8	[0.0179, 0.3412]	0.1322	[0.0136, 0.3776]	0.1298	[0.0548, 0.6851]	0.3079	[0.2583, 0.95]	0.6125	
9	[0.223, 0.8364]	0.5174	[0.1309, 0.6767]	0.3846	[0.2991, 0.8918]	0.6052	[0.5216, 0.986]	0.7888	
10	[0.1287, 0.7458]	0.419	[0.0745, 0.5861]	0.2951	[0.2052, 0.8399]	0.5073	[0.4143, 0.972]	0.7236	
11	[6.5562E-4, 0.2656]	0.0262	[4.5371E-4, 0.0475]	9.94E-03	[2.8409E-3, 0.0597]	0.0202	[9.0211E-3, 0.1084]	0.0458	
12	[1.0845E-6, 0.3243]	0.0341	[2.271E-6, 0.3554]	0.0689	[0.0143, 0.7342]	0.3257	[0.3064, 0.9744]	0.6737	
13	[1.122E-4, 0.2997]	0.052	[1.0793E-4, 0.3694]	0.0919	[0.0295, 0.7276]	0.3363	[0.3024, 0.9654]	0.6615	
14	[1.263E-6, 0.3917]	0.0622	[2.053E-6, 0.3774]	0.0861	[0.0301, 0.7457]	0.3486	[0.319, 0.9725]	0.6827	
15	[1.1481E-6, 0.0721]	9.63E-03	[1.7111E-6, 0.1467]	0.0172	[2.2276E-3, 0.5847]	0.1746	[0.1959, 0.9424]	0.5827	
16	[1.0263E-3, 0.4049]	0.105	[2.4603E-3, 0.4238]	0.1334	[0.0624, 0.7607]	0.3857	[0.3425, 0.9719]	0.6873	
17	[0.0102, 0.39]	0.1334	[0.0148, 0.4331]	0.1629	[0.105, 0.7656]	0.4	[0.3473, 0.9606]	0.6731	
18	[8.3089E-3, 0.3169]	0.1002	[9.2208E-3, 0.3428]	0.1115	[0.0529, 0.6578]	0.3009	[0.2394, 0.9226]	0.5655	
19	[8.4229E-5, 0.0502]	0.0113	[4.3713E-5, 0.0655]	0.0132	[9.3748E-4, 0.3786]	0.0669	[0.0661, 0.8877]	0.4907	
20	[0.0115, 0.1475]	0.0613	[0.0165, 0.1586]	0.0706	[0.0417, 0.2562]	0.1328	[0.0765, 0.3271]	0.1879	

21	[1.5357E-5, 0.0347]	8.40E-03	[1.3093E-4, 0.1079]	0.0197	[2.2916E-3, 0.5082]	0.1064	[0.1209, 0.9206]	0.5448	
22	[1.4649E-3, 0.0853]	0.0251	[4.5364E-3, 0.2589]	0.0667	[0.0232, 0.607]	0.2122	[0.1719, 0.9285]	0.5545	
23	[9.9566E-3, 0.1786]	0.0638	[0.0264, 0.3674]	0.1466	[0.0816, 0.647]	0.2968	[0.1976, 0.8895]	0.5111	
24	[7.3936E-5, 0.0505]	0.0117	[4.1913E-5, 0.2002]	0.0379	[7.3509E-3, 0.5662]	0.1584	[0.1339, 0.9257]	0.5424	
25	[2.8517E-4, 0.0437]	0.0115	[9.4867E-4, 0.1525]	0.0343	[7.6808E-3, 0.5035]	0.1209	[0.0957, 0.9066]	0.5128	
26	[1.7064E-6, 0.0124]	1.99E-03	[1.7378E-6, 0.1037]	8.28E-03	[1.4856E-4, 0.5922]	0.1435	[0.195, 0.9476]	0.6012	
27	[8.7996E-5, 0.0168]	3.81E-03	[3.9076E-4, 0.1955]	0.0227	[3.5027E-3, 0.6266]	0.1989	[0.2199, 0.9581]	0.6113	
28	[6.1319E-5, 8.6183E-3]	1.86E-03	[1.0857E-4, 0.1969]	0.0179	[1.262E-3, 0.6332]	0.2078	[0.2219, 0.9527]	0.6115	
29	[1.2862E-6, 1.7005E-3]	2.36E-04	[1.0822E-6, 0.1801]	8.75E-03	[8.2271E-5, 0.6331]	0.2092	[0.2274, 0.9563]	0.6174	
30	[1.2653E-6, 1.2593E-3]	1.80E-04	[1.1602E-6, 0.1756]	7.46E-03	[1.1642E-4, 0.6334]	0.2067	[0.2393, 0.9659]	0.6229	
31	[4.5954E-5, 0.0124]	2.00E-03	[9.127E-5, 0.2241]	0.0225	[1.4894E-3, 0.6407]	0.2166	[0.2289, 0.9517]	0.6106	
32	[3.6263E-4, 0.4953]	0.0518	[1.0758E-3, 0.3522]	0.0739	[6.6358E-3, 0.6962]	0.292	[0.2862, 0.9776]	0.6611	
33	[1.7051E-4, 0.4716]	0.0416	[4.1558E-4, 0.3257]	0.0596	[4.8832E-3, 0.692]	0.2794	[0.2704, 0.9699]	0.6521	

Supertree 2 relative fossilization rate

prior	**default**		**net diversification rate exp(10)**		**net diversification rate exp(50)**		**relative extinction rate beta(1,4)**		**relative extinction rate beta(2,2)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[1.5472E-6, 0.7047]	0.2038	[8.988E-6, 0.7114]	0.2136	[3.1708E-6, 0.6716]	0.1804	[3.9997E-6, 0.8021]	0.3144	[1.9387E-6, 0.6097]	0.1733
2	[9.6429E-6, 0.7463]	0.2505	[1.1252E-5, 0.7476]	0.25	[1.6258E-6, 0.7461]	0.2513	[1.2966E-5, 0.8212]	0.3409	[1.5914E-5, 0.6694]	0.2147
3	[1.3672E-5, 0.7062]	0.2139	[7.35E-6, 0.707]	0.2124	[2.2193E-5, 0.6937]	0.203	[2.1373E-6, 0.793]	0.3048	[8.6381E-6, 0.5991]	0.1716
4	[4.8114E-3, 0.6826]	0.2289	[4.6077E-3, 0.6818]	0.231	[4.0783E-3, 0.6305]	0.2089	[0.0288, 0.8165]	0.3828	[3.6801E-3, 0.5652]	0.1986

5	[2.4034E-6, 0.6823]	0.1912	[7.6671E-6, 0.6788]	0.1908	[3.5454E-6, 0.6758]	0.186	[1.4985E-6, 0.762]	0.2694	[4.0631E-6, 0.5361]	0.1406
6	[4.5999E-6, 0.6163]	0.132	[1.5422E-6, 0.6111]	0.132	[1.5823E-6, 0.5905]	0.1182	[1.0235E-5, 0.697]	0.2029	[1.3147E-6, 0.4081]	0.0843
7	[1.5338E-6, 0.5692]	0.0954	[2.6564E-6, 0.5708]	0.0947	[3.0747E-6, 0.5487]	0.0891	[7.2664E-6, 0.6531]	0.1657	[1.4212E-6, 0.3287]	0.0563
8	[0.0191, 0.6577]	0.2237	[0.0162, 0.6514]	0.2112	[0.0305, 0.6717]	0.253	[0.0863, 0.8303]	0.4066	[0.0267, 0.5563]	0.2192
9	[0.2773, 0.9502]	0.615	[0.2475, 0.9374]	0.5933	[0.346, 0.9629]	0.6739	[0.463, 0.9875]	0.7766	[0.3075, 0.9253]	0.6292
10	[0.1599, 0.883]	0.4937	[0.1295, 0.8656]	0.4729	[0.2162, 0.9123]	0.5565	[0.3723, 0.9818]	0.7181	[0.2082, 0.883]	0.5323
11	[7.7456E-4, 0.0503]	0.0128	[6.474E-4, 0.0377]	0.0101	[8.6627E-4, 0.2783]	0.0226	[1.3362E-3, 0.0424]	0.0142	[9.4093E-4, 0.0572]	0.015
12	[1.0284E-6, 0.6972]	0.1681	[5.3775E-6, 0.6932]	0.1658	[7.2477E-6, 0.6898]	0.1577	[2.4246E-5, 0.8153]	0.3212	[2.6924E-6, 0.6209]	0.1517
13	[6.9148E-4, 0.6903]	0.2039	[4.7481E-4, 0.6975]	0.2164	[4.8547E-5, 0.6623]	0.1634	[0.0495, 0.8813]	0.4543	[6.9596E-4, 0.639]	0.211
14	[5.9775E-6, 0.7195]	0.2053	[2.9371E-6, 0.7147]	0.2052	[1.7238E-6, 0.7146]	0.1975	[1.2087E-5, 0.8169]	0.3249	[7.018E-6, 0.6395]	0.1786
15	[1.6779E-6, 0.4827]	0.0625	[1.5069E-6, 0.5085]	0.0729	[1.7133E-6, 0.3899]	0.0401	[2.3243E-6, 0.7056]	0.2055	[2.8352E-6, 0.3708]	0.0643
16	[5.9627E-3, 0.7413]	0.2702	[6.9758E-3, 0.7418]	0.2736	[3.2185E-3, 0.724]	0.2505	[0.0597, 0.8844]	0.46	[6.9545E-3, 0.6715]	0.2526
17	[0.0231, 0.7187]	0.28	[0.0284, 0.7347]	0.2897	[0.0188, 0.6737]	0.2402	[0.1727, 0.9239]	0.5346	[0.0386, 0.6658]	0.2951
18	[0.0165, 0.6722]	0.2278	[0.0151, 0.6694]	0.2234	[0.0151, 0.6626]	0.2205	[0.1403, 0.9133]	0.5072	[0.0306, 0.6492]	0.2661
19	[2.443E-5, 0.0943]	0.0156	[1.255E-4, 0.0985]	0.0153	[6.4283E-5, 0.0728]	0.0148	[1.3621E-4, 0.4183]	0.062	[1.0506E-4, 0.1133]	0.0221
20	[0.0126, 0.1798]	0.0737	[0.0111, 0.1718]	0.0684	[0.0142, 0.1947]	0.079	[0.026, 0.2383]	0.1097	[0.0176, 0.2116]	0.0888
21	[1.1828E-4, 0.1314]	0.0239	[1.5269E-4, 0.2304]	0.0322	[9.3656E-5, 0.0586]	0.0138	[8.2866E-4, 0.5085]	0.1018	[1.7507E-4, 0.1222]	0.026
22	[4.6321E-3, 0.5018]	0.1066	[5.6577E-3, 0.5511]	0.1229	[3.4594E-3, 0.2798]	0.0633	[0.0267, 0.7182]	0.2682	[6.3902E-3, 0.3363]	0.0933
23	[0.031, 0.5873]	0.1922	[0.0273, 0.6032]	0.198	[0.0247, 0.5153]	0.1663	[0.0921, 0.7914]	0.3795	[0.0294, 0.4732]	0.1799
24	[0.0138, 0.588]	0.1726	[0.0118, 0.5842]	0.1721	[0.0121, 0.5644]	0.1633	[0.0514, 0.7788]	0.3443	[0.0138, 0.4541]	0.1526

25	[3.4188E-3, 0.4906]	0.1109	[3.9771E-3, 0.5017]	0.1131	[2.9427E-3, 0.4765]	0.106	[0.0171, 0.7053]	0.2507	[3.1475E-3, 0.3452]	0.0947
26	[2.8658E-6, 0.4967]	0.0622	[2.5632E-6, 0.4987]	0.0626	[2.002E-6, 0.4788]	0.0565	[3.503E-6, 0.6107]	0.1306	[1.8904E-6, 0.2242]	0.0332
27	[1.3235E-3, 0.4834]	0.0781	[1.4636E-3, 0.4924]	0.0813	[1.1173E-3, 0.4478]	0.0676	[5.0406E-3, 0.6392]	0.1747	[1.948E-3, 0.2556]	0.0547
28	[1.5028E-3, 0.5275]	0.0906	[1.0065E-3, 0.5328]	0.0916	[1.6323E-3, 0.5099]	0.0805	[6.2039E-3, 0.6588]	0.1872	[1.6089E-3, 0.2614]	0.0542
29	[1.6886E-6, 0.414]	0.0243	[1.0812E-6, 0.425]	0.0254	[1.2629E-6, 0.3917]	0.0198	[2.6604E-6, 0.5001]	0.0532	[1.48E-6, 0.0828]	7.47E-03
30	[1.6177E-6, 0.3963]	0.0166	[1.0734E-6, 0.4017]	0.0174	[1.6354E-6, 0.3813]	0.0142	[2.0258E-6, 0.4711]	0.0389	[1.0835E-6, 0.0487]	3.67E-03
31	[1.9871E-4, 0.4617]	0.0383	[2.6568E-4, 0.4759]	0.0401	[9.2845E-5, 0.4736]	0.0369	[4.3913E-4, 0.528]	0.0674	[8.817E-5, 0.0795]	8.31E-03
32	[7.6616E-5, 0.4122]	0.0224	[4.4134E-5, 0.4018]	0.0206	[7.1319E-5, 0.3999]	0.0203	[9.867E-5, 0.4772]	0.0416	[8.8334E-5, 0.0332]	2.71E-03
33	[2.1321E-4, 0.4949]	0.0489	[1.8487E-4, 0.484]	0.0479	[9.2189E-5, 0.5077]	0.0504	[3.6725E-4, 0.5387]	0.0712	[1.2522E-4, 0.1183]	9.68E-03

prior	**relative extinction rate beta(4,1)**		**relative fossilization rate beta(1,4)**		**relative fossilization rate beta(2,2)**		**relative fossilization rate beta(4,1)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[1.341E-6, 0.3674]	0.0687	[1.1873E-6, 0.3824]	0.0925	[0.0232, 0.7206]	0.3324	[0.2918, 0.9627]	0.6613
2	[1.2071E-6, 0.4541]	0.1027	[1.9851E-6, 0.414]	0.1087	[0.0369, 0.7578]	0.3656	[0.3299, 0.9761]	0.685
3	[2.462E-6, 0.3558]	0.074	[1.3734E-6, 0.3804]	0.095	[0.0295, 0.7302]	0.3386	[0.3057, 0.9706]	0.6636
4	[1.3797E-3, 0.3038]	0.0853	[2.3589E-3, 0.392]	0.1247	[0.0424, 0.6997]	0.3206	[0.2754, 0.9545]	0.6234
5	[2.0422E-6, 0.2877]	0.0582	[5.1588E-6, 0.3704]	0.0886	[0.0221, 0.708]	0.3197	[0.287, 0.9678]	0.6494
6	[2.3669E-6, 0.1662]	0.0303	[1.3926E-6, 0.3149]	0.0627	[0.0107, 0.6788]	0.2759	[0.2523, 0.9567]	0.6298
7	[2.1395E-6, 0.1198]	0.0211	[3.0586E-6, 0.274]	0.0451	[3.7648E-3, 0.6587]	0.2531	[0.2448, 0.9574]	0.6245
8	[0.0168, 0.337]	0.1317	[0.0135, 0.3759]	0.1303	[0.0546, 0.6852]	0.3075	[0.2511, 0.9412]	0.6125

9	[0.2205, 0.8376]	0.5172	[0.1409, 0.6874]	0.3861	[0.2945, 0.8874]	0.6029	[0.5228, 0.9888]	0.7898	
10	[0.1382, 0.756]	0.4165	[0.0767, 0.5911]	0.2982	[0.214, 0.8517]	0.5095	[0.4175, 0.9746]	0.7246	
11	[7.177E-4, 0.2414]	0.0242	[4.952E-4, 0.043]	9.80E-03	[2.3109E-3, 0.0594]	0.0202	[8.6154E-3, 0.1086]	0.0457	
12	[1.0845E-6, 0.3361]	0.0359	[1.4061E-6, 0.3581]	0.0691	[8.0146E-3, 0.725]	0.3271	[0.3047, 0.9728]	0.6747	
13	[3.5689E-5, 0.3193]	0.0586	[1.4498E-4, 0.3693]	0.0944	[0.0326, 0.7282]	0.3382	[0.2997, 0.9667]	0.662	
14	[1.7818E-6, 0.4078]	0.0677	[1.8043E-6, 0.3805]	0.0879	[0.0313, 0.7484]	0.3488	[0.3169, 0.9748]	0.6827	
15	[2.2833E-6, 0.1263]	0.0161	[2.091E-6, 0.219]	0.0291	[4.3317E-3, 0.6308]	0.2262	[0.2306, 0.952]	0.6067	
16	[9.9551E-4, 0.4103]	0.1068	[7.9207E-4, 0.4146]	0.1285	[0.0614, 0.762]	0.378	[0.3369, 0.9691]	0.6862	
17	[9.4088E-3, 0.3857]	0.1305	[0.0122, 0.4123]	0.147	[0.0944, 0.7547]	0.3833	[0.3349, 0.9596]	0.6642	
18	[8.0718E-3, 0.3844]	0.1219	[8.3901E-3, 0.3823]	0.1248	[0.0628, 0.7016]	0.334	[0.2648, 0.9375]	0.6055	
19	[1.3971E-4, 0.0506]	0.0108	[3.8887E-5, 0.0553]	0.0104	[1.0159E-3, 0.3435]	0.0582	[0.0626, 0.8902]	0.4704	
20	[8.8767E-3, 0.1576]	0.0588	[0.0107, 0.1456]	0.0575	[0.0289, 0.2484]	0.1181	[0.0644, 0.3272]	0.1777	
21	[7.5337E-5, 0.0361]	8.34E-03	[6.1935E-5, 0.0832]	0.0172	[1.5439E-3, 0.4058]	0.0794	[0.0791, 0.8901]	0.4925	
22	[1.7692E-3, 0.0901]	0.0267	[4.2287E-3, 0.2615]	0.0677	[0.0271, 0.6305]	0.2248	[0.1805, 0.9297]	0.5662	
23	[8.6488E-3, 0.1724]	0.0609	[0.0211, 0.3509]	0.1317	[0.059, 0.6269]	0.2704	[0.1737, 0.8951]	0.5121	
24	[3.107E-3, 0.1557]	0.048	[0.0115, 0.3368]	0.1125	[0.0431, 0.6334]	0.2595	[0.1835, 0.9179]	0.5406	
25	[7.9941E-4, 0.1015]	0.0282	[3.5465E-3, 0.2664]	0.0739	[0.0211, 0.5992]	0.2074	[0.1586, 0.9259]	0.5422	
26	[1.659E-6, 0.0498]	7.88E-03	[1.5353E-6, 0.2363]	0.0351	[3.3043E-3, 0.6291]	0.2161	[0.2033, 0.9445]	0.6027	
27	[5.664E-4, 0.0572]	0.0147	[1.865E-3, 0.2427]	0.0503	[0.0131, 0.6119]	0.1968	[0.1902, 0.9497]	0.592	
28	[3.8905E-4, 0.0542]	0.0135	[1.3602E-3, 0.2621]	0.0549	[0.0136, 0.632]	0.219	[0.21, 0.9517]	0.6028	

29	[1.1359E-6, 0.0108]	1.61E-03	[1.1918E-6, 0.1855]	0.0134	[5.9155E-4, 0.6288]	0.202	[0.223, 0.9573]	0.6141	
30	[1.1972E-6, 5.2708E-3]	6.98E-04	[1.154E-6, 0.1727]	8.98E-03	[5.9665E-4, 0.6285]	0.2037	[0.2299, 0.9605]	0.6125	
31	[5.8259E-5, 7.5786E-3]	1.57E-03	[2.1443E-4, 0.2185]	0.0212	[2.2312E-3, 0.6377]	0.2124	[0.2336, 0.9633]	0.6156	
32	[2.4243E-5, 2.7509E-3]	6.07E-04	[5.2326E-5, 0.1902]	0.0118	[4.9845E-4, 0.6387]	0.2109	[0.2346, 0.9661]	0.6271	
33	[7.8367E-5, 0.0345]	3.29E-03	[1.2851E-4, 0.2385]	0.0286	[1.7579E-3, 0.6451]	0.2239	[0.2381, 0.9621]	0.6208	

Supertree 3 relative fossilization rate

prior	**default**		**net diversification rate exp(10)**		**net diversification rate exp(50)**		**relative extinction rate beta(1,4)**		**relative extinction rate beta(2,2)**	
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median
1	[2.5599E-6, 0.7028]	0.2036	[2.8317E-6, 0.7147]	0.213	[6.416E-6, 0.6733]	0.1804	[3.8256E-5, 0.8015]	0.3133	[5.139E-6, 0.6105]	0.1741
2	[9.9878E-6, 0.7459]	0.25	[4.8992E-6, 0.7432]	0.2494	[1.9062E-6, 0.7453]	0.2493	[1.0661E-5, 0.8234]	0.3403	[6.5009E-6, 0.6682]	0.2151
3	[1.9181E-6, 0.7074]	0.212	[3.8142E-6, 0.7094]	0.2121	[1.2958E-5, 0.694]	0.2035	[3.9949E-5, 0.7938]	0.3099	[7.4178E-6, 0.6006]	0.1733
4	[2.9961E-3, 0.6753]	0.2293	[5.1579E-3, 0.6838]	0.231	[3.2606E-3, 0.6328]	0.2072	[0.0286, 0.8167]	0.3816	[4.8827E-3, 0.5697]	0.1998
5	[1.1755E-6, 0.6849]	0.193	[1.1251E-5, 0.6792]	0.1915	[1.0679E-5, 0.6739]	0.1872	[1.2776E-6, 0.7571]	0.2682	[4.0403E-6, 0.5365]	0.1399
6	[1.1042E-6, 0.6161]	0.1309	[2.2055E-6, 0.6157]	0.1313	[2.1811E-6, 0.5869]	0.1176	[3.9801E-6, 0.7015]	0.2061	[2.7514E-6, 0.4054]	0.0841
7	[2.8437E-6, 0.5636]	0.0947	[2.4811E-6, 0.5668]	0.0946	[2.0748E-6, 0.5572]	0.0903	[3.62E-6, 0.6604]	0.1663	[1.1935E-6, 0.3277]	0.0564
8	[0.0223, 0.6623]	0.2243	[0.0166, 0.6547]	0.2124	[0.0335, 0.6831]	0.2541	[0.0814, 0.8276]	0.4069	[0.0273, 0.5566]	0.221
9	[0.2765, 0.9453]	0.6158	[0.2438, 0.9333]	0.5916	[0.3478, 0.9633]	0.674	[0.4627, 0.9881]	0.7758	[0.3047, 0.925]	0.6285
10	[0.1487, 0.8777]	0.4969	[0.1373, 0.8735]	0.4729	[0.2205, 0.9139]	0.5635	[0.3752, 0.9811]	0.7205	[0.2015, 0.8742]	0.533
11	[7.0058E-4, 0.0515]	0.0129	[7.7188E-4, 0.0376]	0.0104	[1.0638E-3, 0.3213]	0.0235	[1.6247E-3, 0.0425]	0.0143	[8.5791E-4, 0.0585]	0.0152
12	[4.1256E-6, 0.6876]	0.1631	[1.8028E-6, 0.6929]	0.166	[7.3771E-6, 0.6885]	0.1546	[4.3009E-5, 0.8187]	0.3203	[1.8094E-6, 0.6244]	0.1523

13	[4.3466E-4, 0.6939]	0.2069	[2.6014E-4, 0.6973]	0.216	[9.1609E-6, 0.6573]	0.1623	[0.0526, 0.8866]	0.4599	[1.1358E-3, 0.6377]	0.2127
14	[5.3325E-6, 0.7183]	0.2064	[4.3436E-6, 0.7216]	0.2066	[1.0853E-5, 0.7157]	0.195	[2.6817E-5, 0.8166]	0.327	[4.8608E-6, 0.6458]	0.1827
15	[1.8542E-6, 0.488]	0.064	[2.2421E-6, 0.512]	0.0717	[1.2304E-6, 0.3828]	0.0377	[1.6733E-5, 0.7057]	0.2097	[1.9858E-6, 0.3756]	0.0652
16	[4.6119E-3, 0.7387]	0.2711	[6.182E-3, 0.7455]	0.2742	[2.5953E-3, 0.724]	0.2464	[0.07, 0.8939]	0.4665	[7.121E-3, 0.6689]	0.2537
17	[0.026, 0.724]	0.2791	[0.0258, 0.7367]	0.2904	[0.0157, 0.6643]	0.231	[0.1757, 0.919]	0.5449	[0.0415, 0.6707]	0.2984
18	[8.3406E-3, 0.4892]	0.1319	[7.197E-3, 0.498]	0.1328	[5.8172E-3, 0.4301]	0.117	[0.0608, 0.7981]	0.3746	[0.0145, 0.4958]	0.174
19	[1.3194E-4, 0.0861]	0.0154	[4.2122E-5, 0.0975]	0.0153	[1.3439E-4, 0.0645]	0.0134	[8.116E-5, 0.3758]	0.0578	[2.2432E-4, 0.1048]	0.0214
20	[0.0207, 0.1992]	0.0883	[0.0172, 0.1905]	0.0846	[0.0204, 0.1989]	0.0876	[0.0473, 0.3]	0.1482	[0.0271, 0.2332]	0.1047
21	[1.2829E-4, 0.0623]	0.0149	[1.7265E-4, 0.0748]	0.0164	[1.4987E-4, 0.0449]	0.0115	[2.2291E-4, 0.2662]	0.0462	[1.8842E-4, 0.0677]	0.0167
22	[5.236E-3, 0.4655]	0.1002	[4.9093E-3, 0.5089]	0.1122	[3.9784E-3, 0.3126]	0.0712	[0.0283, 0.7107]	0.2659	[6.2695E-3, 0.3257]	0.0912
23	[0.0425, 0.6219]	0.2252	[0.0389, 0.6387]	0.2331	[0.0344, 0.5647]	0.2013	[0.1219, 0.8191]	0.4222	[0.0409, 0.5168]	0.2108
24	[7.6091E-3, 0.6228]	0.1817	[8.1921E-3, 0.6257]	0.184	[7.1395E-3, 0.608]	0.1711	[0.035, 0.7802]	0.3442	[7.8614E-3, 0.4646]	0.1465
25	[2.821E-3, 0.5103]	0.1106	[3.6402E-3, 0.5111]	0.1122	[2.2293E-3, 0.4872]	0.1011	[0.0166, 0.7062]	0.2436	[4.092E-3, 0.3418]	0.0893
26	[1.5752E-6, 0.4602]	0.0492	[2.0966E-6, 0.4629]	0.0499	[1.3922E-6, 0.4328]	0.0425	[4.5307E-6, 0.5896]	0.1144	[2.211E-6, 0.1955]	0.0269
27	[1.5456E-3, 0.4975]	0.0783	[1.0457E-3, 0.5113]	0.0816	[8.8341E-4, 0.4648]	0.0669	[4.1989E-3, 0.6391]	0.168	[1.3954E-3, 0.2448]	0.0506
28	[7.0469E-4, 0.4412]	0.0513	[9.2289E-4, 0.4545]	0.0532	[3.8812E-4, 0.409]	0.0436	[2.2436E-3, 0.5901]	0.1273	[4.6795E-4, 0.18]	0.0308
29	[1.5025E-6, 0.3897]	0.0175	[2.0929E-6, 0.3851]	0.0178	[1.2862E-6, 0.3666]	0.0144	[3.7802E-6, 0.4946]	0.0488	[1.3868E-6, 0.0636]	5.75E-03
30	[1.0102E-6, 0.4158]	0.0211	[1.2704E-6, 0.4216]	0.0217	[1.4053E-6, 0.4098]	0.0191	[2.0432E-6, 0.4656]	0.0369	[1.0051E-6, 0.0449]	3.06E-03
31	[2.0264E-4, 0.3926]	0.0215	[1.0302E-4, 0.4253]	0.0218	[1.0531E-4, 0.4225]	0.0211	[4.0709E-4, 0.5066]	0.0581	[1.2081E-4, 0.056]	5.39E-03
32	[9.7991E-5, 0.4611]	0.0321	[1.2326E-4, 0.4353]	0.0318	[9.791E-5, 0.4664]	0.0319	[1.6982E-4, 0.4962]	0.0499	[8.4629E-5, 0.0472]	4.04E-03

33	[2.849E-4, 0.4931]	0.0489	[1.5022E-4, 0.4675]	0.0465	[1.8428E-4, 0.5093]	0.0509	[3.0901E-4, 0.5252]	0.0719	[5.3295E-5, 0.1131]	9.69E-03
prior	**relative extinction rate beta(4,1)**		**relative fossilization rate beta(1,4)**		**relative fossilization rate beta(2,2)**		**relative fossilization rate beta(4,1)**			
estimation	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median	95% HPD interval	median		
1	[1.8276E-6, 0.3659]	0.0688	[4.104E-6, 0.3863]	0.093	[0.0205, 0.723]	0.333	[0.296, 0.9655]	0.6586		
2	[1.3454E-6, 0.4579]	0.1041	[2.5503E-6, 0.4149]	0.1085	[0.0347, 0.752]	0.3667	[0.3289, 0.9772]	0.6853		
3	[3.3334E-6, 0.3602]	0.0736	[1.1318E-6, 0.3844]	0.0957	[0.0292, 0.7323]	0.3382	[0.3065, 0.9686]	0.6661		
4	[6.1856E-4, 0.3043]	0.0846	[1.5085E-3, 0.3878]	0.1257	[0.0395, 0.6993]	0.3227	[0.2719, 0.9513]	0.6227		
5	[1.8621E-6, 0.2861]	0.0583	[1.8222E-6, 0.3687]	0.0902	[0.022, 0.7148]	0.3205	[0.2846, 0.9625]	0.6459		
6	[1.7277E-6, 0.1682]	0.0304	[2.0076E-6, 0.3112]	0.0631	[0.0119, 0.6776]	0.2783	[0.2589, 0.9593]	0.631		
7	[1.9925E-6, 0.1215]	0.0212	[2.6706E-6, 0.2777]	0.0448	[6.4551E-3, 0.6542]	0.2524	[0.2425, 0.9551]	0.6248		
8	[0.018, 0.3399]	0.131	[0.0141, 0.3772]	0.1305	[0.0536, 0.6841]	0.3087	[0.2559, 0.9455]	0.6114		
9	[0.2156, 0.8317]	0.517	[0.1391, 0.6833]	0.3884	[0.2957, 0.8902]	0.6036	[0.5235, 0.9852]	0.7908		
10	[0.136, 0.7581]	0.4209	[0.0722, 0.586]	0.2966	[0.2009, 0.8377]	0.5089	[0.4128, 0.9691]	0.7223		
11	[7.6557E-4, 0.2549]	0.025	[6.6293E-4, 0.0432]	9.71E-03	[2.2043E-3, 0.059]	0.0203	[9.37E-3, 0.1085]	0.0458		
12	[1.2234E-6, 0.3276]	0.0348	[2.4692E-6, 0.3565]	0.0694	[7.9354E-3, 0.7264]	0.3274	[0.302, 0.9743]	0.6769		
13	[2.0079E-4, 0.314]	0.0566	[2.9105E-4, 0.3692]	0.0944	[0.0307, 0.7263]	0.3355	[0.3038, 0.9683]	0.6618		
14	[2.2966E-6, 0.4]	0.0657	[1.1394E-6, 0.3764]	0.0869	[0.0273, 0.7447]	0.3474	[0.3181, 0.9725]	0.6814		
15	[1.1205E-6, 0.1188]	0.0154	[1.162E-6, 0.2164]	0.0293	[2.5698E-3, 0.6296]	0.2247	[0.2242, 0.9494]	0.6072		
16	[5.936E-4, 0.4025]	0.1028	[1.5554E-3, 0.4147]	0.1298	[0.0674, 0.7646]	0.3814	[0.3444, 0.9733]	0.6872		

17	[8.3633E-3, 0.3706]	0.1225	[0.0114, 0.4126]	0.1473	[0.0928, 0.7493]	0.3836	[0.3324, 0.9593]	0.662	
18	[4.0343E-3, 0.2407]	0.0714	[5.4437E-3, 0.2666]	0.0779	[0.0353, 0.5857]	0.2376	[0.1743, 0.8669]	0.4825	
19	[7.1488E-5, 0.042]	9.25E-03	[6.0661E-5, 0.0526]	0.0105	[1.1447E-3, 0.3369]	0.0558	[0.0537, 0.8836]	0.4677	
20	[0.011, 0.1463]	0.059	[0.015, 0.159]	0.069	[0.0425, 0.2728]	0.1356	[0.0833, 0.3544]	0.205	
21	[4.9744E-5, 0.0241]	6.16E-03	[8.4529E-5, 0.0454]	0.0112	[1.0907E-3, 0.1915]	0.0428	[0.0313, 0.8443]	0.3515	
22	[1.9632E-3, 0.0888]	0.0274	[4.1453E-3, 0.2402]	0.0654	[0.0259, 0.6087]	0.2149	[0.1828, 0.9343]	0.5658	
23	[0.0127, 0.2034]	0.0751	[0.0287, 0.3816]	0.1546	[0.0807, 0.6535]	0.3052	[0.2133, 0.91]	0.5429	
24	[2.114E-3, 0.1627]	0.0459	[6.4679E-3, 0.3517]	0.1108	[0.038, 0.6635]	0.2816	[0.2247, 0.9413]	0.5852	
25	[9.7532E-4, 0.097]	0.0268	[3.1013E-3, 0.273]	0.0723	[0.0223, 0.6089]	0.2105	[0.1635, 0.9249]	0.5575	
26	[1.47E-6, 0.0403]	6.58E-03	[3.1254E-6, 0.2131]	0.0277	[1.1305E-3, 0.623]	0.2005	[0.2067, 0.9511]	0.6015	
27	[5.0741E-4, 0.0536]	0.0133	[1.0568E-3, 0.2484]	0.0489	[0.0112, 0.6231]	0.2064	[0.2015, 0.9527]	0.5926	
28	[1.7554E-4, 0.032]	7.59E-03	[8.2027E-4, 0.2053]	0.0325	[6.6522E-3, 0.6145]	0.1874	[0.2041, 0.9523]	0.6038	
29	[1.7491E-6, 8.8349E-3]	1.31E-03	[1.0972E-6, 0.1669]	0.0101	[7.5023E-4, 0.6245]	0.195	[0.217, 0.9504]	0.612	
30	[1.0729E-6, 4.2444E-3]	5.46E-04	[1.2354E-6, 0.1867]	0.0112	[5.454E-4, 0.6301]	0.2095	[0.2355, 0.9585]	0.6175	
31	[5.4262E-5, 4.9351E-3]	1.05E-03	[1.0955E-4, 0.1865]	0.0124	[9.7443E-4, 0.6212]	0.2053	[0.2273, 0.9432]	0.613	
32	[4.5673E-5, 3.7977E-3]	8.51E-04	[4.0363E-5, 0.2129]	0.0182	[8.7453E-4, 0.6479]	0.2176	[0.2316, 0.9504]	0.6158	
33	[8.5264E-5, 0.0359]	3.47E-03	[6.4374E-5, 0.2425]	0.0285	[1.3789E-3, 0.6414]	0.2189	[0.2383, 0.9661]	0.6231	

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